INFORMATION TO USERS

This manuscript has been reproduced from the microfilm master. UMI films the text directly from the original or copy submitted. Thus, some thesis and dissertation copies are in typewriter face, while others may be from any type of computer printer.

The quality of this reproduction is dependent upon the quality of the copy submitted. Broken or indistinct print, colored or poor quality illustrations and photographs, print bleedthrough, substandard margins, and improper alignment can adversely affect reproduction.

In the unlikely event that the author did not send UMI a complete manuscript and there are missing pages, these will be noted. Also, if unauthorized copyright material had to be removed, a note will indicate the deletion.

Oversize materials (e.g., maps, drawings, charts) are reproduced by sectioning the original, beginning at the upper left-hand corner and continuing from left to right in equal sections with small overlaps. Each original is also photographed in one exposure and is included in reduced form at the back of the book.

Photographs included in the original manuscript have been reproduced xerographically in this copy. Higher quality 6" x 9" black and white photographic prints are available for any photographs or illustrations appearing in this copy for an additional charge. Contact UMI directly to order.

UMI

A Bell & Howell Information Company
300 North Zeeb Road, Ann Arbor MI 48106-1346 USA
313/761-4700 800/521-0600

GREEN'S FUNCTION MONTE CARLO CALCULATIONS OF FEW NUCLEON SYSTEMS

BY

BRIAN SCOTT PUDLINER

B.S., Pennsylvania State University, 1991
M.S., University of Illinois, 1992

THESIS

Submitted in partial fullfilment of the requirements
for the degree of Doctor of Philosophy in Physics
in the Graduate College of the
University of Illinois at Urbana-Champaign, 1996

Urbana, Illinois

UMI Number: 9702643

UMI Microform 9702643
Copyright 1996, by UMI Company. All rights reserved.

This microform edition is protected against unauthorized copying under Title 17, United States Code.

UMI
300 North Zeeb Road
Ann Arbor, MI 48103

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN

THE GRADUATE COLLEGE

APRIL 1996

WE HEREBY RECOMMEND THAT THE THESIS BY

BRIAN SCOTT PUDLINER

ENTITLED GREEN'S FUNCTION MONTE CARLO CALCULATIONS OF FEW NUCLEON SYSTEMS

BE ACCEPTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR

THE DEGREE OF DOCTOR OF PHILOSOPHY

V. R. Pandharipande
Director of Thesis Research

Head of Department

Committee on Final Examination†

Chairperson

† Required for doctor's degree but not for master's.

O-517

GREEN'S FUNCTION MONTE CARLO CALCULATIONS OF FEW NUCLEON SYSTEMS

Brian Scott Pudliner, Ph.D.
Department of Physics
University of Illinois at Urbana-Champaign, 1996
Vijay Pandharipande, Advisor

Recently, a realistic nucleon-nucleon potential has been developed with the goal of accurately reproducing isospin symmetry breaking effects in the nuclear interaction. We briefly discuss the motivation for this potential, called Argonne v_{18}, as well as the Urbana models of three-nucleon interactions. Our approach for modeling the ground states of nuclear systems begins with the development of a variational wave function. We discuss the basis for our variational wave functions and the Monte Carlo techniques used to evaluate expectation values with these wave functions. To determine the exact ground state, Green's function Monte Carlo (GFMC) techniques are used. We review the method of GFMC in general and then discuss its application to the problem of nuclear ground states in some detail.

We study a range of nuclear ground states using GFMC. A baseline calculation is done for the ^3H and ^4He ground states to fit the parameters of the three-nucleon potential and to evaluate the ^3He-^3H mass splitting. A spectrum of six-body states, including the lowest energy $J^\pi = 0^+, 1^+, 2^+$, and 3^+ states of ^6Li and 0^+ states of ^6He and ^6Be, is then investigated. Isospin symmetry breaking effects are probed by measuring the ^6Be-^6He isovector and $\frac{1}{2}(^6Be+^6He)$-^6Li isotensor mass splittings. The calculations are then extended to the ^7Li $J^\pi = \frac{1}{2}^-, \frac{3}{2}^-, \frac{5}{2}^-, \frac{7}{2}^-$ states, and the isovector mass splitting ^7Be-^7Li is evaluated. To provide constraints on Skyrme-type effective interactions in neutron-rich systems, GFMC calculations for ^{8}n($J^\pi = 0^+$) and ^{7}n($J^\pi = 1/2^-$ and $3/2^-$) neutron drop clusters are carried out. The spin-orbit splitting of the ^{7}n drops and the central density of ^{8}n are used to offer improvements to energy-density functionals.

Acknowledgments

When in the course of graduate education it becomes necessary for a student to dissolve the obligatory bands which have connected him with his advisor, and to assume among the powers of the earth, the separate and equal station to which the laws of nature entitle him, a decent respect to the opinions of his peers requires that he should write a thesis. Having accomplished that, it is time to acknowledge the folks who have had a hand in making the culmination of my 21 year formal education a reality.

I would like to thank Vijay Pandharipande for providing direction, time and understanding over the past five years. Of course, I could not have performed these calculations without Joe Carlson's help and instruction in the area of Green's function Monte Carlo. I would also like to thank Bob Wiringa for his help with the Variational Monte Carlo, his superb wave functions, and for expanding my cinematic horizons. The neutron drop work would have been impossible without the work of Geoff Ravenhall, Steve Pieper, and Augusto Smerzi. I would also like to thank Steve Pieper for his assistance with the IBM SP1 at Argonne.

It would have been very hard to maintain insanity without the denizens of 361 Loomis, Arya Akmal, Solomon Feinberg, Jun Forest, Roger Loucks, Mark Paris, Chuck Plager and the ghost of 361, Dan Lewart. I will really miss the practical jokes, heated arguments, and general immaturity which reigned in our office.

I would like to acknowledge Ron Stanley, a classmate who has inspired and challenged me since my undergraduate days.

Inspiring teachers are at a premium, thus I must thank Alfred Actor, Richard Robinett, and Dan Reed. Alfred Actor, is the first physicist I ever met and was the inspiration for my foray into the field. Richard Robinett taught me the wonders of quantum mechanics and Dan Reed taught me the wonders of the microprocessor.

I would like to thank NSF PHY 94-21309 for providing the funds for this research and the Mathematics and Computer Science Division, Argonne National Laboratory and Cornell

Theory Center for their generous grants of computer time on the IBM SP1 and IBM SP2 respectively.

Finally, and most important, I thank my wife, Kim, for her willingness to live in central Illinois, for her support in my career choices, and for always being there for me.

Contents

1 Introduction **1**

2 The Hamiltonian **3**

2.1 The NN Potential 3

2.2 The NNN Potential 5

3 The Wave Function and Variational Monte Carlo **7**

3.1 The Variational Wave Function 8

3.2 Variational Monte Carlo 14

4 Green's Function Monte Carlo **20**

4.1 The General Idea 20

4.2 GFMC Calculations of Nuclear Ground States 28

4.2.1 Choosing and Sampling $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$ 29

4.2.2 Building $G(\mathbf{R}, \mathbf{R}')$ 31

5 Results **39**

5.1 Computational Considerations 39

5.2 Three- and Four-body Results 42

5.3 Six-body Results 46

5.4 Seven-body Results 64

5.5 Neutron Drop Results 70

6 Conclusions **80**

6.1 Tying It All Together 80

6.2 Fermion Sign Problem 84

6.3 Musings on the Future 84

Appendix: Reduced Isospin States 88

Bibliography 93

Vita 97

List of Figures

2.1 Simplest two-pion exchange three-body diagram 6

3.1 Correlation functions for ^6Li (1^+). Note that the non-central correlations have been multiplied by 5 to amplify their features 13

5.1 E(τ) and fit for ^6Li (1^+) and ^6Li (2^+) states using Argonne v_{18} + Urbana IX and $\triangle\tau = 0.0005$ MeV^{-1} 50

5.2 E(τ) and fit for ^6Li (3^+) and ^6Li (0^+) states using Argonne v_{18} + Urbana IX and $\triangle\tau = 0.0005$ MeV^{-1} 51

5.3 E(τ) and fit for ^4He (0^+) and ^6He (0^+) states using Argonne v_{18} + Urbana IX and $\triangle\tau = 0.0005$ MeV^{-1} 52

5.4 Short time E(τ) and fit for ^4He using Argonne v_8 + Urbana IX and $\triangle\tau = 0.0003$ MeV^{-1} 54

5.5 χ^2 for $^6Li(0^+)$, $^6Li(1^+)$, $^6Li(2^+)$, $^6Li(3^+)$, and $^6He(0^+)$ 56

5.6 Point nucleon and pair densities for ^6Li (1^+) where the solid lines represent VMC results. 62

5.7 Point nucleon and pair densities for ^6He (0^+) where the solid lines represent VMC results. 63

5.8 E(τ) and fit for ^7Li $(\frac{3}{2}^-)$ and ^7Li $(\frac{1}{2}^-)$ states using Argonne v_{18} + Urbana IX and $\triangle\tau = 0.0005$ MeV^{-1} 67

5.9 E(τ) and fit for ^7Li $(\frac{5}{2}^-)$ and ^7Li $(\frac{7}{2}^-)$ states using Argonne v_{18} + Urbana IX and $\triangle\tau = 0.0005$ MeV^{-1} 68

5.10 Single particle wave functions and Woods Saxon wells for 8 neutron variational wave function 73

5.11 The transient energy E(τ) for the GFMC calculations of ^{8}n(0^+) and ^{7}n($1/2^-$ and $3/2^-$) as a function of the imaginary time τ. 75

5.12 Neutron density distribution for ^{8}n, according to methods described in the text. The curves labeled A and B come from modified versions of the effective interaction FPS21. 78

6.1 The GFMC energy spectrum for 4- to 7-body nuclei compared to experiment. 81

6.2 Normalization as a function of τ for systems studied in this work 85

List of Tables

3.1 Size of wave function in number of spin-isospin basis elements 17

5.1 GFMC code performance on the Cornell IBM SP2 Wide Nodes. N is the size of the wave function in spin-isospin states and P is the number of pairs . . . 41

5.2 Variational Parameters for Three- and Four- body nuclei 43

5.3 Three- and Four-body GFMC Results without TNI. All results are in MeV . 44

5.4 Three- and Four-body Isospin Symmetry Breaking Contributions. All results are in KeV. 45

5.5 Variational Parameters for Six- and Seven- body nuclei 48

5.6 Mixing parameters for 6-body nuclei 48

5.7 VMC expectation values for 3- to 6-body nuclei using Argonne v_{18} + Urbana IX potential. All energies are in MeV. 49

5.8 Fit results for 6-body data. Energies are in MeV 55

5.9 Fit results for 6-body data using "physical" energies. Energies are in MeV . 57

5.10 GFMC expectation values for 3- to 5-body nuclei using Argonne v_{18} + Urbana IX potential. All energies are in MeV. 58

5.11 GFMC expectation values for 6-body nuclei using Argonne v_{18} + Urbana IX potential. All energies are in MeV. 59

5.12 Energy differences in MeV within isospin multiplets 60

5.13 VMC expectation values for 7-body nuclei using Argonne v_{18} + Urbana IX potential. All energies are in MeV. 65

5.14 Fit results for 7-body data. Energies are in MeV 66

5.15 Fit results for 7-body data using "physical" energies. Energies are in MeV . 66

5.16 Energy differences in MeV within 7-body isospin doublets 70

5.17 Comparison of microscopic and Skyrme-model energies. 77

6.1 Ground state energies calculated using the Argonne v_{18} + Urbana IX interactions 82
6.2 Fits to the experimental spectrum. All energies are in MeV 83

Chapter 1

Introduction

Models of nuclear interactions have been around since the discovery of the neutron. As computational techniques have improved, so has the sophistication of these models. Since the true test of any model is its ability to accurately reflect the physical system which it models, improvements in nuclear interaction models are dependent on our ability to reliably calculate physical observables. The complex nature of nuclear interactions has required that advances in interaction models go hand in hand with advances in nuclear structure calculations. The purpose of this work is to study the ground states of few-body nuclear systems in an exact fashion using a realistic model of nuclear interactions.

The model of interest in this work is the Argonne v_{18} (AV18) nucleon-nucleon potential used in conjunction with the Urbana IX (UIX) three nucleon interaction. The Argonne v_{18} interaction, fit to the two-nucleon scattering data and deuteron properties, was developed by Wiringa, Stoks and Schiavilla to address the apparent isospin symmetry breaking in nuclear interactions. The Urbana models of three-nucleon interactions are phenomenologically based and have been used successfully in past calculations. The form of AV18+UIX Hamiltonian is discussed in Chapter 2.

The focus of this work is the application of Monte Carlo techniques, specifically Variational (VMC) and Green's function Monte Carlo (GFMC), to determine the ground state properties of few-nucleon systems using this Hamiltonian. Variational Monte Carlo, discussed in Chapter 3, makes use of a variational wave function and Monte Carlo techniques to perform the multi-dimensional integrals required to evaluate expectation values. Since VMC can only provide an upper bound on the ground state, Green's function Monte Carlo is used to project out the exact ground state wave function from the variational results. The first section of Chapter 4 discusses the technique of GFMC in general and the second section

is devoted to GFMC as it applies to our nuclear ground state calculations.

With the technique in hand, we have investigated several few-body nuclear systems. The purpose here is two fold: we would like to test the limits of this Hamiltonian by performing exact calculations of the structure of few-nucleon systems, and we would also like to learn something about these systems through these calculations. The nuclear systems considered here range from three-body nuclei (^3H) to seven-body nuclei (^7Li) to clusters of seven and eight neutrons. Chapter 5 details the methods used in studying these systems and the results of our calculations. Finally, the conclusions we can draw from this body of work are discussed in Chapter 6.

Chapter 2

The Hamiltonian

Fundamentally, interactions between nucleons are the result of QCD color interactions between the quarks they are made up from. In the absence of a suitable theory of nuclear interactions based on the QCD degrees of freedom, it is expedient to model nuclear interactions using the nucleon degrees of freedom. In this work, we use a non-relativistic, many-body Hamiltonian of the following form:

$$\hat{H} = (\sum_{i} \frac{p_i^2}{2m_i} + \sum_{i<j} v_{ij} + \sum_{i<j<k} V_{ijk} + \ldots) \tag{2.1}$$

Note that, in this representation, the composite nature of nucleons dictates the presence of three-body and higher-order interactions. To date, up to three- body terms in the interaction have been included in our calculations. It is also important to note that, in this non-relativistic formulation, terms of order $(p/m)^4$ and higher have been neglected. Calculations of three- and four-body ground states which include relativistic corrections have shown these corrections to be of the same order as the three-body interaction [1, 2]. Work is under way to develop a relativistic nuclear Hamiltonian [3, 4].

2.1 The NN Potential

Several realistic, phenomenological models of NN interactions (Bonn[5], Paris[6], Argonne[7][8], Urbana[9]) have been developed based on the idea of using boson exchange interactions to fit the NN scattering data. It has been discovered that the long-range features of the NN interaction could be explained in terms of one pion exchanges between the nucleons. In order to properly fit the observed NN scattering data, this idea was extended to include $\eta, \rho, \omega, \sigma$, and σ' mesons (Bonn), where the σ, and σ' are hypothetical scalar mesons

introduced to approximate two pion exchange interactions. Since the range of the interactions due to the heavier mesons is of the order of the proton radius, the approach of the Argonne and Urbana models has been to retain the one pion exchange (OPE) interaction term and to fit the rest of the interaction directly to the NN scattering data. This approach has yielded several successful models, the most recent of which is the Argonne v_{18}.

Early Argonne[7] and Urbana[9] models of NN interactions were fit to the n-p scattering data in the T = 0, 1 channels. These fits produced potentials which did not accurately reproduce the p-p scattering data since the n-p interaction is slightly more attractive than the p-p interaction. To address this isospin symmetry breaking, Wiringa, Stoks and Schiavilla[8] developed the Argonne v_{18} model which is fit to the n-p and the p-p scattering data as well as the nn scattering length and the deuteron properties.

The Argonne and Urbana models break the NN interaction into three terms

$$v_{NN} = v_\pi + v_I + v_S, \tag{2.2}$$

where the v_π is the OPE term, v_I is the attractive, intermediate range, two-pion exchange part and the v_S is the repulsive short range part. The interaction is expanded in terms of operators which are functions of the spin, isospin and angular momentum

$$v_{ij} = \sum_p v_p(r_{ij}) O^p_{ij}. \tag{2.3}$$

In the Argonne v_{18} model, the interaction is expressed in terms of fourteen isoscalar operators (like the earlier Argonne and Urbana models) and four isospin symmetry breaking terms. The fourteen isoscalar operators are given by

$$O^{p=1,14}_{ij} = [1, (\sigma_\mathbf{i} \cdot \sigma_\mathbf{j}), S_{ij}, (\mathbf{L} \cdot \mathbf{S}), \mathbf{L}^2, \mathbf{L}^2(\sigma_\mathbf{i} \cdot \sigma_\mathbf{j}), (\mathbf{L} \cdot \mathbf{S})^2] \otimes [1, (\tau_\mathbf{i} \cdot \tau_\mathbf{j})] \tag{2.4}$$

where

$$S_{ij} = 3(\sigma_\mathbf{i} \cdot \hat{\mathbf{r}}_\mathbf{ij})(\sigma_\mathbf{j} \cdot \hat{\mathbf{r}}_\mathbf{ij}) - (\sigma_\mathbf{i} \cdot \sigma_\mathbf{j}) \tag{2.5}$$

is the tensor operator. The isospin symmetry is broken by three isotensor and one isovector operators

$$O^{p=15,18}_{ij} = T_{ij}, T_{ij}(\sigma_\mathbf{i} \cdot \sigma_\mathbf{j}), T_{ij} S_{ij}, (\tau_{iz} + \tau_{jz}), \tag{2.6}$$

where T_{ij} is the isotensor operator given by

$$T_{ij} = 3\tau_{iz}\tau_{jz} - (\tau_\mathbf{i} \cdot \tau_\mathbf{j}). \tag{2.7}$$

The isotensor terms include corrections for the mass difference between charged and neutral pions, while the isovector term, fit to the nn scattering length, is assumed to arise from $\rho - \omega$ and $\pi^o - \eta - \eta'$ mixing. Finally, the isospin symmetry breaking due to the proton-neutron mass difference is accounted for by using the kinetic energy operator

$$\hat{T} = -\frac{\hbar^2}{4}\sum_i (\frac{1}{m_p} + \frac{1}{m_n})\nabla_i^2 + (\frac{1}{m_p} - \frac{1}{m_n})\tau_{iz}\nabla_i^2. \tag{2.8}$$

In addition to the two-body nuclear potential, we also include fourteen two-body electromagnetic terms, outlined in detail in Ref. [8], in the Hamiltonian. For p-p interactions these include one- and two-photon Coulomb terms, a Darwin Foldy term, a vacuum polarization term, and two magnetic terms. For n-p interactions we have a one-photon Coulomb term, and three magnetic terms. Finally, the n-n interactions have only three magnetic terms. These terms are correct up to $\frac{1}{m^2}$ and include corrections for finite nucleon size and short-range effects.

2.2 The NNN Potential

Exact calculations of nuclear ground states (Fadeev[10, 11], GFMC[12]) have shown that two-body potentials alone are insufficient to obtain observed binding energies of light nuclei (Table 5.3). The inclusion of the three-body potential is of particular importance in the study of nuclear ground states since there is a large cancelation between the kinetic energy and the two-body potential energy. In the Urbana models, the three-body force is broken into two terms

$$V_{ijk} = V_{ijk}^{2\pi} + V_{ijk}^{R}, \tag{2.9}$$

where $V_{ijk}^{2\pi}$ corresponds to the attractive two-pion exchange three-nucleon interaction. The Urbana model assumes $V_{ijk}^{2\pi}$ to have the Fujita-Miyazawa form

$$V_{ijk}^{2\pi} = \sum_{cyc} A_{2\pi}(\{X_{ij}, X_{jk}\}\{(\tau_i \cdot \tau_j), (\tau_j \cdot \tau_k)\} + \frac{1}{4}[X_{ij}, X_{jk}][(\tau_i \cdot \tau_j), (\tau_j \cdot \tau_k)]) \tag{2.10}$$

$$X_{ij} = Y_\pi(r_{ij})(\sigma_i \cdot \sigma_j) + T_\pi(r_{ij})S_{ij}, \tag{2.11}$$

where $Y_\pi(r)$ and $T_\pi(r)$ are the Yukawa and Tensor functions respectively. These functions are taken from the pion exchange part of the two-nucleon interaction and contain short-range

cutoffs. The Fujita-Miyazawa term assesses contributions from diagrams such as Fig. (2.1) where the two pions excite and de-excite a delta resonance.

The second term, V_{ijk}^R, is purely phenomenological. It is assumed to be spin-independent and of intermediate range:

$$V_{ijk}^R = \sum_{cyc} U_o T_\pi^2(r_{ij}) T_\pi^2(r_{jk}). \tag{2.12}$$

At present only a few observables can be calculated accurately enough to determine unknown parameters in V_{ijk}. The simple Urbana model has only two parameters, $A_{2\pi}$ and U_o. The $A_{2\pi}$ is adjusted to reproduce the binding energy of ^3H in exact calculations, and the parameter U_o is adjusted to reproduce saturation density of nuclear matter. It is hoped that studies of five- to eight-body nuclei will give additional information needed to estimate the strength of the other possible terms in V_{ijk}.

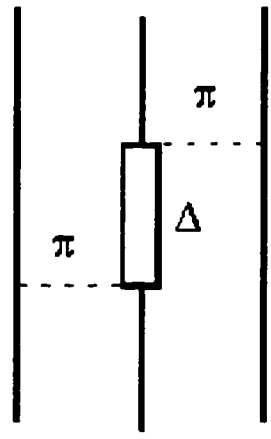


Figure 2.1: Simplest two-pion exchange three-body diagram

Chapter 3

The Wave Function and Variational Monte Carlo

The problem to be solved is the many-body non-relativistic Schrödinger's equation.

$$\hat{H}\Psi_o = (\sum_i \frac{p_i^2}{2m_i} + \sum_{i<j} v_{ij} + \sum_{i<j<k} V_{ijk})\Psi_o = E_o\Psi_o. \tag{3.1}$$

where Ψ_o is our many-body ground state wave function. Solving this equation for Ψ_o quickly becomes a daunting task as we move from two to three or four nucleons. Exact solutions for three- and four-body systems have been obtained by solving the Fadeev and Fadeev-Yakubovsky[13, 11] equations for a variety of Hamiltonians. These techniques can attain a high degree of accuracy ($< 1\%$), however they are currently limited to systems where $A < 5$.

Rather than a direct solution, we can introduce a variational wave function, Ψ_V and evaluate expectation values in configuration space of the following form:

$$\langle \hat{O} \rangle = \frac{\int d\mathbf{R}\Psi_V^\dagger(\mathbf{R})\hat{O}\Psi_V(\mathbf{R})}{\int d\mathbf{R}\Psi_V^\dagger(\mathbf{R})\Psi_V(\mathbf{R})} \tag{3.2}$$

where $\mathbf{R} = (\mathbf{r_1}, \mathbf{r_2}, \cdots)$ is the multi-dimensional position vector for all of the particles in the nucleus. Such integrals have 3A dimensions, making direct integration computationally impossible. For this reason we use the Monte Carlo method to evaluate the integral (3.2) , in particular, we evaluate the expectation value of the Hamiltonian to determine the variational energy. The parameters of the variational wave function are varied to minimize the energy expectation value. This process encompasses the technique of Variational Monte Carlo (VMC). In this section, I will discuss the form of our variational wave function and give a brief outline of of VMC as it applies to few-body nuclei. Later in this work, we will

discuss the explicit form of the variational wave function for each particular system we intend to study.

3.1 The Variational Wave Function

Any two-body wave function can be represented as a correlation operator acting on an uncorrelated state. We can generalize this idea to multi-particle wave functions by writing our wave function as a product of correlation operators acting on an uncorrelated state, Φ.

$$\Psi_V = (\mathcal{S} \prod_{i<j} F_{ij})\Phi \tag{3.3}$$

where F_{ij} is the two-body correlation operator which induces correlations according to the operator terms in our interaction. The uncorrelated state is an antisymmetric Slater determinant of single particle states. The $\mathcal{S}$ indicates a symmetrized product over pairs since, in general, the F_{ij}'s may not commute. When Φ is antisymmetric, the product of F_{ij}'s must be symmetric to maintain the antisymmetry of the the overall wave function. For nuclei where A$\leq$4, Φ is just the spatially independent Slater determinant of the spin and isospin states. For example, in ^4He Φ is given by,

$$\Phi = \begin{vmatrix} n\uparrow & n\downarrow & p\uparrow & p\downarrow \\ n\uparrow & n\downarrow & p\uparrow & p\downarrow \\ n\uparrow & n\downarrow & p\uparrow & p\downarrow \\ n\uparrow & n\downarrow & p\uparrow & p\downarrow \end{vmatrix} \tag{3.4}$$

When A$\geq$5, spatial dependence must be included in the uncorrelated state in order to obtain a fully antisymmetric wave function. In such cases, an antisymmetric combination of an alpha core and nucleons in p-wave orbitals with the appropriate overall angular momentum J, isospin T, and projections J_z, and T_z is used. Note that Φ is chosen to be independent of the center of mass so that Ψ_v has no center of mass motion.

It is customary to write Eq. (3.3) in terms of a Jastrow wave function in the following way

$$|\Psi\rangle = \mathcal{S} \prod_{i<j} (1 + U_{ij})|\Psi_J\rangle \tag{3.5}$$

where the Jastrow wave function comprises the uncorrelated state acted on by the central correlations, $f_c(r_{ij})$:

$$|\Psi_J\rangle = \prod_{i<j} f_c(r_{ij})|\Phi\rangle. \tag{3.6}$$

The operator correlations are expressed as a sum over the operators contained in the potential.

$$U_{ij} = \sum_{p\geq 2} u_p(r_{ij}) O^p_{ij}. \tag{3.7}$$

In this work, only the first six static operators of the potential are retained in the two-body correlations.

Suitable approximations for the central, $f_c(r_{ij})$, and non-central, $u_p(r_{ij})$, pair correlations can be found by solving the two-body Schrödinger's equation in the presence of a modified potential, $\bar{v} + \lambda$, where

$$\bar{v}(r_{ij}) = \sum_p \alpha_p v_p(r_{ij}) O^p_{ij}. \tag{3.8}$$

The α_p's are introduced as variational parameters to simulate the quenching of spin-isospin interactions due to the presence of other particles. The modifying term, $\lambda(r_{ij}) = \sum_{p=1,6} \lambda_p(r_{ij}) O^p_{ij}$ is added to simulate screening effects and contains parameters; some of which are used to match the boundary conditions, others are variational.

It has been suggested by Wiringa [14] that suitable correlations for light nuclei can be determined by taking the $k_F \to 0$ limit in the Euler-Lagrange equations for correlations in fixed density infinite nuclear matter [9]. Following this prescription, Wiringa arrived at a set of eight differential equations for correlations according to the first eight operators in the potential. Since only the first six operator correlations are used in this work, we set the $\mathbf{L}\cdot\mathbf{S}$ correlations and the corresponding λ-functions to zero to obtain six differential equations. In the two-nucleon spin, S, and isospin, T, channels, the six correlations are expressed as four central correlations, $f_{S,T}$ and two tensor correlations, $f_{t,T}$; and obey the following set of equations:

$$-\frac{\hbar^2}{m}[(f_{0,1}r)''] + [\bar{v}_{0,1} + \lambda_{0,1}](f_{0,1}r) = 0 \tag{3.9}$$

$$-\frac{\hbar^2}{m}\left[(f_{0,0}r^2)'' - \frac{2}{r^2}(f_{0,0}r^2)\right] + [\bar{v}_{0,0} + \lambda_{0,0} + 2\bar{v}_{q0,0}](f_{0,0}r^2) = 0 \tag{3.10}$$

$$-\frac{\hbar^2}{m}[(f_{1,0}r)''] + [\bar{v}_{1,0} + \lambda_{1,0}](f_{1,0}r) + 8[\bar{v}_{t,0} + \lambda_{t,0}](f_{t,0}r) = 0 \tag{3.11}$$

$$-\frac{\hbar^2}{m}\left[(f_{t,0}r)'' - \frac{6}{r^2}(f_{t,0}r)\right] + [\bar{v}_{t,0} + \lambda_{t,0}](f_{1,0}r)$$
$$+ [\bar{v}_{1,0} + \lambda_{1,0} - 2(\bar{v}_{t,0} + \lambda_{t,0}) - 3\bar{v}_{b,0} + 6\bar{v}_{q1,0} + 9\bar{v}_{bb,0}](f_{t,0}r) = 0 \qquad (3.12)$$

$$-\frac{\hbar^2}{m}\left[(f_{1,1}r^2)'' - \frac{2}{r^2}(f_{1,1}r^2)\right] + [\bar{v}_{1,1} + \lambda_{1,1} + 2(\bar{v}_{q1,1} + \frac{2}{3}\bar{v}_{bb,1})](f_{1,1}r^2)$$
$$+ 8[\bar{v}_{t,1} + \lambda_{t,1} - \frac{1}{6}\bar{v}_{bb,1}](f_{t,1}r^2) = 0 \qquad (3.13)$$

$$-\frac{\hbar^2}{m}\left[(f_{t,1}r^2)'' - \frac{8}{r^2}(f_{t,1}r^2)\right] + [\bar{v}_{t,1} + \lambda_{t,1} - \frac{1}{6}\bar{v}_{bb,1}](f_{1,1}r^2) + [\bar{v}_{1,1} + \lambda_{1,1}$$
$$-2(\bar{v}_{t,1} + \lambda_{t,1}) - 3\bar{v}_{b,1} + 8\bar{v}_{q1,1} + \frac{32}{3}\bar{v}_{bb,1}](f_{t,1}r^2) = 0 \qquad (3.14)$$

where the subscripts, t, b, q, and bb refer to the tensor, $(\mathbf{L}\cdot\mathbf{S})$, $\mathbf{L}^2$, and $(\mathbf{L}\cdot\mathbf{S})^2$ terms of the potential respectively. As always, Fermi statistics dictate that the quantum numbers of any particle pair must obey the relation, $L + S + T =$ odd , in order to maintain antisymmetry under particle exchanges. This set of six equations is comprised of two uncoupled and four coupled equations. The first uncoupled equation, Eq. (3.9), is simply the two-body Schrödinger's equation in the $^1S_0$ channel and would provide the wave function in this state if $\lambda_{0,1} = -E$. Similarly, Eq. (3.10) provides the correlation in the $^1P_1$ channel. Since the tensor operator couples states of L=J-1 to states of L=J+1 in the S=1 channels, there are two sets of coupled equations in the S=1 channels. The first set of coupled equations, Eqs. (3.11-3.12), acts between the $^3S_1$ and $^3D_1$ channels. Solutions to these equations provide the deuteron wave function when $\lambda_{1,0} = -E_d$ and $\lambda_{t,0} = 0$. The final set of equations, Eqs. (3.13-3.14), couple the $^3P_2$ and $^3F_2$ channels. The nuclear matter equations are derived by summing over all L and J channels and then minimizing the two-body cluster energy. Such a procedure causes the potential contributions in the L=odd states to be a weighted average over all possible J states when the $k_F \to 0$ limit is taken.

The boundary conditions for Eqs. (3.9-3.14) have been determined [15, 16, 14] by observing the behavior of the full wave function when two particles are extremely close or when one particle is very far from the remaining nucleons. The form of the boundary conditions is given by,

$$f_{S,T}(r = 0) \;=\; constant$$

$$
\begin{aligned}
f_{S,T}(r \to \infty) &= constant \left[\frac{exp(-k_{S,T}r)}{r}\right]^{1/(A-1)} \\
f_{t,T}(r=0) &= 0 \\
f_{t,T}(r \to \infty) &= \eta_{t,T} T(r) f_{1,T}(r)
\end{aligned} \tag{3.15}
$$

where,

$$
T(r) = \left[1 + \frac{3}{k_{S,T}r} + \frac{3}{(k_{S,T}r)^2}\right](1 - exp(-(r/d_{t,T})^2)) \tag{3.16}
$$

$$
k_{S,T} = \left[\frac{A-1}{A}\frac{2m}{\hbar^2}E_{S,T}\right] \tag{3.17}
$$

The parameters, $\eta_{t,T}$ and $d_{t,T}$ are variational parameters. $E_{S,T}$ is the separation energy for removing one nucleon from the nucleus, it is also treated as a variational parameter. In light nuclei, this separation energy can depend upon the two-nucleon quantum numbers S,T. For example, a neutron in ^3He is more likely to be in S=1, T=0 state than a proton.

The λ functions are parametrized in the following way,

$$
\lambda_{S,T} = \Gamma_{S,T}\left[1 + exp\left[\frac{r - R_{S,T}}{a_{S,T}}\right]\right]^{-1} + \Lambda_{S,T}(r)(1 - exp(-(r/c_{S,T})^2)) \tag{3.18}
$$

$$
\lambda_{t,T} = \Gamma_{t,T}\left[1 + exp\left[\frac{r - R_{t,T}}{a_{t,T}}\right]\right]^{-1} + \Lambda_{t,T}(r)(1 - exp(-(r/c_{t,T})^2)). \tag{3.19}
$$

Here the long-range functional form, given by the functions $\Lambda_{t,T}(r)$ and $\Lambda_{S,T}(r)$, is determined from the large r limits of $f_{S,T}$ and $f_{t,T}$. This long-range part is multiplied by a short-range cutoff which contains the variational parameters c_x (x=S,T and t,T). The short-range part is assumed to have a Woods-Saxon shape and the parameters, Γ_x, are fixed from the r=0 boundary conditions. The Woods-Saxon shape parameters, R_x and a_x are determined variationally.

Once the functional form of the correlations has been obtained in the channels, the correlations are scaled relative to each other by a variational parameter, $f^{scal}_{S,T}$, and then projected into the operator basis according to,

$$
\sum_{S,T} f_{S,T}(r_{ij}) P_S P_T + \sum_T f_{t,T}(r_{ij}) S_{ij} P_T = f_c(r_{ij})\left[1 + \sum_{p=2,6} u_p(r_{ij}) O^p_{ij}\right]. \tag{3.20}
$$

where P_S and P_T are the projection operators in spin and isospin space. For example, the projection operators for the S states are,

$$
P_{S=0} = \frac{1}{4}(1 - \sigma_i \cdot \sigma_j) \tag{3.21}
$$

$$P_{S=1} = \frac{1}{4}(3 + \sigma_i \cdot \sigma_j) \tag{3.22}$$

The T state projection operators have the same form with σ is replaced by τ. The projection given by Eq. (3.20) leads us to.

$$\begin{aligned}
f_c &= \frac{1}{16}(f_{0,0} + 3f_{0,1} + 3f_{1,0} + 9f_{1,1}) \\
u_\tau &= \frac{1}{16}(-f_{0,0} + f_{0,1} - 3f_{1,0} + 3f_{1,1})/f_c \\
u_\sigma &= \frac{1}{16}(-f_{0,0} - 3f_{0,1} + f_{1,0} + 3f_{1,1})/f_c \\
u_{\sigma\tau} &= \frac{1}{16}(f_{0,0} - f_{0,1} - f_{1,0} + f_{1,1})/f_c \\
u_t &= \frac{1}{4}(f_{t,0} + 3f_{t,1})/f_c \\
u_{t\tau} &= \frac{1}{4}(-f_{t,0} + f_{t,1})/f_c
\end{aligned} \tag{3.23}$$

Figure 3.1 shows an example of the correlations for the ^6Li (J^π=1^+) state using the parameters from Table 5.5.

In principle more operator correlations can be included from the two-body potential. Wiringa[14] has successfully carried out VMC calculations using two-body spin-orbit correlations. Since the ultimate goal of our VMC calculations is to provide a starting point for Green's function Monte Carlo calculations, we deemed the inclusion of spin-orbit correlations to be an unnecessary complication.

Using a product of two-body correlations to represent a state with more than two particles is inherently an approximation. In comparisons between variational and exact Fadeev three-body wave functions, it has been found that three-body correlations can substantially reduce the difference between the variational and exact wave functions[17]. In our work, we use the three-nucleon correlations which have the most significant effect. Our wave function using these three-body central and operator correlations is as follows:

$$|\Psi\rangle = [1 + \sum_{i<j<k} U_{ijk}] \{\mathcal{S} \prod_{i<j}(1 + \prod_{k\neq i,j} f_{ijk} U_{ij})\}\{\prod_{i<j<k} f^c_{ijk}\}|\Psi_J\rangle \tag{3.24}$$

where U_{ijk} are three-body correlations induced by V_{ijk}, while f_{ijk} and f^c_{ijk} are central three-body correlations which depend on the geometric configuration the the triplet.

The central three-body correlation

$$f^c_{ijk} = (1 + q_{c1}\cos(\theta_{ij})\cos(\theta_{jk})\cos(\theta_{ki})e^{-q_{c2}R_{ijk}}) \tag{3.25}$$

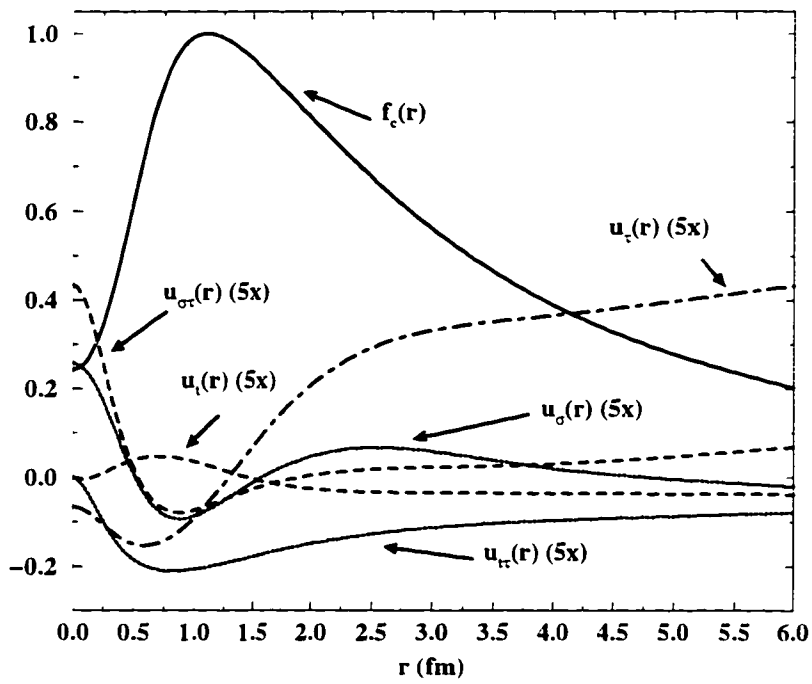


Figure 3.1: Correlation functions for ^6Li (1^+). Note that the non-central correlations have been multiplied by 5 to amplify their features

is parametrized such that equilateral arrangements of the triplet are emphasized over linear arrangements. Here $R_{ijk} = |r_{ij}| + |r_{jk}| + |r_{ki}|$ and θ_{ij} is the angle formed between particles i and j with the vertex at particle k. The central three-body correlations which multiply the two-body operator correlations are given by.

$$f_{ijk} = (1 + q_1(1 - \cos(\theta_{ij}))e^{-q_2 R_{ijk}}) \tag{3.26}$$

This parameterization correlates the spin-isospin correlations of particles i and j with the particle k, depending on how much particle k intercedes between particles i and j.

The three-body operator correlations are assumed to have the same form as our three-body force.

$$U_{ijk} = \epsilon V_{ijk}(\bar{r}_{ij}, \bar{r}_{jk}, \bar{r}_{ki}) \tag{3.27}$$

where ϵ is typically a small negative number. The particle separation in V_{ijk} is scaled, $\bar{r}_{ij} = br_{ij}$, where b is a scale factor (b < 1). This spreads the correlation out relative to the interaction as one would expect from elementary quantum mechanics. It has been found that, by doubling the strength of the anti-commutator terms, the commutator terms can be neglected in the three-body correlation without seriously effecting the quality of the variational wave function. This simplification can speed up our energy calculation by as much as a factor of two. The parameters of these three-body correlations, $q_{c1}, q_{c2}, q_1, q_2, \epsilon$ and b are treated as additional variational parameters.

3.2 Variational Monte Carlo

The expectation values of the form Eq. (3.2) contain multi-dimensional integrals over all particle positions. Since direct integration is computationally prohibitive, Monte Carlo methods are used to evaluate the expectation values.

The wave function contains the symmetrized product of two-body correlations. Each order of the operators in $\mathcal{S}\prod_{i<j}(1 + U_{ij})$ is denoted by p or q. Thus, Ψ_p is that component of Ψ in which the operators act in the order p. Expectation values are given by.

$$\langle \hat{O} \rangle = \frac{\sum_{p,q} \int d\mathbf{R} \Psi_p^\dagger(\mathbf{R}) \hat{O} \Psi_q(\mathbf{R})}{\sum_{p,q} \int d\mathbf{R} \Psi_p^\dagger(\mathbf{R}) \Psi_q(\mathbf{R})}. \tag{3.28}$$

The Monte Carlo integration of Eq. (3.28) is facilitated by introducing a probability distribution, $W_{pq}(\mathbf{R})$, which approximates the distribution represented by the wave function.

$$\langle \hat{O} \rangle = \frac{\sum_{p,q} \int d\mathbf{R} \frac{\Psi_p^\dagger(\mathbf{R})\hat{O}\Psi_q(\mathbf{R})}{W_{pq}(\mathbf{R})} W_{pq}(\mathbf{R})}{\sum_{p,q} \int d\mathbf{R} \frac{\Psi_p^\dagger(\mathbf{R})\Psi_q(\mathbf{R})}{W_{pq}(\mathbf{R})} W_{pq}(\mathbf{R})}. \tag{3.29}$$

The probability function is usually taken to be, $W_{pq}(\mathbf{R}) = Re(\Psi_p^\dagger(\mathbf{R})\Psi_q(\mathbf{R}))$, however a simpler Ψ might be used in the $W_{pq}(\mathbf{R})$ to reduce the computational effort. For example, in our calculations the U_{ijk} correlations are omitted in the Ψ used to define $W_{pq}(\mathbf{R})$.

The crux of the Monte Carlo integration of Eq. (3.28) is to stochastically sample the probability distribution $W_{pq}(\mathbf{R})$ and obtain a collection of N uncorrelated or independent configurations, $\{\mathbf{R}_i\}$. By the 'Law of Large Numbers' [18], as $N \rightarrow \infty$ Eq. (3.28) can be reduced to a sum over independent configurations as follows:

$$\langle \hat{O} \rangle = \frac{\frac{1}{N}\sum_{i=1}^{N} \Psi_p^\dagger(\mathbf{R}_i)\hat{O}\Psi_q(\mathbf{R}_i)/W_{pq}(\mathbf{R}_i)}{\frac{1}{N}\sum_{i=1}^{N} \Psi_p^\dagger(\mathbf{R}_i)\Psi_q(\mathbf{R}_i)/W_{pq}(\mathbf{R}_i)}. \tag{3.30}$$

Since we cannot evaluate an infinite number of samples, our expectation value, $\langle \hat{O} \rangle$ has a sampling error which we approximate as the standard deviation,

$$\sigma = \left[\frac{\langle \hat{O}^2 \rangle - \langle \hat{O} \rangle^2}{N-1} \right]^{\frac{1}{2}}. \tag{3.31}$$

A treatment of various techniques for sampling probability distributions can be found in Kalos and Whitlock [18]. A technique of great utility in sampling complicated distributions such as ours is the Metropolis technique developed by Metropolis, Rosenbluth, Rosenbluth, Teller and Teller [19]. Below the steps in the Metropolis algorithm are outlined:

1. We begin with a set of particle positions $\mathbf{R}$, operator orders p and q; and the corresponding weight, $W_{pq}(\mathbf{R})$.
2. A new point, $\mathbf{R}'$ is generated from $\mathbf{R}$ by randomly moving each particle within a step size l,

$$\begin{aligned} x_i' &= x_i + (\zeta_1 - \frac{1}{2})l \\ y_i' &= y_i + (\zeta_2 - \frac{1}{2})l \\ z_i' &= z_i + (\zeta_3 - \frac{1}{2})l. \end{aligned} \tag{3.32}$$

where each ζ_i is a random number between 0 and 1. New operator orders, p′ and q′ are randomly generated and the weight $W_{p'q'}(\mathbf{R}')$ of the new configuration is constructed.

3. The ratio of the new weight to the old weight is compared to a random number between 0 and 1. If the ratio is greater than the random number, the new configuration is kept. Otherwise, the old is regarded as the new configuration.

4. This process (steps 2-3) is repeated several times ($\sim$ 10) until an independent configuration is generated.

The step size, l, is adjusted such that roughly 50% of the configurations are accepted. If the step size is too small, most of the steps will be accepted, but the full configuration space will not be adequately sampled. If the step size is too big, most of the steps will be outside the nucleus and thus be rejected; and once again the distribution will be inadequately sampled. In our calculations, typically ten steps are tried before generating an independent configuration.

At each step of the Metropolis walk we must construct the wave function for the two operator orders p and q. The wave function is a complex vector in spin and isospin space,

$$\Psi_V = \sum_{n_s, n_t} v(\mathbf{R}, n_s, n_t)|n_s, n_t\rangle \tag{3.33}$$

where the n_s indexes the 2^A spin states and $v(\mathbf{R}, n_s, n_t)$ is the amplitude for each spin-isospin state. The n_t labels all allowed isospin states. Typically there are $\frac{A!}{Z!(A-Z)!}$ charge conserving isospin states, however this number can be greatly reduced by retaining only those isospin states which have the correct total isospin of the nucleus. We can do this because the correlations do not couple states of different overall isospin. This set of isospin states is referred to as the reduced isospin basis. A table of the reduced isospin states for the nuclei studied in this work can be found in the Appendix. Typical sizes of the wave functions used for the nuclei studied in this work can be found in Table 3.1.

In order to construct Ψ_V we start with the Jastrow wave function, Ψ_J, and successively operate on it with the two-body correlation operators in the chosen order. Building the wave function and evaluating the action of the operators in the potential acting on the wave function requires a means of projecting those operators into the wave function's spin-isospin basis. To facilitate this projection there are some very useful matrices which aid us in

Table 3.1: Size of wave function in number of spin-isospin basis elements

Nucleus	Number of complex elements
^3H	16
^4He	32
$^6Li(1^+, 2^+, 3^+)$	320
$^6Li, ^6He, ^6Be(0^+)$	576
$^7Li(\frac{3}{2}^-, \frac{1}{2}^-, \frac{7}{2}^-, \frac{5}{2}^-)$	1792
$^7n(\frac{3}{2}^-, \frac{1}{2}^-)$	128
$^8n(0^+)$	256

manipulating our spin-like basis. The first matrix, $m(i, n_s)$, provides the z component of σ_i in the A particle spin state, $|n_s\rangle$. Another useful matrix is $n_{exch}(k_{ij}, n_s)$ which gives the number of the A particle spin state which results from the exchange of spins of the particles i and j in the state $|n_s\rangle$. Here k_{ij} is the pair number which labels the pair ij. The matrix, $n_{flip}(i, n_s)$ gives the number of the spin state obtained by flipping the spin of particle i in the spin state $|n_s\rangle$. Together, these three matrices enable us to evaluate the action of all of our spin-dependent operators.

The easiest operator to evaluate is $(\sigma_i \cdot \sigma_j)$ since its action can be represented as

$$(\sigma_i \cdot \sigma_j) = 2P_{ij} - 1 \tag{3.34}$$

where P_{ij} is the pair exchange operator. In terms of our wave function, the action of $(\sigma_i \cdot \sigma_j)$ can be expressed as

$$(\sigma_i \cdot \sigma_j)\Psi = 2\Psi' - \Psi \tag{3.35}$$

where

$$\psi'(n_s, n_t) = 2\psi(n_{exch}(k_{ij}, n_s), n_t). \tag{3.36}$$

By representing $(\sigma_i \cdot \sigma_j)$ in this manner, we need only evaluate 2^A operations for each pair compared to $2^A \times 2^A$ operations if we simply represented it as a matrix in spin space.

The tensor operator is slightly more complicated to evaluate. We have already seen how to evaluate the $(\sigma_i \cdot \sigma_j)$ portion of the tensor operator given in Eq. (2.5). To evaluate the remaining portion, we make the following rearrangement,

$$(\sigma_i \cdot \hat{r}_{ij}) = \sigma_i^+ \hat{r}_{ij}^- + \sigma_i^- \hat{r}_{ij}^+ + \sigma_i^z \hat{r}_{ij}^z \tag{3.37}$$

where

$$\begin{aligned}
\hat{r}^z_{ij} &= \hat{z}_{ij} \\
\hat{r}^+_{ij} &= \frac{1}{2}(\hat{x}_{ij} + i\hat{y}_{ij}) \\
\hat{r}^-_{ij} &= \frac{1}{2}(\hat{x}_{ij} - i\hat{y}_{ij})
\end{aligned} \tag{3.38}$$

and σ_i^+, σ_i^-, and σ_i^z are the well known spin raising, lowering and z-projection operators. In terms of our matrix representation, the operation of these spin operators is as follows:

$$\begin{aligned}
\sigma_i^z|n_s\ n_t\rangle &= m(i, ns)|n_s\ n_t\rangle \\
\sigma_i^+|n_s\ n_t\rangle &= \frac{1}{2}\left[1 - m(i, n_s)\right]|n_{flip}(i, n_s)\ n_t\rangle \\
\sigma_i^-|n_s\ n_t\rangle &= \frac{1}{2}\left[1 + m(i, n_s)\right]|n_{flip}(i, n_s)\ n_t\rangle
\end{aligned} \tag{3.39}$$

Each factor of $(\sigma_i \cdot \hat{r}_{ij})(\sigma_j \cdot \hat{r}_{ij})$ can be successively evaluated or, in a more efficient implementation, the full product can be expanded and expressed in terms of raising, lowering and z-projection operators.

The action of the isospin dependent operators is complicated by the fact that the wave function is represented in the reduced isospin basis. It is important to remember that the isospin states can also be represented by the spin-like basis states, where the 2^A states specify the τ_z of each nucleon. Since it is straight-forward to express the operators in the spin-like basis, the isospin operators are expressed in that basis and then projected into the reduced basis. In this way, operators used to construct the wave function, like $(\tau_i \cdot \tau_j)$, and operators used in expectation values, like $\tau_{zi}\tau_{zj}$ and $(\tau_{zi} + \tau_{zj})$, are expressed as matrices in the reduced basis states.

As an example, we consider the ground state of ^6Li. In VMC, it has total isospin T=0 and the reduced isospin basis contains the five orthogonal T=0 states of three protons and three neutrons. In contrast, the charge conserving basis has twenty states, requiring a four times more multiplications to evaluate the operation of spin and tensor operators. Clearly, the reduced basis greatly reduces the computational effort required to build the wave function and evaluate expectation values.

Expectation values are calculated by having the operators act entirely on the right hand side of the expectation value shown in Eq. (3.30). The matrix machinery used to construct the wave function can be also be used to evaluate the corresponding expectation values.

Derivatives of the wave function with respect to the coordinates of particle i are evaluated as

$$\frac{\partial \Psi_V(\mathbf{R})}{\partial x_i} = \frac{\Psi_V(\mathbf{R} + \epsilon \hat{x}_i) - \Psi_V(\mathbf{R} - \epsilon \hat{x}_i)}{2\epsilon}. \tag{3.40}$$

where ϵ is chosen to be much smaller than the grid on which the correlations are tabulated. Second derivatives are evaluated according to

$$\frac{\partial^2 \Psi_V(\mathbf{R})}{\partial x_i^2} = \frac{\Psi_V(\mathbf{R} + \epsilon \hat{x}_i) + \Psi_V(\mathbf{R} - \epsilon \hat{x}_i) - 2\Psi_V(\mathbf{R})}{\epsilon^2}. \tag{3.41}$$

Thus, we need 6A+1 evaluations of Ψ_V to determine the kinetic energy. Using a trick described in Ref. [43], we can evaluate the action of the angular momentum dependent terms in the potential with an additional 3P evaluations, where P is the number of pairs. As one can see, not only does the size of the wave vector grow with the number of nucleons, but also the number of evaluations necessary to calculate the energy. For this reason, calculations of energy expectation values dominate the computer time required to carry out a variational Monte Carlo calculation for systems with more than 4 nucleons.

The variational wave functions discussed later in this work can contain more than 20 variational parameters. Currently, variational searches to find the minimum energy are carried out by hand. One Metropolis walk is conducted with a starting set of parameters. The parameters are varied slightly and expectation values are re-calculated using configurations generated from the original Metropolis walk. Once an optimum set of parameters is found, a new walk is carried out using those parameters and the whole process is repeated. By using the same Metropolis walk to calculate the energies of different wave functions, the errors in our expectation values are correlated and thus the statistical noise is reduced when evaluating the optimum parameters.

Chapter 4

Green's Function Monte Carlo

The variational techniques, outlined in the previous chapter, can provide useful upper bounds on nuclear ground states. However, variational techniques alone cannot provide the exact ground state. Using the variational result as a starting point, the powerful technique of GFMC can be used to project out the exact ground state solution.

The application of Green's function Monte Carlo to many-body quantum systems has enjoyed great success, particularly for bosonic systems like bulk liquid ^{4}He[21, 22]. Because of the required anti-symmetry of the fermion ground state, fermionic systems have posed a greater challenge in the application of these techniques, as will be discussed later. Regardless, significant progress has been made in the study of a variety of fermionic systems such as bulk ^{3}He [22][23], nuclear shell model systems[24, 25, 26], many-electron systems [27], atoms and molecules[28]. In the area of few-body nuclei, GFMC has been applied successfully in the past using spin-independent potentials [29] and has been adapted by Carlson [30, 39, 12] for use with realistic state dependent potentials.

My intent in this chapter is to review first the basic ideas behind the GFMC technique in general and then as it applies to the study of nuclear ground states.

4.1 The General Idea

An instructive and frequently used exercise to introduce the idea behind GFMC is to expand the starting variational wave function, Ψ_V, in terms of eigenvectors of the Hamiltonian with eigenvalue, E_i,

$$\Psi_V = \sum_i \alpha_i \Psi_i. \tag{4.1}$$

Acting on Ψ_V with the imaginary time evolution operator, $e^{-\hat{H}\tau}$ gives,

$$e^{-\hat{H}\tau}\Psi_V = \sum_i e^{-E_i\tau}\alpha_i\Psi_i. \tag{4.2}$$

For large values of τ, only the ground state will be left on the right hand side of Eq. (4.2). The assumption made here is that $E_o = 0$, which is easily achieved by subtracting a constant from the Hamiltonian. This simplistic picture forms the basis of the GFMC technique used in this work. It is important to note at this point that the wave function is not, in general, a real scalar. In the case of nuclear ground states, the wave function is a complex vector in spin and isospin space. Throughout the following analysis Ψ will be treated as a complex vector since the simplification to a scalar wave function is straightforward.

In order to apply Monte Carlo techniques to this problem, it is useful to re-cast the operation of the imaginary time evolution operator in terms of an integral equation in configuration space. First, the evolution operator is written as a product of small-time evolution operators,

$$e^{-\hat{H}\tau} = \prod e^{-\hat{H}\Delta\tau}. \tag{4.3}$$

The small-time evolution operator can then be written in terms of a Green's function in configuration space, $G(\mathbf{R}', \mathbf{R})$,

$$G(\mathbf{R}, \mathbf{R}') = \langle \mathbf{R}|e^{-H\Delta\tau}|\mathbf{R}'\rangle \tag{4.4}$$

Note that if the exact Green's function could be constructed as a function of τ, there would be no point in doing GFMC since all of the eigenstates would have to be known beforehand. The small-time construction allows us to make use of small $\Delta\tau$ approximations in the construction of the Green's function.

Using this configuration space Green's function, Eq. (4.2) can be written in the form of an integral equation ,

$$\Psi^{(n)}(\mathbf{R}) = \int G(\mathbf{R}, \mathbf{R}')\Psi^{(n-1)}(\mathbf{R}')d\mathbf{R}', \tag{4.5}$$

where $\Psi^{(n)}(\mathbf{R})$ is the nth generation of the GFMC wave function. Starting with $\Psi^{n=0} = \Psi_V$ and a technique for sampling $G(\mathbf{R}, \mathbf{R}')$, Eq. (4.5) could be iterated until the left-hand side converges to the ground state wave function.

A direct Monte Carlo integration of Eq. (4.5) could, however, be crippled by large statistical errors as configurations diffuse into regions where they make very little contribution to

the ground state wave function. Since the purpose is to project out a specific quantum state, information regarding that state can be built into the calculation. An importance function, Ψ_I, is introduced for this purpose. Multiplying Eq. (4.5) from the left hand side by $\Psi_I^\dagger(\mathbf{R})$ yields,

$$\Psi_I^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R}) = \int \Psi_I^\dagger(\mathbf{R})G(\mathbf{R},\mathbf{R}')\Psi^{(n-1)}(\mathbf{R}')d\mathbf{R}'. \tag{4.6}$$

Frequently, the importance function is taken to be the variational wave function since Ψ_V is usually the best guess at the ground state wave function. For the moment, no assumptions will be made about the relationship of the importance function and the variational wave function.

After some rearranging Eq. (4.6) becomes

$$\Psi_I^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R}) = \int \left[\Psi_I^\dagger(\mathbf{R})G(\mathbf{R},\mathbf{R}')\frac{1}{\Psi_I^\dagger(\mathbf{R}')}\right]\Psi_I^\dagger(\mathbf{R}')\Psi^{(n-1)}(\mathbf{R}')d\mathbf{R}' \tag{4.7}$$

where we have multiplied and divided by $\Psi_I^\dagger(\mathbf{R}')$ under the integral sign. For the moment, we will put aside the interpretation of this operation when $\Psi_I(\mathbf{R}')$ is a vector. The quantity in brackets is referred to as the importance sampled Green's function, $G_I(\mathbf{R}',\mathbf{R})$. By sampling both the importance function and the Green's function, we use the knowledge of the state we wish to project out to sample the new configurations.

Note that Eq. (4.7) is now an integral equation for the density $\Psi_I^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R})$. This density distribution can approximated by a set of Monte Carlo configurations if it is positive definite. In fermion systems, however, this is not the case since the fermion ground state wave function has nodal surfaces which can differ from those of the importance function and, hence, $\Psi_I^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R})$ can have negative values for certain configurations. For the calculations presented here, we avoid this difficulty by sampling the distribution, $|\Psi_I^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R})|$.

An estimate of our Monte Carlo population size, $N^{(n)}$, at the nth generation can be obtained by integrating the density distribution,

$$N^{(n)} = \int \Psi_I^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R})d\mathbf{R} = \sum_{i=1}^{N} \frac{\Psi_I^\dagger(\mathbf{R}_i)\Psi^{(n)}(\mathbf{R}_i)}{|\Psi_I^\dagger(\mathbf{R}_i)\Psi^{(n)}(\mathbf{R}_i)|} \tag{4.8}$$

where the sum is carried out over the N Monte Carlo configurations $\{\mathbf{R}_i\}$. The quality of the choice of $|\Psi_I^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R})|$ as the sampling distribution can be monitored by checking how much this estimate of the population size deviates from the actual number of configurations. As configurations diffuse across nodal surfaces in fermion systems, the quantity $N^{(n)}/N$,

becomes less than one since the accumulated minus signs in the numerator of Eq. (4.8) cancel positive contributions. As this deviation becomes significant ($\sim 20\%$) the calculation will be dominated by the statistical noise. As we will discuss later, GFMC calculations of fermion systems are frequently limited by this condition.

Using the definition given in Eq. (4.8), it is easy to demonstrate how the introduction of the importance function can reduce the statistical errors in the calculation. The importance function can be expanded in terms of eigen vectors of the Hamiltonian as we have done in Eq. (4.1) and the Green's function can be expressed in terms of eigenvectors as follows:

$$G(\mathbf{R},\mathbf{R}') = \sum_i \Psi_i(\mathbf{R})e^{-E_i\Delta\tau}\Psi_i^\dagger(\mathbf{R}') \tag{4.9}$$

Inserting these into Eq. (4.6) and integrating both sides of the equation with respect to $\mathbf{R}$ we are left with the following:

$$\sum_i \int \alpha_i^* \Psi_i^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R})d\mathbf{R} = \sum_i \int \alpha_i^* \Psi_i^\dagger(\mathbf{R}')e^{-E_i\Delta\tau}\Psi^{(n-1)}(\mathbf{R}')d\mathbf{R}'. \tag{4.10}$$

If the importance function is very close to the true ground state wave function, then $\alpha_0^* \approx 1$ and $\alpha_{i\neq 0}^* \approx 0$ and Eq. (4.10) becomes

$$\int \Psi_0^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R})d\mathbf{R} \approx \int \Psi_0^\dagger(\mathbf{R}')\Psi^{(n-1)}(\mathbf{R}')d\mathbf{R}'. \tag{4.11}$$

According to the definition (Eq. (4.8)) of the population size, Eq. (4.11) indicates that if the importance function is close to the ground state wave function, fluctuations in the population size from one time step to the next will be limited. A reduction in population size fluctuations means a reduction in statistical error in the Monte Carlo integration.

To this point, the assumption has been made that the Hamiltonian has been adjusted such that the ground state energy is zero. In practice, the exact ground state energy is not known beforehand. In this case, an approximate value, E_T, is subtracted from the Hamiltonian. Assuming that the importance function is sufficiently close to the ground state, Eq. (4.11) becomes

$$N^{(n)} \approx e^{-(E_o-E_T)\Delta\tau}N^{(n-1)} \tag{4.12}$$

and E_T can be adjusted to keep the population size stable. Eq. (4.12) also provides a simple way to estimate the exact ground state energy from the change in population size. Solving for E_o in Eq. (4.12) gives

$$E_o \approx \frac{1}{\Delta\tau} ln\left[\frac{N^{(n-1)}}{N^{(n)}}\right] + E_T. \tag{4.13}$$

This is referred to as the growth estimate and it provides an important consistency check on the GFMC calculation. Statistical variations in the growth estimate are caused by fluctuations in the population size. An importance function which is very close the exact ground state reduces the variance in the growth estimate.

Constructing and sampling $G_I(\mathbf{R}', \mathbf{R})$ can be a non-trivial task for complicated interactions. The Green's function can be simplified by separating the complications of the interaction from the kinetic energy in the Hamiltonian. The short-time evolution operator, $e^{-\hat{H}\Delta\tau}$, can be approximated to second order in $\Delta\tau$ by separating the kinetic energy and potential terms in symmetrical fashion as follows:

$$\begin{aligned} e^{-\hat{H}\Delta\tau} &= e^{-(\hat{T}+\hat{V})\Delta\tau} \\ &\approx e^{-\hat{V}\Delta\tau/2} e^{-\hat{T}\Delta\tau} e^{-\hat{V}\Delta\tau/2}. \end{aligned} \tag{4.14}$$

where the error in the approximation is of order $(\Delta\tau)^3$. We can now make the following simplification in the construction of the configuration space Green's function:

$$\begin{aligned} G(\mathbf{R}, \mathbf{R}') &= \langle \mathbf{R} | e^{-\hat{V}\Delta\tau/2} e^{-\hat{T}\Delta\tau} e^{-\hat{V}\Delta\tau/2} | \mathbf{R}' \rangle \\ &= e^{-V(\mathbf{R})\Delta\tau/2} \langle \mathbf{R} | e^{-\hat{T}\Delta\tau} | \mathbf{R}' \rangle e^{-V(\mathbf{R}')\Delta\tau/2}. \end{aligned} \tag{4.15}$$

The term within the angle brackets is the free-particle propagator in imaginary time, $G_o(\mathbf{R}', \mathbf{R})$, which, for non-relativistic systems, can be expressed as a simple Gaussian in configuration space

$$G_o(\mathbf{R}, \mathbf{R}') = \langle \mathbf{R} | e^{-\hat{T}\Delta\tau} | \mathbf{R}' \rangle = \left[\frac{1}{8\pi} \sqrt{\frac{2\pi m}{\Delta\tau \hbar^2}} \right]^{3A} exp \left[\frac{-(\mathbf{R} - \mathbf{R}')^2}{\frac{2\hbar^2 \Delta\tau}{m}} \right]. \tag{4.16}$$

In practice, $V(\mathbf{R})$ could be a complicated operator making it exceedingly difficult to sample $G(\mathbf{R}, \mathbf{R}')$ in a straightforward fashion. In the nuclear physics calculations conducted here, $V(\mathbf{R})$, and thus $G(\mathbf{R}, \mathbf{R}')$ is a matrix in spin-isospin space. In light of this fact, a scalar approximation to the importance sampled Green's function, $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$, is introduced. It is important to remember that we are approximating the sampling distribution which cannot change our expectation values but may effect our statistical errors.

With some rearrangement, Eq. (4.7) becomes

$$\Psi_I^\dagger(\mathbf{R}) \Psi^{(n)}(\mathbf{R}) = \int \left[\frac{\Psi_I^\dagger(\mathbf{R}) G(\mathbf{R}, \mathbf{R}') \Psi^{(n-1)}(\mathbf{R}')}{\Psi_I^\dagger(\mathbf{R}') \tilde{G}_I(\mathbf{R}, \mathbf{R}') \Psi^{(n-1)}(\mathbf{R}')} \right] \tilde{G}_I(\mathbf{R}, \mathbf{R}') \Psi_I^\dagger(\mathbf{R}') \Psi^{(n-1)}(\mathbf{R}') d\mathbf{R}'. \tag{4.17}$$

The quantity sampled in this formulation is also a density as in Eq. (4.7), but now each new sampled configuration is given a weight,

$$w^{(n)} = \left[\frac{\Psi_I^\dagger(\mathbf{R})G(\mathbf{R},\mathbf{R}')\Psi^{(n-1)}(\mathbf{R}')}{\Psi_I^\dagger(\mathbf{R}')\tilde{G}_I(\mathbf{R},\mathbf{R}')\Psi^{(n-1)}(\mathbf{R}')}\right] \tag{4.18}$$

where $G(\mathbf{R},\mathbf{R}')$ is an operator acting on $\Psi^{(n-1)}(\mathbf{R}')$. Making the following definition for $\Psi^{(n)}(\mathbf{R})$,

$$\Psi^{(n)}(\mathbf{R}) \equiv \left[\frac{G(\mathbf{R},\mathbf{R}')}{\tilde{G}_I(\mathbf{R},\mathbf{R}')}\right]\Psi^{(n-1)}(\mathbf{R}') \tag{4.19}$$

the weight can be expressed as

$$w^{(n)} = \left[\frac{\Psi_I^\dagger(\mathbf{R})\Psi^{(n)}(\mathbf{R})}{\Psi_I^\dagger(\mathbf{R}')\Psi^{(n-1)}(\mathbf{R}')}\right]. \tag{4.20}$$

Iterations of Eq. (4.17) amount to multiple matrix operations to build $\Psi^{(n)}$ for the current generation,

$$\Psi^{(n)}(\mathbf{R}^{(n)}) = \left[\frac{G(\mathbf{R}^{(n)},\mathbf{R}^{(n-1)})}{\tilde{G}_I(\mathbf{R}^{(n)},\mathbf{R}^{(n-1)})}\right]\cdots\left[\frac{G(\mathbf{R}^{(2)},\mathbf{R}^{(1)})}{\tilde{G}_I(\mathbf{R}^{(2)},\mathbf{R}^{(1)})}\right]\left[\frac{G(\mathbf{R}^{(1)},\mathbf{R}^{(0)})}{\tilde{G}_I(\mathbf{R}^{(1)},\mathbf{R}^{(0)})}\right]\Psi_V(\mathbf{R}^0). \tag{4.21}$$

A frequent point of confusion is the interpretation of $\Psi^{(n)}$. $\Psi^{(n)}$ is not the exact ground state wave function, but instead represents a vector of amplitudes which, when taken in product with the hermitian conjugate of the importance function gives an overlap for each component of the wave function. It is the subtle changes in these overlaps which pushes the distribution towards that of the true ground state.

The point of any GFMC calculation is to actually measure some phenomena, which means we would like to evaluate expectation values throughout the calculation. As we have stated before, at any stage in the calculation, the configurations are distributed according to $|\Psi_I(\mathbf{R})\Psi^{(n)}(\mathbf{R})|$. Thus, the expectation values must be evaluated as mixed estimates,

$$\langle\hat{O}\rangle_{mixed} = \frac{\langle\Psi_I|\hat{O}|\Psi^{(n)}\rangle}{\langle\Psi_I|\Psi^{(n)}\rangle} = \frac{\sum_{i=1}^{N}\frac{\Psi_I^\dagger(\mathbf{R}_i')\hat{O}\Psi^{(n)}(\mathbf{R}_i')}{|\Psi_I^\dagger(\mathbf{R}_i')\Psi^{(n)}(\mathbf{R}_i')|}}{\sum_{i=1}^{N}\frac{\Psi_I^\dagger(\mathbf{R}_i')\Psi^{(n)}(\mathbf{R}_i')}{|\Psi_I^\dagger(\mathbf{R}_i')\Psi^{(n)}(\mathbf{R}_i')|}} \tag{4.22}$$

The mixed estimate is sufficient for evaluating the energy, since the expectation value of the Hamiltonian in Eq. (4.22) is exact, provided that τ is sufficiently large for the calculation to have converged. It is easy to see this by the following analysis, (for simplicity we have taken $\Psi_I = \Psi_V$)

$$\frac{\langle\Psi_V|\hat{H}e^{-\hat{H}\tau}|\Psi_V\rangle}{\langle\Psi_V|e^{-\hat{H}\tau}|\Psi_V\rangle} = \frac{\langle\Psi_V|e^{-\hat{H}\tau/2}\hat{H}e^{-\hat{H}\tau/2}|\Psi_V\rangle}{\langle\Psi_V|e^{-\hat{H}\tau/2}e^{-\hat{H}\tau/2}|\Psi_V\rangle} \geq \frac{\langle\Psi_o|\hat{H}|\Psi_o\rangle}{\langle\Psi_o|\Psi_o\rangle}; \tag{4.23}$$

where the equality is obtained in the limit $\tau \to \infty$. Unfortunately, the mixed estimate is only exact for operators which commute with the Hamiltonian. Other ground state expectation values must be evaluated by extrapolating from the mixed estimate to first order in the difference between the variational wave function and the GFMC wave function.

$$\langle \hat{O} \rangle \approx 2\frac{\langle \Psi_V | \hat{O} | \Psi_o \rangle}{\langle \Psi_V | \Psi_o \rangle} - \frac{\langle \Psi_V | \hat{O} | \Psi_V \rangle}{\langle \Psi_V | \Psi_V \rangle}. \tag{4.24}$$

In the nuclear systems studied in this work, the variational wave functions are deemed to be good enough for this approximation to be valid.

After several iterations of Eq. (4.17) many of the configurations will diffuse into regions where their weight becomes very small. These configurations will make practically no contribution to the calculation. Eventually, only a handful of configurations will be making most of the contribution to the calculation. The result of such a process is an increase in the statistical noise in GFMC calculation. Furthermore, it is computationally inefficient to keep such small weight configurations in the calculation. To remedy this problem a technique called branching is used. Every few iterations of Eq. (4.17) the configurations are redistributed by keeping n_i unit weight copies of the ith configuration where

$$n_i = int(|w_i| + \zeta_i) \tag{4.25}$$

and ζ_i is a random number between 0 and 1. Using this technique, the total sum of the weights is reproduced on average. However, configurations with small weights will be more likely to be discarded while configurations with large weights will be reproduced. In this way, statistical noise is reduced by keeping an adequate population of contributing configurations.

There are several points which will require further discussion. How to construct $G(\mathbf{R}, \mathbf{R}')$, good choices for $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$, techniques for sampling $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$ and the form of the importance function will all be discussed in the next section in the context of our particular nuclear Hamiltonian. At this point, however, we can outline a general algorithm for conducting a GFMC calculation.

1. A collection of configurations, $\mathbf{R}'_i$, distributed with probability $|\Psi_I^\dagger \Psi_V|$, is generated using the Metropolis method. The starting GFMC wave function is taken to be $\Psi^{(n=0)} = \Psi_V$. The importance function, Ψ_I, is usually taken to be the variational wave function.

2. For each configuration, $\mathbf{R}'_i$, a new configuration, $\mathbf{R}_i$, is sampled from the approximate Green's function, $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$. Typically, the new point is sampled from the free-particle propagator, $G_o(\mathbf{R}, \mathbf{R}')$. Frequently, the statistical error can be reduced by using an approximate technique to incorporate the importance function in the sampling.

3. Using the initial point, $\mathbf{R}'$, and the final point, $\mathbf{R}$, the full operator Green's function, $G(\mathbf{R}, \mathbf{R}')$, is constructed and operated on $\Psi^{(n-1)}$.

4. Steps 2 and 3 are repeated until branching. At which point, the the weight for each configuration, $w_i^{(n)}$ is calculated according to Eq. (4.18) and the configurations are redistributed using Eq. (4.25).

5. Expectation values are calculated periodically using Eq. (4.22).

To this point, we have made no assumption about the nature of the system we are studying. It is important to consider the distinction between fermionic and bosonic systems. Fermion systems require that the ground state be antisymmetric under the exchange of any pair of particles. One can imagine starting with a set of configurations representing a fermionic system where 50% of those configurations are in regions where the wave function is negative and 50% where the wave function in positive as required by antisymmetry. If we perform a Monte Carlo walk where each configuration moves independently of the others, such that configurations can walk across nodal surfaces in the wave function, the asymptotic distribution of configurations will be symmetric (bosonic). This problem arises because there is no way for the global property of antisymmetry to be maintained when the configurations are moved independently of each other. This problem is referred to as the 'sign problem' and is a long standing problem in the calculation of fermion ground states. The net result of the sign problem in our GFMC calculation is a diminishing overlap with the antisymmetric importance function which will cause large statistical fluctuations. The fluctuations can grow quickly with τ since quite often the symmetric ground state has a lower energy than the antisymmetric ground state.

There are several techniques for minimizing or delaying the sign problem. The simplest technique is to allow the population size to grow quickly enough to accommodate the increase in fluctuations. This is impractical for large τ since the population size will have to grow exponentially to unwieldy proportions. However, very stringent upper bounds to the

ground state energy can be obtained from the small τ results using Eq. (4.22). For a given computational effort, the accuracy of the results will improve with a good choice of variational wave function. This technique is referred to as 'transient estimation'[32]. Another way to obtain an upper bound on the ground state energy is to prevent the configurations from diffusing across nodal boundaries. This is called the 'fixed node' method [27]. In most systems of interest, the exact nodes are not known a priori and fixing the nodal surfaces can result in discontinuities in the first derivative of the wave function, thus only an upper bound can be determined. Typically, the configurations which diffuse into regions where there is a sign change in the overlap of the importance function and the GFMC wave function are discarded. In this way, each nodal surface is defined by the nodes in the importance function and the distribution within each nodal surface is bosonic. It is possible, in some systems, to vary the nodes in the importance function to minimize the ground state energy obtained with this technique. Unfortunately, most systems are too complex to allow a direct parameterization of the nodal surfaces in the importance function. There are other techniques which involve correlating the walks of the configurations from positive and negative regions of the wave function. Thus far, these approaches have only been successfully applied in simple systems[33, 34, 35].

4.2 GFMC Calculations of Nuclear Ground States

The complicated spin-isospin dependence of the nuclear potential and the complex nature of the nuclear ground state wave functions makes the application of GFMC techniques to this problem especially challenging. The general technique as outlined in the previous section remains, for the most part, unchanged. However, the devil is in the details and this section is devoted to the details of GFMC calculations of nuclear ground states. The intent here is to provide the reader with sufficient information to reproduce the calculations discussed in the remaining sections of this work. The intestinal fortitude to take on such a task is left as an exercise for the reader.

In this section, we will expand upon the algorithm for GFMC calculations presented in the previous section. Specifically, we intend to discuss our choice and sampling technique for $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$, and the two techniques which we have used to construct the $G(\mathbf{R}, \mathbf{R}')$ operator.

4.2.1 Choosing and Sampling $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$

The importance sampled Green's function was introduced in Eq. (4.7), however, this definition is ill-defined when we consider spin-dependent interactions and wave functions. As we have discussed before, it is frequently necessary to introduce an approximate Green's function for which we have a straight-forward sampling technique. The difference between the exact $G_I(\mathbf{R}, \mathbf{R}')$ and the approximate $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$ is taken into account with the weights (Eq. 4.18). We would like to have a form for $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$ which is convenient to work with and also close to $G_I(\mathbf{R}, \mathbf{R}')$ so that the weights are close to unity. A computationally simple choice for $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$ is the free particle propagator, $G_o(\mathbf{R}, \mathbf{R}')$. The technique for sampling a Gaussian is well known, but we will outline the steps here for completeness.

Given two random variables, x and y, distributed according to e^{-x} and e^{-y} respectively, then $x = -\ln(\zeta)$ and $y = -\ln(\zeta')$ where ζ and ζ' are random numbers between one and zero. By noting that the probability that

$$y > \frac{1}{2}(x-1)^2 \text{is} \int_{\frac{1}{2}(x-1)^2}^{\infty} e^{-y'} dy' = e^{-\frac{1}{2}(x-1)^2} \tag{4.26}$$

and

$$x < z \text{is} \int_0^z e^{-x'} dx' \tag{4.27}$$

means that the joint probability is given by

$$\int_0^z e^{-x'} e^{-\frac{1}{2}(x'-1)^2} dx' = \sqrt{e} \int_0^z e^{-\frac{x'^2}{2}} dx'. \tag{4.28}$$

This means that if y and x are chosen to simultaneously satisfy Eq. (4.26) and Eq. (4.27), then x will have a Gaussian distribution, independent of z. Using this idea it is possible to sample a Gaussian distribution using the following algorithm:

1. Generate two random numbers, ζ and ζ', between zero and one.
2. Let $x = -\ln(\zeta)$ and $y = -\ln(\zeta')$.
3. If $y < \frac{1}{2}(x-1)^2$ then go back to step 1.
4. Otherwise, the sampled point is given by $x_s = x$.

The above technique can be applied to the free-particle propagator, where a new coordinate position, x_i' is sampled for each of the 3A Gaussians which make up $G_o(\mathbf{R}, \mathbf{R}')$.

Accordingly, each new position, x_i', is given by

$$x_i' = x_i + \frac{\zeta - 0.5}{|\zeta - 0.5|} x_s \sqrt{\frac{\Delta\tau\hbar^2}{m}} \tag{4.29}$$

where ζ is a random number between zero and one to account for the arbitrary sign. Thus, a new point, $\mathbf{R}_i$ for configuration i can be randomly sampled from the free-particle propagator using $\mathbf{R}_i'$.

Of course, using the free-particle propagator to sample the new point completely disregards the importance function and the exponent of the potential in our Green's function. In our work, the spin-isospin dependence of the wave function and the potential, make it particularly difficult to sample them directly. We can, however, employ some simple approximations to incorporate the potential and the importance function in our choice of a new configuration. Recalling the form of the importance sampled Green's function from Eq. (4.7),

$$G_I(\mathbf{R}, \mathbf{R}') = \left[\Psi_I(\mathbf{R}) e^{-V(\mathbf{R})\Delta\tau/2} G_o(\mathbf{R}, \mathbf{R}') e^{-V(\mathbf{R}')\Delta\tau/2} \frac{1}{\Psi_I(\mathbf{R}')}\right] \tag{4.30}$$

we introduce a scalar approximation, $\tilde{\Psi}_I$, for Ψ_I which approximates the central parts of the wave function. In the case of three- and four-body nuclei, this is just a product of the central correlations, $\tilde{\Psi}_I = \prod_{i<j} f_c(r_{ij})$. For nuclei with greater than four particles, the wave function contains p-state orbitals. To include this information, we use the magnitude of the Jastrow wave function at that point, $\tilde{\Psi}_I = \sqrt{\Psi_J^\dagger \Psi_J}$. We also need a scalar approximation of the potential. For this we use the average of the central parts of the potentials in the $^1S_0$ and $^3S_1$ channels. With these approximations in mind, Eq. (4.30) now becomes our equation for $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$

$$\tilde{G}_I(\mathbf{R}, \mathbf{R}') = \left[\tilde{\Psi}_I(\mathbf{R}) e^{-(V_{10}(\mathbf{R}) + V_{01}(\mathbf{R}))\Delta\tau/4} G_o(\mathbf{R}, \mathbf{R}') e^{-(V_{10}(\mathbf{R}') + V_{01}(\mathbf{R}')\Delta\tau/4} \frac{1}{\tilde{\Psi}_I(\mathbf{R}')}\right], \tag{4.31}$$

where V_{01} and V_{10} are the central parts of the potential in the $^1S_0$ and $^3S_1$ channels respectively.

Now that we have a form for our approximate importance sampled Green's function, how do we sample it? Our approximations for the importance function and potential still contain tabulated functions which form a non-trivial distribution to sample. We can sample $\tilde{G}_I(\mathbf{R}, \mathbf{R}')$ in an approximate way by first sampling a point from the free-particle propagator

and then weighting our result by using local information about Ψ_I around the point $\mathbf{R}'$. To see how this is done, first factor out the free-particle propagator from $\tilde{G}_I(\mathbf{R},\mathbf{R}')$.

$$\tilde{G}_I(\mathbf{R},\mathbf{R}') = P(\mathbf{R},\mathbf{R}')G_o(\mathbf{R},\mathbf{R}') \tag{4.32}$$

where $P(\mathbf{R},\mathbf{R}')$ contains the ratio of the importance functions and the exponential of the potentials. When a configuration is sampled from $G_o(\mathbf{R},\mathbf{R}')$, there is an equal probability of choosing any point on the 3A dimensional sphere where $G_o(\mathbf{R},\mathbf{R}')$ has the same value. In principle we can eliminate the first derivative of P in the expansion about the point $\mathbf{R}'$, by averaging P over the 6A points around $\mathbf{R}'$ which correspond to the two possible choices for each coordinate of each particle. In practice, we only use two points in our average, $\mathbf{R}_1$ sampled from $G_o(\mathbf{R},\mathbf{R}')$ and the mirror symmetric configuration given by $\mathbf{R}_2 = \mathbf{R}' + (\mathbf{R}' - \mathbf{R}_1)$. This choice of points eliminates the first derivative along the direction sampled in configuration space. We choose one these points to be $\mathbf{R}$ with the probability,

$$\mathcal{P} = \frac{P(\mathbf{R},\mathbf{R}')}{P(\mathbf{R}_1,\mathbf{R}') + P(\mathbf{R}_2,\mathbf{R}')} \tag{4.33}$$

and weight:

$$\frac{\frac{1}{2}(P(\mathbf{R}_1,\mathbf{R}') + P(\mathbf{R}_2,\mathbf{R}'))}{P(\mathbf{R},\mathbf{R}')}. \tag{4.34}$$

4.2.2 Building $G(\mathbf{R},\mathbf{R}')$

Once we have sampled a new point in configuration space, we can construct the full operator Green's function. The complicated operator dependence of the nuclear potential makes it especially difficult to evaluate the Green's function. In this section, we will detail two of the approaches we have used to approximate the operator Green's function in our calculations. The first approach utilizes the second order approximation to $G(\mathbf{R},\mathbf{R}')$ given in Eq. (4.15). This approach has the advantage that the Green's function can be easily constructed, however small $\Delta\tau$ (< 0.0003 MeV^{-1}) is necessary for our approximations to be valid. Furthermore, terms in the potential which are quadratic in the angular momentum are very difficult to evaluate with this technique. In the second approach, the two-body portion of the Green's function is expressed in terms of a product of exact two-body pair propagators. Cooling techniques are used in each operator channel to allow larger values of $\Delta\tau$. In principle, the exact two-body propagators can include the the momentum dependent terms from the

potential. The three-body terms are, once again, treated to first order in $\Delta\tau$. Note that a $G(\mathbf{R},\mathbf{R}')$ exact to order $\Delta\tau$ is sufficient for GFMC. The error in propagation up to τ is then of order $(\Delta\tau)^2\tau/\Delta\tau = (\Delta\tau)\tau$ and can be made negligibly small by choosing small $\Delta\tau$. However, the computation time is proportional to $1/\Delta\tau$, hence it is useful to use as large a $\Delta\tau$ as possible while keeping the error small.

First-order Propagator

According to Eq. (4.15), the operator Green's function can be approximated to second order by

$$G(\mathbf{R},\mathbf{R}') \approx e^{-V(\mathbf{R})\Delta\tau/2} G_o(\mathbf{R},\mathbf{R}')\mathbf{R} e^{-V(\mathbf{R}')\Delta\tau/2} \tag{4.35}$$

where the potentials in the exponents represents the total potential for that configuration. The exponentials of the two-body potentials can be approximated to first order by turning the sums over pairs in the exponent into a symmetrized product of exponentials of the individual pair potentials.

$$e^{-V(\mathbf{R})\Delta\tau/2} e^{-V(\mathbf{R}')\Delta\tau/2} \approx \mathcal{S} \prod_{i<j} e^{-\sum_p (v_p(r_{ij}) + v_p(r'_{ij}))O^p_{ij}\Delta\tau/2} \tag{4.36}$$

where the sum is over the operator terms in the potential. The first six terms of the potential can be easily exponentiated. In order to carry out the exponentiation, it is useful to write the potential for each pair in terms of the contributions, v_{ST}, to each T, S channel as follows...

$$\begin{aligned} v(r_{ij}) &= v_{00}(r_{ij})P_{S=0,T=0} + v_{11}(r_{ij})P_{S=1,T=1} + v_{10}(r_{ij})P_{S=1,T=0} \\ &\quad + v_{01}(r_{ij})P_{S=0,T=1} + v_{t,0}(r_{ij})S_{ij}P_{T=0} + v_{t,1}(r_{ij})S_{ij}P_{T=1} \end{aligned} \tag{4.37}$$

where $v_{t,T}$ is the tensor potential and $P_{S,T}$ is the projection operator for the spin S, isospin T channel. The tensor operators only contribute in the S=1 channels, so the $(\sigma_i \cdot \sigma_j)$ portion of the tensor operator can be absorbed into $v_{S=1,T}$ by defining:

$$\tilde{v}_{S=1,T} = (v_{S=1,T} - v_{t,T}) \tag{4.38}$$

and

$$\tilde{S}_{ij} = (\sigma_\mathbf{i} \cdot \hat{\mathbf{r}}_\mathbf{ij})(\sigma_\mathbf{j} \cdot \hat{\mathbf{r}}_\mathbf{ij}). \tag{4.39}$$

Using these definitions we can re-write Eq. (4.37) as

$$\begin{aligned} v(r_{ij}) &= v_{00}(r_{ij})P_{S=0,T=0} + v_{01}(r_{ij})P_{S=0,T=1} \\ &+(\bar{v}_{10}(r_{ij}) + 3v_{t,0}(r_{ij})\bar{S}_{ij})P_{S=1,T=0} \\ &+(\bar{v}_{11}(r_{ij}) + 3v_{t,1}(r_{ij})\bar{S}_{ij})P_{S=1,T=1} \end{aligned} \tag{4.40}$$

Exponentiating and using the fact that $\bar{S}_{ij}$ is a unitary operator, we have

$$\begin{aligned} & e^{-\sum_{p=1}^{6}(v_p(r_{ij})+v_p(r'_{ij}))O^p\Delta\tau/2} = \\ & e^{-(v_{00}(r_{ij})+v_{00}(r'_{ij}))\frac{\Delta\tau}{2}}P_{S=0,T=0} + e^{-(v_{01}(r_{ij})+v_{01}(r'_{ij}))\frac{\Delta\tau}{2}}P_{S=0,T=1} \\ & +e^{-(\bar{v}_{10}(r_{ij})+\bar{v}_{10}(r'_{ij}))\frac{\Delta\tau}{2}}[\cosh\left(3v_{t,0}(r_{ij})\frac{\Delta\tau}{2}\right) + \sinh\left(3v_{t,0}(r_{ij})\frac{\Delta\tau}{2}\right)\bar{S}_{ij}] \times \\ & [\cosh\left(3v_{t,0}(r'_{ij})\frac{\Delta\tau}{2}\right) + \sinh\left(3v_{t,0}(r'_{ij})\frac{\Delta\tau}{2}\right)\bar{S}'_{ij}]P_{S=1,T=0} \\ & +e^{-(\bar{v}_{11}(r_{ij})+\bar{v}_{11}(r'_{ij}))\frac{\Delta\tau}{2}}[\cosh\left(3v_{t,1}(r_{ij})\frac{\Delta\tau}{2}\right) + \sinh\left(3v_{t,1}(r_{ij})\frac{\Delta\tau}{2}\right)\bar{S}_{ij}] \times \\ & [\cosh\left(3v_{t,1}(r'_{ij})\frac{\Delta\tau}{2}\right) + \sinh\left(3v_{t,1}(r'_{ij})\frac{\Delta\tau}{2}\right)\bar{S}'_{ij}]P_{S=1,T=1} \end{aligned} \tag{4.41}$$

One final rearrangement using the first-order approximation, $\bar{S}_{ij}\bar{S}'_{ij} \approx 1$ and resolving the action of $\bar{S}_{ij}$ on $P_{S=1}$ gives us,

$$\begin{aligned} e^{-\sum_{p=1}^{6}(v_p(r_{ij})+v_p(r'_{ij}))O^p\Delta\tau/2} &\approx g_{00}(r_{ij},r'_{ij})P_{S=0,T=0} + g_{01}(r_{ij},r'_{ij})P_{S=0,T=1} \\ &+g_{10}(r_{ij},r'_{ij})P_{S=1,T=0} + g_{11}(r_{ij},r'_{ij})P_{S=1,T=1} \\ &+(g_{t,0}(r_{ij},r'_{ij})\bar{S}_{ij} + g'_{t,0}(r_{ij},r'_{ij})\bar{S}'_{ij})P_{T=0} \\ &+(g_{t,1}(r_{ij},r'_{ij})\bar{S}_{ij} + g'_{t,1}(r_{ij},r'_{ij})\bar{S}'_{ij})P_{T=1} \end{aligned} \tag{4.42}$$

where

$$\begin{aligned} g_{0T}(r_{ij},r'_{ij}) &= e^{-(v_{0T}(r_{ij})+v_{0T}(r'_{ij}))\frac{\Delta\tau}{2}} \times [\sinh\left(3v_{t,T}(r'_{ij})\frac{\Delta\tau}{2}\right)\cosh\left(3v_{t,T}(r_{ij})\frac{\Delta\tau}{2}\right) \\ &+\sinh\left(3v_{t,T}(r_{ij})\frac{\Delta\tau}{2}\right)\cosh\left(3v_{t,T}(r'_{ij})\frac{\Delta\tau}{2}\right)] \end{aligned} \tag{4.43}$$

$$g_{1T}(r_{ij},r'_{ij}) = e^{-(v_{1T}(r_{ij})+v_{1T}(r'_{ij}))\frac{\Delta\tau}{2}} \times [\sinh\left(3v_{t,T}(r'_{ij})\frac{\Delta\tau}{2}\right)\sinh\left(3v_{t,T}(r_{ij})\frac{\Delta\tau}{2}\right)$$

$$+\cosh\left(3v_{t,T}(r'_{ij})\frac{\Delta\tau}{2}\right)\cosh\left(3v_{t,T}(r_{ij})\frac{\Delta\tau}{2}\right)] \tag{4.44}$$

$$g_{t,T}(r_{ij},r'_{ij}) = e^{-(v_{1T}(r_{ij})+v_{1T}(r'_{ij}))\frac{\Delta\tau}{2}} \times [\cosh\left(3v_{t,T}(r'_{ij})\frac{\Delta\tau}{2}\right)\sinh\left(3v_{t,T}(r_{ij})\frac{\Delta\tau}{2}\right)] \tag{4.45}$$

$$g'_{t,T}(r_{ij},r'_{ij}) = e^{-(v_{1T}(r_{ij})+v_{1T}(r'_{ij}))\frac{\Delta\tau}{2}} \times [\sinh\left(3v_{t,T}(r'_{ij})\frac{\Delta\tau}{2}\right)\cosh\left(3v_{t,T}(r_{ij})\frac{\Delta\tau}{2}\right)] \tag{4.46}$$

$$\tag{4.47}$$

This expression for the exponent of the pair potential can be projected back into operator channels to form an operator.

$$\begin{aligned} g(r_{ij},r'_{ij}) &= \sum_{p=1}^{4} g_p(r_{ij},r'_{ij})O^p_{ij} + (g_5(r_{ij},r'_{ij}) + g_6(r_{ij},r'_{ij})(\tau_i\cdot\tau_j))S_{ij} \\ &\quad +(g'_5(r_{ij},r'_{ij}) + g'_6(r_{ij},r'_{ij})(\tau_i\cdot\tau_j))S'_{ij} \end{aligned} \tag{4.48}$$

Now we can perform the operation of $g(r_{ij},r'_{ij})$ on the GFMC wave function in the same way as the two-body operator correlations.

In this approach, the momentum dependent terms cannot be treated satisfactorily since they contain derivatives which act on the wave function. Terms quadratic in the angular momentum could be grouped with the kinetic energy and treated as a direction and state dependent effective mass. At this point, the details of such an approach have not yet been explored. A less accurate, but easier to implement technique is to expand the exponential of the momentum dependent terms to first order in $\Delta\tau$ and use integration by parts to let the derivatives act on the free-particle Green's function. This approach can only be successfully applied for the terms in the potential which are linear in momentum, such as $\mathbf{L}\cdot\mathbf{S}$. Terms quadratic in the momentum cannot be evaluated to first order in this manner. For this reason we use an approximation to the v_{18} called the v'_8 which contains only the first eight operators. The difference (v_{18}-v'_8) is treated in perturbation theory.

Using the first order expansion of the spin-orbit terms and the three-body potential, we can write the full operator Green's function as follows:

$$G(\mathbf{R},\mathbf{R}') \approx e^{E_o\Delta\tau} I_{NNN}(\mathbf{R},\mathbf{R}') I_{\mathbf{LS},ij}(\mathbf{R},\mathbf{R}')\mathcal{S}\prod_{i<j}\left[g(r_{ij},r'_{ij})\right] exp\left[\frac{-m(\mathbf{R}-\mathbf{R}')^2}{2\hbar^2\Delta\tau}\right]. \tag{4.49}$$

where,

$$I_{\mathbf{LS},ij}(\mathbf{R},\mathbf{R}') = \mathcal{S}\prod[1-\frac{\Delta\tau}{2}(v_{\mathbf{LS},ij}(\mathbf{R})+v_{\mathbf{LS},ij}(\mathbf{R}'))]. \tag{4.50}$$

$$I_{NNN}(\mathbf{R},\mathbf{R}') = (1 - \frac{\Delta\tau}{2}\sum(V^{2\pi}_{ijk}(\mathbf{R}) + V^{2\pi}_{ijk}(\mathbf{R}'))exp[-\sum(V^{R}_{ijk}(\mathbf{R}) + V^{R}_{ijk}(\mathbf{R}'))\frac{\Delta\tau}{2}]). \tag{4.51}$$

The order of the operators in the symmetrized products is randomly sampled. Note that the repulsive part of the three-body force has no operator dependence and is therefore left in the exponent. Within this framework, the procedure for building the Green's function is straight-forward, though tedious and the operation of $G(\mathbf{R},\mathbf{R}')$ on $\Psi^{(n-1)}$ involves the same matrix algebra used to construct the variational wave function.

Exact Two-body Propagator

An alternative is to build the Green's function operator as a product of exact two-body propagators,

$$G(\mathbf{R},\mathbf{R}') \approx \mathcal{S}\prod_{i<j}\frac{g_{ij}(\mathbf{r}_{ij},\mathbf{r}'_{ij})}{g^0_{ij}(\mathbf{r}_{ij},\mathbf{r}'_{ij})}G_o(\mathbf{R},\mathbf{R}') \tag{4.52}$$

where $g_{ij}(\mathbf{r}_{ij},\mathbf{r}'_{ij})$ is the exact two-body propagator and $g^0_{ij}(\mathbf{r}_{ij},\mathbf{r}'_{ij})$ is the two-body free-particle propagator. Unfortunately, constructing the exact two-body propagator, $g_{ij}(\mathbf{r}_{ij},\mathbf{r}'_{ij})$, is nontrivial for nuclear interactions.

In our work, the pair propagator for particles i and j, g_{ij}, is expanded to second order in $\Delta\tau$ as in Eq. (4.14). The propagator with a time step of size $N\Delta\tau$ is given by,

$$g_{ij} = [e^{-\hat{V}_{ij}\Delta\tau/2}e^{-\hat{T}_{ij}\Delta\tau}e^{-\hat{V}_{ij}\Delta\tau/2}]^N. \tag{4.53}$$

As before, the error in such an expansion is of order $\Delta\tau^3$. After N operations, the accumulated error is proportional to $1/N^2$. Schmidt and Lee [36] have developed a highly accurate scheme for eliminating time step errors in propagators like Eq. (4.53) where the error is an even power of $1/N$. Their technique involves building g_{ij} on a $\mathbf{r}_{ij}$, $\mathbf{r}'_{ij}$ grid for different values of N, and then using a polynomial extrapolation in $1/N^2$ to extract the exact result. Such an approach allows us to use larger time steps and thus, fewer iterations in our calculation.

To implement Schmidt and Lee's technique, we first express the two-body propagator as a sum over two-body eigenfunctions, in the case of nuclear interactions, the summation amounts to a sum over channels,

$$g_{ij}(\mathbf{r}_{ij},\mathbf{r}'_{ij}) = \sum_{J,m_J}\sum_{L,m_L}\sum_{L',m_{L'}}\sum_{S,m_S}\sum_{m'_S}\sum_{T,m_T}|L,m_L,S,m_S,T,m_T\rangle \times$$

$$\langle L, m_L, S, m_S | J, m_J \rangle g_{ij,JST,L,L'}(\mathbf{r}_{ij}, \mathbf{r}'_{ij}) \langle J, m_J | L', m_{L'}, S, m'_S \rangle \times$$
$$\langle L', m_{L'}, S, m'_S, T, m_T |. \tag{4.54}$$

Typically, channels up to L=55 are used in the sum. In this expansion we have used the fact that the nuclear interaction conserves J and is diagonal in S, T, and m_T. Furthermore, we know that the the tensor part of the nuclear interaction couples $L = J \pm 1$ channels. Hence,

$$g_{ij,JST,L,L'}|_{UncoupledChannels} = g_{ij,JST}\ \delta_{L,L'} \tag{4.55}$$

in uncoupled channels and

$$g_{ij,JST,L,L'}|_{CoupledChannels} = \begin{bmatrix} g_{ij,JST,L=J-1,L'=J-1} & g_{ij,JST,L=J-1,L'=J+1} \\ g_{ij,JST,L=J+1,L'=J-1} & g_{ij,JST,L=J+1,L'=J+1} \end{bmatrix} \tag{4.56}$$

in the coupled channels. In the uncoupled channels there is the usual symmetry,

$$g_{ij,JST,L,L'}(\mathbf{r}_{ij}, \mathbf{r}'_{ij}) = g_{ij,JST,L,L'}(\mathbf{r}'_{ij}, \mathbf{r}_{ij}), \tag{4.57}$$

while in the coupled channels the following symmetry is obeyed:

$$g_{ij,JST,L=J\pm1,L'=J\pm1}(\mathbf{r}_{ij}, \mathbf{r}'_{ij}) = g_{ij,JST,L=J\pm1,L'=J\pm1}(\mathbf{r}'_{ij}, \mathbf{r}_{ij})$$
$$g_{ij,JST,L=J\pm1,L'=J\mp1}(\mathbf{r}_{ij}, \mathbf{r}'_{ij}) = g_{ij,JST,L=J\mp1,L'=J\pm1}(\mathbf{r}'_{ij}, \mathbf{r}_{ij}). \tag{4.58}$$

It is important to note that the sum in Eq. (4.54) is carried out over both physical (L+T+S=odd) and unphysical (L+T+S=even) channels. Summing over the physical channels only will give a symmetric propagator for T=0 and an antisymmetric propagator for T=1. If this is the case, then the new point must be sampled with a free-particle propagator of the appropriate symmetry. In the present work, this type of complication is avoided by summing over all channels.

The propagator is evaluated on a grid such that $\mathbf{r}'_{ij}$ lies along the z-axis and $\mathbf{r}_{ij} - \mathbf{r}'_{ij}$ lies in the $x - z$ plane such that $z - z' > 0$ and $x > 0$. In this way, only three dimensions (x, z, z') are required to specify the propagator and, combined with the symmetries from Eqs. (4.57) and (4.58), the propagator at all $\mathbf{r}'_{ij}$ and $\mathbf{r}_{ij}$ values can be obtained from rotations. Since $\mathbf{r}'_{ij}$ lies along the z-axis, spin-projections are quantized along $\mathbf{r}'_{ij}$. Appropriate SU(3) rotations must be used to project $g_{ij,JST,L,L'}$ into a spin basis quantized along the z-axis.

Within each channel, the propagator $g_{ij,JST,L,L'}$ for time step $\Delta\tau$ is constructed from N smaller time step propagators according to Eq. (4.53). To take advantage of the properties

of the polynomial extrapolation in N, several even values of N are used until the extrapolation converges within the desired error. Building the propagator from smaller time step propagators involves integrating the integral equation formulated from Eq. (4.53).

$$g_{ij,JST,L,L'}(r_{ij}, r'_{ij}; n\Delta\tau/N) = \sum_{L''} \int e^{-v_{JSTL}(r_{ij})\Delta\tau/2} e^{-\hat{T}_{ij}\Delta\tau} e^{-v_{JSTL''}(r''_{ij})\Delta\tau/2} \times g_{ij,JST,L'',L'}(r''_{ij}, r'_{ij}; (n-1)\Delta\tau/N) dr''_{ij} \quad (4.59)$$

until the propagator for step size $\Delta\tau$ has been constructed. Iteration of Eq. (4.59) is accomplished by beginning with a delta function, $\delta(r''_{ij} - r'_{ij})$, for $g_{ij,JST,L'',L'}$. Each iteration begins with the exponential $e^{-v_{JSTL''}(r''_{ij})\Delta\tau/2}$, acting on $g_{ij,JST,L'',L'}$ and then Fourier transforming into k-space from r''_{ij}-space. In momentum space, the kinetic energy propagator is easily expressed as $e^{-\Delta\tau\frac{\hbar^2 k^2}{2\mu}}$. After multiplying by kinetic energy propagator, we then Fourier transform back to r_{ij}-space and multiply by $e^{-v_{JSTL}(r_{ij})\Delta\tau/2}$. Finally, we integrate over r''_{ij} and sum over the allowed values of L'' for that channel. For the uncoupled channels there is only one allowed value of L'', while in the coupled channels, the allowed values of L are $J \pm 1$.

Once the propagator is constructed for each value of N, the exact propagator in that channel is determined by using a polynomial extrapolation in powers of $\frac{1}{N^2}$ to extrapolate to the limit $\frac{1}{N^2} \to 0$, ie. $N \to \infty$. By taking advantage of the fact that the errors in the expansion (4.53) are even powers of $\frac{1}{N}$, the polynomial extrapolation converges very quickly.

The final propagator, $g_{ij}(\mathbf{r}_{ij}, \mathbf{r}'_{ij})$, is constructed by summing over all channels and projecting into an operator basis. Currently, the pair propagator at each grid point is expressed as a complex $2 \times 4 \times 4$ matrix for the 2 possible isospin states (T=0,1) and the four possible initial and final spin states ($\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$) for the pair. The propagator is constructed once and stored in a file. The GFMC program reads the grid data from the file and interpolates in z', z and x to construct the propagator at the specific points $\mathbf{r}_{ij}$ and $\mathbf{r}'_{ij}$.

Unlike the first order propagator, terms quadratic in the angular momentum can be incorporated into the exact pair propagator. However, the inclusion of these non-local terms requires that the sampled distribution have the same locality structure if the statistical error is to be kept to a acceptable level. As we have said before, this could be accomplished with a direction and state- dependent effective mass in the free particle propagator. In this work, we use the v'_8 version of the potential in the exact pair propagator just as we have in the first-order propagator. The difference here is that the exact pair propagator makes

no approximations for the $(\mathbf{L}\cdot\mathbf{S}_{ij})$ terms. As in the first-order propagator case, the three nucleon interaction is expanded to first-order, but is treated in a more symmetrical fashion as follows:

$$G(\mathbf{R},\mathbf{R}') \approx e^{E_o\Delta\tau} I_{NNN}(\mathbf{R})\mathcal{S}\prod_{i<j}\frac{g_{ij}(\mathbf{r}_{ij},\mathbf{r}'_{ij})}{g^0_{ij}(\mathbf{r}_{ij},\mathbf{r}'_{ij})}G_o(\mathbf{R},\mathbf{R}')I_{NNN}(\mathbf{R}') \tag{4.60}$$

where

$$I_{NNN}(\mathbf{R}) \;=\; (1-\frac{\Delta\tau}{2}\sum V^{2\tau}_{ijk}(\mathbf{R}))exp[-\sum V^{R}_{ijk}(\mathbf{R})\frac{\Delta\tau}{2}]). \tag{4.61}$$

Although this technique allows us to use larger time steps in our calculation, the size of the time step used is still limited by the fact that we are approximating the full propagator as a product of two-body propagators. The error in such an approximation is of order $[v_{ij}, v_{jk}](\Delta\tau)^2$. Clearly, the error becomes significant when three of the particles come very close, an event which does not happen frequently in our calculations. In this work, we have used a step size of $\Delta\tau$=0.0005 MeV^{-1}.

Chapter 5

Results

Aided by advances in supercomputing power, the extension of exact calculations of nuclear ground states beyond two nucleon systems has proceeded very rapidly over the past decade. Faddeev and Faddeev-Yakubovosky methods have been applied in the study of three-body [13, 37] and four-body systems [11], methods making use of hyperspherical functions have been used to study low-energy three- and four- nucleon states [38], and Quantum Monte Carlo calculations have been applied to $A \leq 5$ nuclei [39, 40]. Parallel computation has opened the door to a whole new range of nuclear systems to study. In this section, we will discuss the application of GFMC techniques to nuclei with $A \leq 7$ as well as 7 and 8 neutron clusters. Unlike the $A = 2$ to 4 nuclei, the $A = 6$ to 8 nuclei have a spectrum of bound states indicating shell structure and spin-orbit coupling. Exact calculations of these nuclei will be useful to probe the physics of the shell model, to construct realistic models of three-nucleon interactions, and will provide wave functions to study electron-scattering observables and reactions of interest in astrophysics. Furthermore, calculations of neutron drop ground states will provide important information regarding the strength of the spin-orbit interaction in neutron matter and provide data for fitting Energy Density Functionals (EDF's) in neutron-rich systems.

5.1 Computational Considerations

The complicated nature of the nuclear interaction and the computational complexity of the calculations presented here require high performance computing. In the past, vector supercomputers were used for the first ^4He and ^5He GFMC calculations. As we have stated before the size of the wave function grows exponentially with the number of nucleons and the

number of matrix operations grows with the number of pairs. In making the step from five to six body calculations at least an order of magnitude increase in computational performance was required. The clear means of achieving this performance goal was parallel computation.

A frequent method of achieving performance gains in Monte Carlo calculations is to distribute the configurations over several processors and let each processor carry out its own independent Monte Carlo calculation. Such an "embarrassingly parallel" implementation is sufficient for simple calculations where each processor can handle a calculation with a minimum acceptable number of configurations in a reasonable amount of time. For the six and seven body systems considered here this is not the case.

The heart of the variational Monte Carlo calculation is the Metropolis algorithm which is an inherently serial algorithm. Since the bulk of the work in our variational calculations lies in the energy expectation value, the straight-forward division of labor is to have one master processor perform the Metropolis walk while several servant processors calculate the energy and other interesting expectation values for the configurations that the master generates. Provided that the time it takes to generate a new configuration is much less than the time to calculate an energy expectation value, this method is an efficient means of carrying out a VMC calculation on a parallel computer. Unfortunately, for ^4He the time it takes to calculate expectation values is only a few times the time required to generate a configuration, so only a few servant processors can be kept busy. On the other hand, typically 24 servant processors on the Cornell IBM SP2 can be used efficiently by one master in a ^7Li VMC.

Implementing the GFMC algorithm on a parallel architecture provides some special challenges. There is no clear division of labor as in the VMC calculation and the number of configurations can change throughout the calculation. An "embarrassingly parallel" implementation works very well for ^4He, since one processor on current machines is capable of handling enough ^4He configurations ($\sim$ 1000) for an independent GFMC calculation. In ^7Li, on the other hand, only a few ($\sim$ 50) configurations can be propagated on each processor to achieve an acceptable turnaround time for an independent calculation. With such a small configuration set on each node, the population fluctuations on each processor could leave some processors with few configurations and others with too many. To avoid such an inefficient use of resources, periodic load balancing between processors is required.

In our implementation, one master processor is responsible for initially distributing the configurations, collecting and averaging energy expectation values, and determining load

Table 5.1: GFMC code performance on the Cornell IBM SP2 Wide Nodes. N is the size of the wave function in spin-isospin states and P is the number of pairs

	N	P	Energy Calculation	Propagation	Total Node Hours
^4He	32	6	0.02 sec.	0.005 sec.	1.4
^6Li	320	15	1.07 sec.	0.09 sec.	15.0
^7Li	1792	21	12.6 sec.	0.82 sec.	150.0

balancing distributions. Each of the servant processors is responsible for a block of configurations. The servants perform propagation, energy calculations, and branching for their own particular block of configurations. During load balancing each servant reports its current load to the master, which then performs load balancing by instructing each over-loaded servant to send its excess configurations to under-loaded servants. In this way, all servants have, within tolerances, same number of configurations. When a servant completes a block of energy calculations for a particular time step, the energies are sent back to the master. After all of the energy results for a time step are received by the master, they are averaged. This program structure scales well with the number of processors since there are no major communication bottlenecks in the course of the calculation and load balancing keeps the servants somewhat synchronized.

The current version of our GFMC code is written using Fortran 90 and makes use of the MPI message passing library. Table 5.1 lists shows the performance of our GFMC code on the Cornell SP2 using their "wide nodes." The data is for 1000 configurations propagated for 80 times steps ($\Delta\tau$=0.0005 MeV^{-1}) using the exact two-body propagator with energy evaluations and load balancing every 20 time steps. As one can see, the total node hours (execution time $\times$ number of processors) grows by roughly a factor of 10 from ^4He to ^6Li to ^7Li. The total computational effort approximately scales as the product of the wave function size N and the number of pairs P. This rapid increase in computation time is a serious obstacle in extending these calculations to larger nuclei. We will return to this point in our conclusions.

5.2 Three- and Four-body Results

Exact calculations of three-body nuclei have long been possible using Fadeev techniques and recently these calculations have been extended to four-body nuclei[11]. We have carried out GFMC calculations for ^3H and ^4He here to illustrate the importance of the three-body interaction, fit a new TNI, and provide us with a baseline to compare with larger nuclei to study shell effects. Furthermore, we would like to take advantage of the isospin symmetry breaking of the Argonne v_{18} potential to evaluate the ^3He - ^3H mass splitting.

The variational wave functions used for the three- and four-body systems has the form given by Eq. (3.24), where the uncorrelated basis wave function is just the anti-symmetric Slater determinant of the single particle spin-isospin states (eg. Eq. (3.4)). Table 5.2 shows the variational parameters used for these wave functions. These parameters do not minimize the variational energy, but since these wave functions serve as starting point for GFMC an optimum wave function is not required.

Since our propagator only contains the first 8 terms of the potential, the GFMC calculations were carried out with a simpler isoscalar Hamiltonian than the one described in Chapter 2:

$$\hat{H} = K + \sum_{i<j} v_8'(ij) + \sum_{i<j<k} V_{ijk}. \tag{5.1}$$

The interaction $v_8'(ij)$ has eight terms, with operators $[1, \tau_i \cdot \tau_j] \otimes [1, \sigma_i \cdot \sigma_j, S_{ij}, \mathbf{L} \cdot \mathbf{S}]$, chosen such that it equals the v_{14} in all S- and P-waves as well as in the $^3D_1$ wave and its coupling to the $^3S_1$. The eigenstates of $\hat{H}$ are computed, and the contributions of all other terms, namely $(v_{14} - v_8')$, v_{vb}, v_{csb}, v_{em}, and K_{csb}, to the energy eigenvalue are calculated in first order according to Eq. (4.24). The error in the binding energy of ^3H due to the perturbative treatment of $v_{14} - v_8'$ has been estimated by Kamada and Glöckle [41] to be ≤ 0.02 MeV.

Using the first-order propagator described in Chapter 4, the ^3H and ^4He variational wave functions were propagated to an imaginary time $\tau = 0.08$ MeV^{-1} using a time step of $\Delta\tau = 0.0001$ MeV^{-1}. Calculations with time steps $\Delta\tau < 0.0001$ MeV^{-1} showed no statistically significant change in ^4He energies. Transient energy estimates, E(τ) were evaluated every twenty time steps where

$$E(\tau) = \frac{\langle \Psi_v | \hat{H} | \Psi(\tau) \rangle}{\langle \Psi_v | \Psi(\tau) \rangle} \tag{5.2}$$

Table 5.2: Variational Parameters for Three- and Four- body nuclei

Nucleus	^3H	^4He
$E_{0.0}$	3.0	11.0
$E_{0.1}$	5.8	18.0
$E_{1.0}$	12.8	27.0
$E_{1.1}$	5.8	17
$f^{scal}_{0.0}$	0.88	0.91
$f^{scal}_{0.1}$	1.0	1.0
$f^{scal}_{1.0}$	0.88	0.91
$f^{scal}_{1.1}$	1.0	1.0
$\eta_{t.0}$	0.026	0.036
$\eta_{t.1}$	-0.007	-0.012
$a_{0.T}$	0.4	0.4
$a_{1.T}$	0.4	0.4
a_t	0.4	0.4
$c_{0.T}$	1.2	1.2
$c_{1.T}$	3.0	3.0
c_t	3.0	3.0
$R_{0.T}$	0.8	0.8
$R_{1.T}$	2.8	2.8
R_t	3.6	3.6

Table 5.3: Three- and Four-body GFMC Results without TNI. All results are in MeV

Nucleus	^3H	^4He
Configurations	38,000	20,000
E(Expt.)	-8.48	-28.3
E(Calc.)	-7.70(2)	-24.1(1)
$\langle K \rangle$	40.1(6)	100.8(18)
$\langle v_{14} \rangle$	-48.1(7)	-127.1(18)

The transient estimate provides an upper bound which approaches the lowest-energy eigenvalue with the same quantum numbers as Ψ_v in the limit $\tau \rightarrow \infty$. In the ^3H and ^4He calculations described below, the transient estimate had converged by $\tau \sim 0.02$ MeV^{-1}.

In an effort to evaluate the importance of the three-body interaction, we carried out GFMC calculations for the ^3H and ^4He ground states without a three-body interaction. The results are shown in Table 5.3. As one can see, ^3H is under bound by -0.78 MeV and ^4He is under bound by -4.2 MeV. These results agree with the results of Fadeev calculations using other two-body potentials for ^3H (Argonne v_{14} and Paris with -7.68 MeV and -7.47 MeV respectively [10]) and ^4He (Argonne v_{14} and Paris with -24.62 MeV and -24.26 MeV, respectively [11]). We can conclude from these results that exact calculations which use realistic potentials require three-nucleon interactions to bind light nuclei with observed energies.

In the Urbana VIII three-nucleon interaction, which was used in conjunction with the Argonne v_{14} two-body potential, the $A_{2\pi}$ parameter was adjusted to reproduce the binding energy of ^3H and ^4He in exact calculations. The U_o parameter is fixed by reproducing the saturation density of nuclear matter. Unlike the Argonne v_{14}, which was fit to just the np scattering data, the Argonne v_{18} potential has been fit to pp, np, and nn data and thus has a better treatment of the isospin dependence. Since the two-body potential has changed, a refit of the parameters of the three-nucleon interaction was required. The refit version of the three-body potential is referred to as the Urbana IX. The $A_{2\pi}$ parameter of the Urbana IX potential was fixed such that the ground state energies of the ^3H, and ^4He are within the error bounds of the experimental values. The final values of the Urbana IX parameters are $A_{2\pi} = -0.0293$ MeV and $U_o = 0.0048$ MeV. These numbers compare favorably with previous Urbana Model (Urbana VIII $A_{2\pi} = -0.028$ MeV and $U_o = 0.005$ MeV)[42][43], and

Table 5.4: Three- and Four-body Isospin Symmetry Breaking Contributions. All results are in KeV.

Nucleus	^3H	^3He	^3He - ^3H	^4He
$\langle K_{csb} \rangle$	-7.0(1)	7.0(1)	14.0(1)	0.0
$\langle v_{csb} \rangle$	-32.9(7)	32.9(7)	65.8(9)	0.0
$\langle v_{em} \rangle$	39.3(4)	716.5(2)	677.2(4)	879.0(39)
Total:			757(1)	877.5(39)
Break down of electromagnetic terms:				
v_{c1}(pp)	0	647.8(46)	647.8(46)	755.5(35)
v_{df}(pp)	0	-2.0	-2.0	-2.9
v_{c2}(pp)	0	-0.55	-0.55	-0.7
v_{vp}(pp)	0	5.59(5)	5.59(5)	6.7
v_{oo}(pp)	0	8.3(1)	8.3(1)	8.3(1)
v_{c1}(np)	9.3(1)	9.3(1)	0.0	23.6(3)
v_{mm}(pp)	0	32.3(5)	32.3(5)	45.6(7)
v_{mm}(np)	15.8(2)	15.8(2)	0.0	32.0(3)
v_{mm}(nn)	14.2(3)	0	-14.2(3)	19.1(3)

the value of $A_{2\pi}$ is consistent with Fujita and Miyazawa's estimate of -0.03 MeV for A_2- [44]. The results for the ^3H and ^4He GFMC calculations with the Argonne v_{18} + Urbana IX Hamiltonian are shown in Table 5.10.

The mass splitting between ^3H and ^3He provides the simplest test for the isospin symmetry breaking portion of the potential. Table 5.4 shows the contribution of the various isospin symmetry breaking terms in the nuclear and electromagnetic portions of the potential to the ^3H, ^3He and ^4He binding energies. The calculated value for the $^3He - ^3H$ mass splitting is 0.757(1) MeV compared to the experimental value of 0.764 MeV. Apart from being close to the experimental value, these results are in good agreement with a model independent calculation of the $^3He - ^3H$ mass splitting by Y. Wu, S. Ishikawa, and T. Sasakawa[45]. The present calculations underestimate the mass splitting by only $\sim$ 1%; without the v_{csb} and K_{csb} terms the difference with experiment is 10 times bigger.

The calculated values of the rms proton point radius, $R(p)$, (Table 5.10) compare well with the those extracted from observed charge radii [46]. The experimental $R(p)$ of ^3He is 1.77 fm, and is in reasonable agreement with the calculated $R(n) = 1.72(2)$ of ^3H.

5.3 Six-body Results

Six-body nuclei are a rich testing ground for our Hamiltonian. There are several ^6Li states of different quantum numbers, ($J^\pi = 0^+, 1^+, 2^+$,and 3^+), that can be calculated accurately using GFMC techniques. Their spectrum provides important information regarding the accuracy of our Hamiltonian. There is also the isobaric triplet ^6Li (0^+), ^6Be (0^+), and ^6He (0^+) which will allow us test the isospin symmetry breaking terms in our potential by evaluating the ^6Be-^6He isovector and $\frac{1}{2}(^6Be+^6He)$-^6Li isotensor mass splittings.

The variational wave functions used in these calculations have the form given by Eq. (3.24). The Jastrow wave functions are constructed to provide the appropriate quantum numbers for the state being considered. The basic form is given by

$$|\Psi_J\rangle = \mathcal{A}[\prod_{i<j<k\leq 4} f^c_{sss}(\mathbf{r}_{ij}, \mathbf{r}_{ik}, \mathbf{r}_{jk}) \prod_{\substack{i<j\leq 4\\ k=5,6}} f^c_{ssp}(\mathbf{r}_{ik}, \mathbf{r}_{jk}) \prod_{i\leq 4} f^c_{spp}(\mathbf{r}_{i5}, \mathbf{r}_{i6})$$
$$\times \prod_{i<j\leq 4} f^c_{ss}(r_{ij}) \prod_{1\leq k\leq 4} f^c_{sp}(r_{k5}) f^c_{sp}(r_{k6})\, f^c_{pp}(r_{56})\, |\Phi(JMTT_z)_{1234:56}\rangle\,] . \quad (5.3)$$

where $\mathcal{A}$ is the antisymmetrization operator. There are three possible two-body central correlations, f^c_{ss}, f^c_{sp}, f^c_{pp}, where the s subscript refers to s-shell and p refers to p-shell nucleons. The f^c_{ss} central correlations are assumed to be the f_c's obtained by solving Eqs. (3.9-3.14) for ^4He. Various forms of f^c_{sp} and f^c_{pp} were tried. The lowest energy was obtained by using the following form:

$$f^c_{sp}(r) = f^c_{ss}(r)/[1 + \exp(\frac{r - r_0}{a})] + b[1 - \exp(-(\frac{r}{c})^2)] . \quad (5.4)$$

The r_0, a, b, and c, are variational parameters with optimized values of 4.0 fm, 1.0 fm, 0.9, and 3.2 fm, respectively. Several p-shell correlations were studied with $f^c_{pp}(r) = f^c_{ss}(r)$ providing the lowest variational energy.

The three possible three-body central correlations are denoted by f^c_{sss}, f^c_{ssp}, and f^c_{spp}. These three-body correlations have the same form as Eq. (3.25), but the parameters depend on the specific type of triple under consideration. In the current calculations, the best results are obtained by setting f^c_{ssp}=f^c_{spp}=0 and using f^c_{sss} from ^4He.

The uncorrelated wave function, $\Phi(JMTT_3)_{1234:56}$, is a single particle wave function with total angular momentum J, projection M, overall isospin T, and isospin projection T_z,

$$|\Phi(JMTT_z)_{1234:56}\rangle = |\Phi_\alpha(0000)_{1234}\, \phi_p(R_{\alpha 5})\phi_p(R_{\alpha 6})$$

$$\times \sum_{LS} \beta_{LS} \left\{ [Y_1^{m_l}(\Omega_{\alpha 5}) Y_1^{m_l'}(\Omega_{\alpha 6})]_{LM_L} \times [\chi_5(\frac{1}{2}m_s)\chi_6(\frac{1}{2}m_s')]_{SM_S} \right\}_{JM}$$
$$\times [\nu_5(\frac{1}{2}t_z)\nu_6(\frac{1}{2}t_z')]_{TT_z}\rangle \ . \tag{5.5}$$

Here Φ_α is an uncorrelated, antisymmetrized alpha particle state and $o_p(R_\alpha)$ is a p-shell single particle wave function relative to the center of mass of the alpha core, and it is chosen as the p-wave solution of a particle in a Woods-Saxon well,

$$V_{WS}(r) = \frac{V_0}{1 + e^{-(r-R_v)/a_v}}. \tag{5.6}$$

where V_0, R_v, and a_v are variational parameters.

The fifth and sixth particles are placed into these p-shell orbitals using LS coupling with $\mathbf{J} = \mathbf{S}+\mathbf{L}$. The overall TT_z is constructed from the isospin functions ν under the requirement that $L+S+T$ be odd. The reduced isospin basis used for these states are tabulated in the Appendix. The variational parameters β_{LS} describe the mixing of the LS states and are adjusted to obtain an optimal wave function. For the ^6Li (1^+) T=0 ground state we mix β_{01}, β_{10}, and β_{21} terms. Both of the ^6Li (2^+) and ^6Li (3^+) T=0 states are "stretched" states and use only β_{21}. In the ^6Li (0^+) T=1 excited state, the β_{00} and β_{11} amplitudes contribute.

Table 5.7 shows the results of variational Monte Carlo calculations for the 6-body nuclei using these wave functions with the parameters listed in Tables 5.5 and 5.6. These parameters were chosen under the constraint that the variational wave function reproduce the experimental rms radius. An unconstrained variation with the present form of Ψ_V results in six-nucleon states with a deuteron far away from an α. Such a Ψ_V is not a good starting point for our GFMC. The Ψ_V's with the correct rms radius provide a better starting point. The VMC energy for ^6Li (0^+) given in Table 5.7 is $\sim$ 6 MeV above the experimental value. Even the best available variational wave functions for ^6Li, using two-body spin-orbit operator correlations as described in Ref. [14], underbind the ground state by $\sim$ 4 MeV. The kinetic and potential energies in ^6Li are very large (see Table 5.7) and compared to these the underbinding obtained in VMC calculations is small.

GFMC calculations have been conducted for ^6Li (1^+), ^6Li (0^+), ^6Li (3^+), ^6Li (2^+), and ^6He (0^+). The ^6Be (0^+) state was determined by treating the neutrons like protons and visa-versa in the evaluation of the isospin symmetry breaking terms for the ^6He (0^+) calculation. These GFMC calculations used the exact two-body propagator described in Chapter 4, with a time step of $\Delta\tau$ = 0.0005 MeV^{-1}. The step size of $\Delta\tau$ = 0.0005 MeV^{-1} was chosen

Table 5.5: Variational Parameters for Six- and Seven- body nuclei

Nucleus (J^π)	$^6He(0^+)$	$^6Li(0^+)$	$^6Li(1^+)$	$^6Li(2^+)$	$^6Li(3^+)$	$^7Li(all)$
$E_{0.0}$	12.0	17.0	12.0	17.0	12.0	12.0
$E_{0.1}$	16.0	16.0	16.0	16.0	16.0	16.0
$E_{1.0}$	24.0	23.5	24.0	23.5	24.0	24.0
$E_{1.1}$	15.0	16.5	15.0	16.5	15.0	15.0
$\eta_{t.0}$	0.036	0.035	0.036	0.035	0.036	0.036
$\eta_{t.1}$	-0.012	-0.015	-0.012	-0.015	-0.012	-0.012
$a_{0.T}$	0.4	0.35	0.4	0.35	0.4	0.4
$a_{1.T}$	0.4	0.4	0.4	0.4	0.4	0.4
a_t	0.4	0.4	0.4	0.4	0.4	0.4
$c_{0.T}$	1.0	1.1	1.0	1.1	1.0	1.0
$c_{1.T}$	3.0	3.0	3.0	3.0	3.0	3.0
c_t	2.0	2.0	2.0	2.0	2.0	2.0
$R_{0.T}$	0.8	0.75	0.8	0.75	0.8	0.8
$R_{1.T}$	2.8	2.8	2.8	2.8	2.8	2.8
R_t	3.6	3.7	3.6	3.7	3.6	3.6
V_o	-18.5	-40.0	-18.5	-17.0	-16.5	-18.5
R_v	4.0	2.0	4.0	4.0	4.0	4.0
a_v	1.5	1.6	1.5	1.5	1.5	1.5

Table 5.6: Mixing parameters for 6-body nuclei

$^6He(0^+)$ β_{00} = 1.00, β_{11} = -0.25

$^6Li(0^+)$ β_{00} = 1.00, β_{11} = -0.25

$^6Li(1^+)$ β_{01} = 1.00, β_{10} = 0.15, β_{21} = 0.2

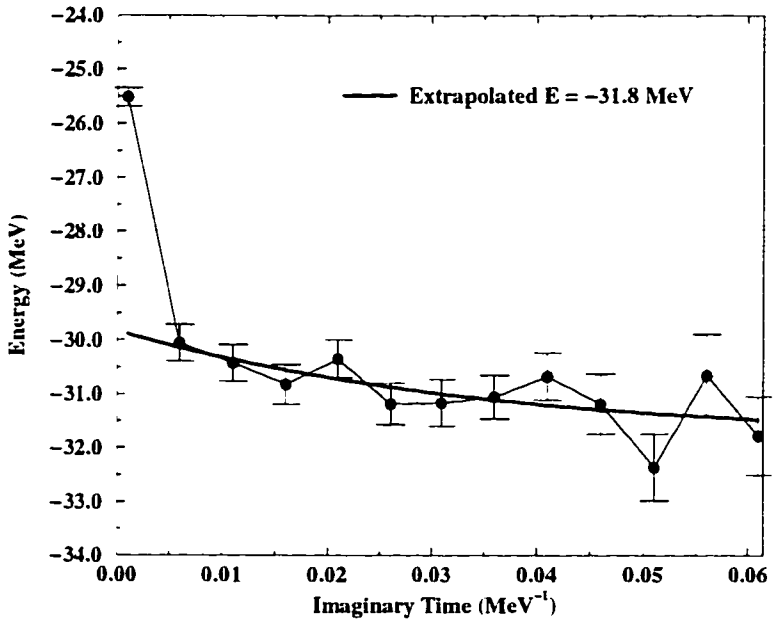
Table 5.7: VMC expectation values for 3- to 6-body nuclei using Argonne v_{18} + Urbana IX potential. All energies are in MeV.

Nucleus (J)	$^3H(\frac{1}{2})$	$^4He(0)$	$^6He(0^+)$	$^6Li(1^+)$	$^6Li(3^+)$	$^6Li(0^+)$	$^6Li(2^+)$
N	80.000	40.000	10.000	10.000	10.000	10.000	10.000
E	-8.18(2)	-27.04(8)	-21.7(2)	-25.6(1)	-22.6(2)	-21.1(2)	21.5(2)
$\langle K \rangle$	50.9(4)	110.(1)	143.(2)	154.(2)	151.(3)	146.(2)	149.(2)
$\langle v_{ij} \rangle$	-59.1(4)	-137.(1)	-165.(2)	-180.(2)	-173.(2)	-167.(2)	-170.(2)
$\langle v_{LS} \rangle$	-0.06(1)	0.31(4)	0.4(1)	0.5(1)	-0.2(1)	0.48(6)	0.97(7)
$\langle v_\tau \rangle$	-45.1(2)	-102.7(6)	-119.(1)	-143.(1)	-133.(1)	-118.(1)	-132.(1)
$\langle V_{ijk} \rangle$	-1.04(4)	-5.5(2)	-7.6(3)	-7.4(3)	-7.6(4)	-7.8(4)	-7.1(3)
R(n)	1.74(1)	1.42(1)	2.64(8)	2.39(7)	2.5(1)	2.43(7)	2.41(8)
R(p)	1.59(1)	1.42(1)	1.92(4)	2.39(7)	2.5(1)	2.43(7)	2.41(8)

since ^4He calculations with smaller time steps showed no statistically significant change in the binding energy. In each case, the variational configurations were propagated to an imaginary time of $\tau = 0.06$ MeV^{-1} with transient energy estimates evaluated every 20 time steps. Each calculation, except the ^6Li (1^+) and ^6Li (2^+) started with 10.000 variational configurations. E(τ) for the ^6Li (1^+) and ^6Li (2^+) states exhibited a great deal of noise beyond $\tau = 0.04$ MeV^{-1} and thus 20.000 variational configurations were used to reduce the statistical errors. Figures 5.1, 5.2 and 5.3 show E(τ) for the 4- and 6-body states.

In addition to the 3-, 4- and 6-body results reported here, we also include Joe Carlson's results for the ^5He scattering state GFMC calculation using our Hamiltonian for the sake of completeness. It is well known that there are no $A = 5$ bound states. The $\frac{3}{2}^-$ and $\frac{1}{2}^-$ resonances in ^5He have been studied via $n-^4He$ scattering experiments, and the phase shifts in these partial waves have been extracted [47]. The GFMC calculations for these states are carried out with an external $\alpha - n$ potential that vanishes for $|\mathbf{r}_\alpha - \mathbf{r}_n| < 12.5$ fm and is infinitely repulsive for $|\mathbf{r}_\alpha - \mathbf{r}_n| > 12.5$ fm. These conditions also imply that the external potential is $0(\infty)$ for $|\mathbf{r}_n| < 10(> 10)$ fm in the center of mass frame. The calculated Ψ_t corresponds to a scattering solution with a node at $|\mathbf{r}_\alpha - \mathbf{r}_n| = 12.5$ fm, however, only the interesting part with $|\mathbf{r}_\alpha - \mathbf{r}_n| < 12.5$ fm is retained in the calculation. Assuming that the interaction between α and n is negligible for $|\mathbf{r}_\alpha - \mathbf{r}_n| > 12.5$ fm the true energy of this state can be obtained from the phase shifts [48].

The ^4He E(τ) decreases from the variational energy $E_v = E(\tau = 0)$ by ~ 1.2 MeV in the



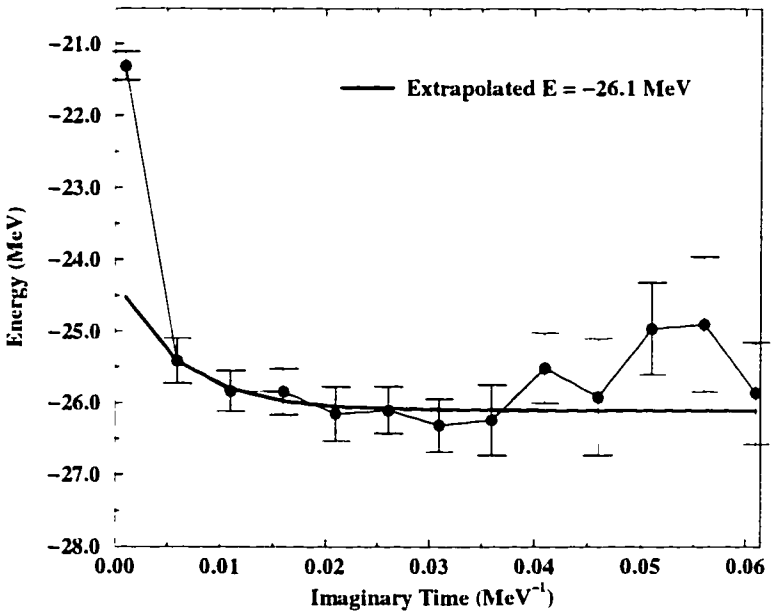
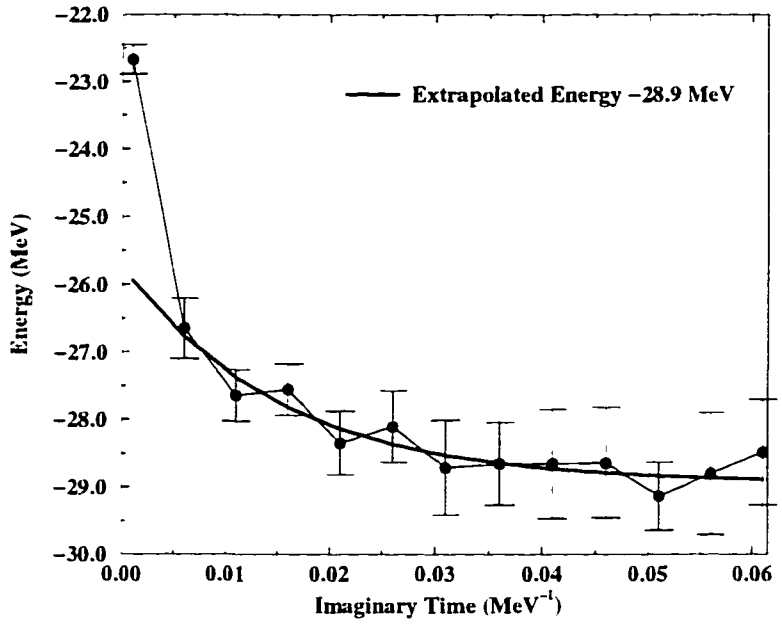


Figure 5.1: E(τ) and fit for ^6Li (1^+) and ^6Li (2^+) states using Argonne v_{18} + Urbana IX and $\Delta\tau = 0.0005$ MeV^{-1}



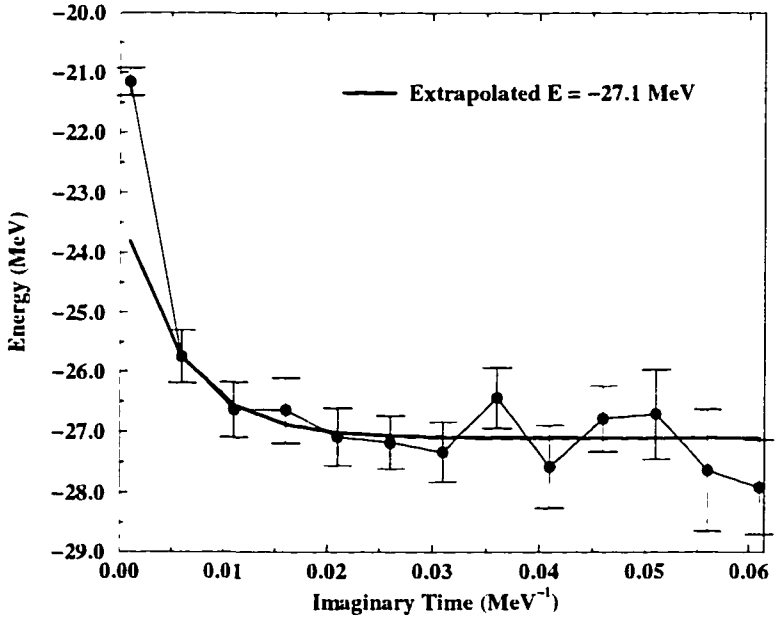
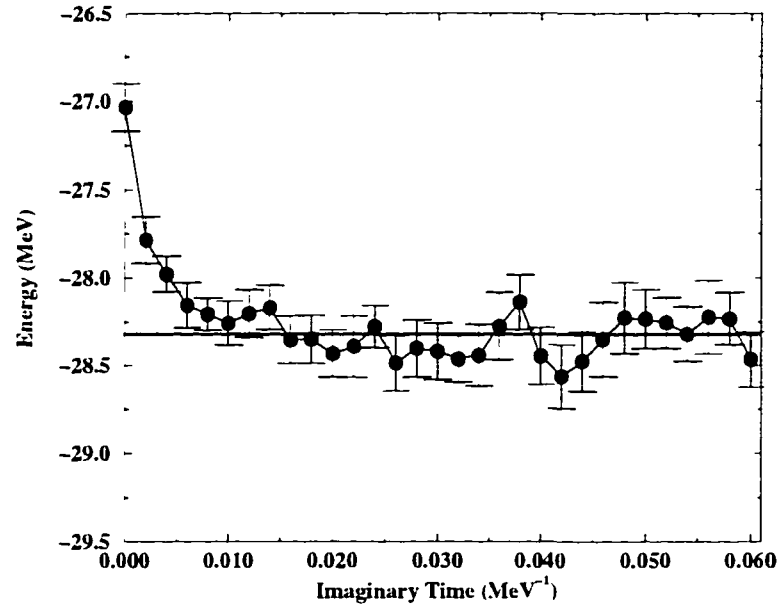


Figure 5.2: E(τ) and fit for ^6Li (3^+) and ^6Li (0^+) states using Argonne v_{18} + Urbana IX and $\Delta\tau = 0.0005$ MeV^{-1}



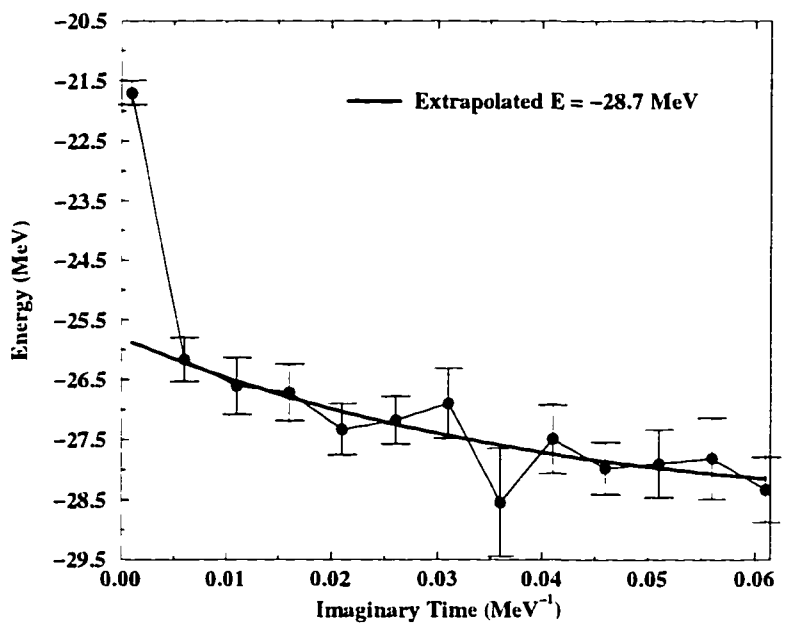


Figure 5.3: E(τ) and fit for ^4He (0^+) and ^6He (0^+) states using Argonne v_{18} + Urbana IX and $\Delta\tau = 0.0005$ MeV^{-1}

first ~ 0.02 MeV^{-1}, and is almost independent of τ thereafter.

The E(τ) of the 6-body states, in contrast, decrease by $\sim$ 5 MeV from E_v in the first $\tau \sim 0.02$ MeV^{-1}, and do not appear to have reached the $\tau \rightarrow \infty$ asymptotic value at $\tau = 0.06 MeV^{-1}$. The statistical error, governed by variance of $\hat{H}\Psi_v(\mathbf{R})/\Psi_v(\mathbf{R})$ is larger in the 6-body E(τ). This indicates that the 6-body Ψ_v are not as accurate as the ^4He variational wave function, and that they need to be improved for more accurate GFMC calculations.

Since the fermion "sign problem" begins to become significant by $\tau = 0.06$ MeV^{-1}, it would not be fruitful continue the propagation of the 6-body states to convergence. We can, however, make a fit to the E(τ) curves to try to determine the final converged result. Let Ψ_V, normalized such that $|\langle\Psi_o|\Psi_V\rangle|^2 = 1$, contain admixtures of states with excitation energy E of strength $\alpha(E)$. We then have

$$E(\tau) = \frac{\langle\Psi|\hat{H}e^{-(\hat{H}-E_o)\tau}|\Psi\rangle}{\langle\Psi|e^{-(\hat{H}-E_o)\tau}|\Psi\rangle} = \frac{E_o + \int_{E_o}^{\infty} E\alpha(E)e^{-(E-E_o)\tau}dE}{1 + \int_{E_o}^{\infty} \alpha(E)e^{-(E-E_o)\tau}dE}. \tag{5.7}$$

If $\alpha(E)$ can be described with a simple function, then its parameters can be determined from the calculated E(τ) and the extrapolated E_o can be extracted. A simple approximation for $\alpha(E)$ is as a sum of many delta functions:

$$\alpha(E) = \sum_i \alpha_i \delta(E_i - E). \tag{5.8}$$

In the case of ^4He, the main variation in E(τ) is within the first 0.01 MeV^{-1}. Figure 5.4 shows the small τ (< 0.006 MeV^{-1}) E(τ) for ^4He using Argonne v_8' + Urbana IX Hamiltonian and the corresponding fit using two terms in Eq. (5.8). For this particular fit, E_o was fixed at its converged value of -30.70 MeV (for the v_8' interaction) and the remaining parameters varied to minimize χ^2,

$$\chi^2 = \frac{1}{N}\sum_i (E_{fit}(\tau_i) - E(\tau_i))^2/\delta E(\tau_i)^2, \tag{5.9}$$

where,

$$E_{fit}(\tau_i) = \frac{E_o + \alpha_1^2 E_1 e^{-(E_1-E_o)\tau_i} + \alpha_2^2 E_2 e^{-(E_2-E_o)\tau_i}}{1 + \alpha_1^2 e^{-(E_1-E_o)\tau_i} + \alpha_2^2 e^{-(E_2-E_o)\tau_i}} \tag{5.10}$$

The best fit (χ^2=0.44) resulted with the parameters, $\alpha_1 = 0.0036$, $\alpha_2 = 0.00084$, $E_1 = 197$ MeV, and $E_2 = 1864$ MeV. This figure shows how the high energy components modes in the ^4He variational wave function are rapidly damped out by the action of the imaginary time propagator.

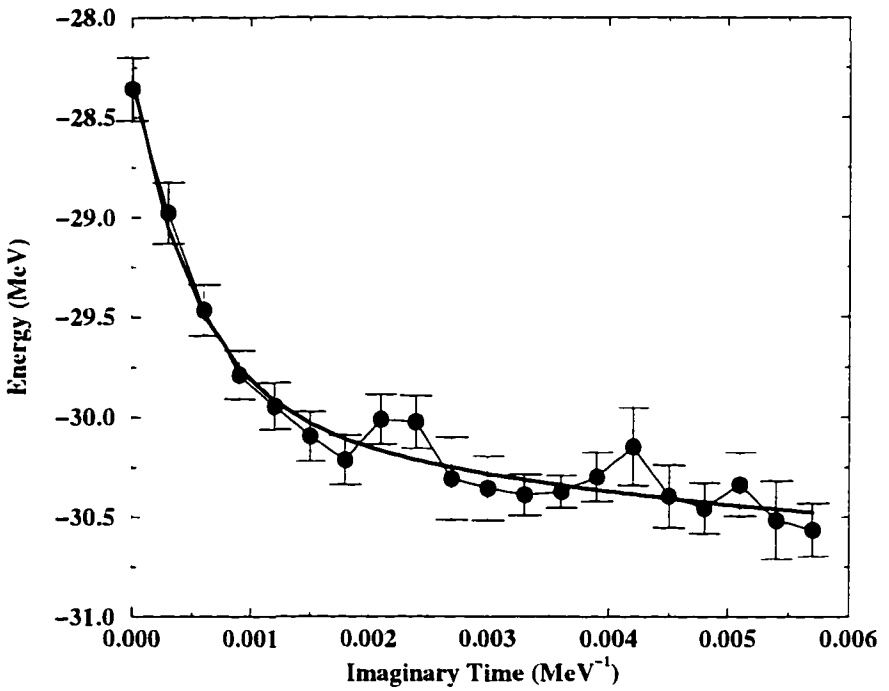


Figure 5.4: Short time E(τ) and fit for ^4He using Argonne v_8 + Urbana IX and $\Delta\tau = 0.0003$ MeV^{-1}

Table 5.8: Fit results for 6-body data. Energies are in MeV

Nucleus	E_o	χ^2	α_1	E_1
$^6He(0^+)$	-28.7(8)	0.255	0.111	-0.335
$^6Li(0^+)$	-27.1(7)	0.395	0.018	154.
$^6Li(1^+)$	-31.8(8)	0.662	0.070	-2.19
$^6Li(2^+)$	-26.1(6)	0.267	0.0095	141.
$^6Li(3^+)$	-28.9(7)	0.187	0.047	38.7

We are interested in extrapolating the converged energy eigenvalue from the calculated 6-body $E(\tau)$. The data at $\tau < 0.005$ MeV^{-1} is not useful for this purpose since the high energy components in Ψ_V, which strongly influence this data, are damped out by $\tau \sim 0.01$ MeV^{-1}. We therefore only fit the $E(\tau_i > 0.005$ MeV$^{-1})$. They can be fit with only one δ-function in Eq. (5.8) representing the total of low-energy excitations in Ψ_V and their average energy. Table 5.8 shows the results of such fits. Figure 5.5 shows χ^2 as a function of E_o for the 6-body spectrum. The minimum of χ^2, in some cases, is quite shallow and in all cases the fit provides a minimum which is within error bars of the average over the last 6 points. In each case we determined the error bar on E_o based on the quality of the χ^2 minimum. Caution should be used when trying to attach meaning to the excitation energies listed in Table 5.8 since a 10% change in χ^2 can, in many cases, can change the excitation energy by tens of MeV. As one can see, χ^2 in all cases is less than one indicating that our fit is over determined.

As a check on the fits performed above, we also tried fitting the data using more "physical" excitation energies. These fits were made to $E(\tau \geq 0.02$ MeV$^{-1})$ using two excited states. It is experimentally known that the lowest T=0, $J^\pi = 1^+$ (3^+) resonances in ^6Li are 5.65 (13.6) MeV above the 1^+ (3^+) ground states [49]. Accordingly we assumed that E_1-E_o = 5.65 (13.6) MeV for the 1^+ (3^+) states. Lacking experimental data, we assumed that E_1-E_o = 13.6 MeV for the T=0 2^+ state and the T=1 0^+ states. The second excited state was set such that E_2-E_o = 30.0 MeV. Table 5.9 shows the results of these fits. Note that there is no statistically significant difference between these extrapolated energies and those from the previous fit.

Tables 5.10 and 5.11 summarizes the 6-body results as well as the ^5He scattering

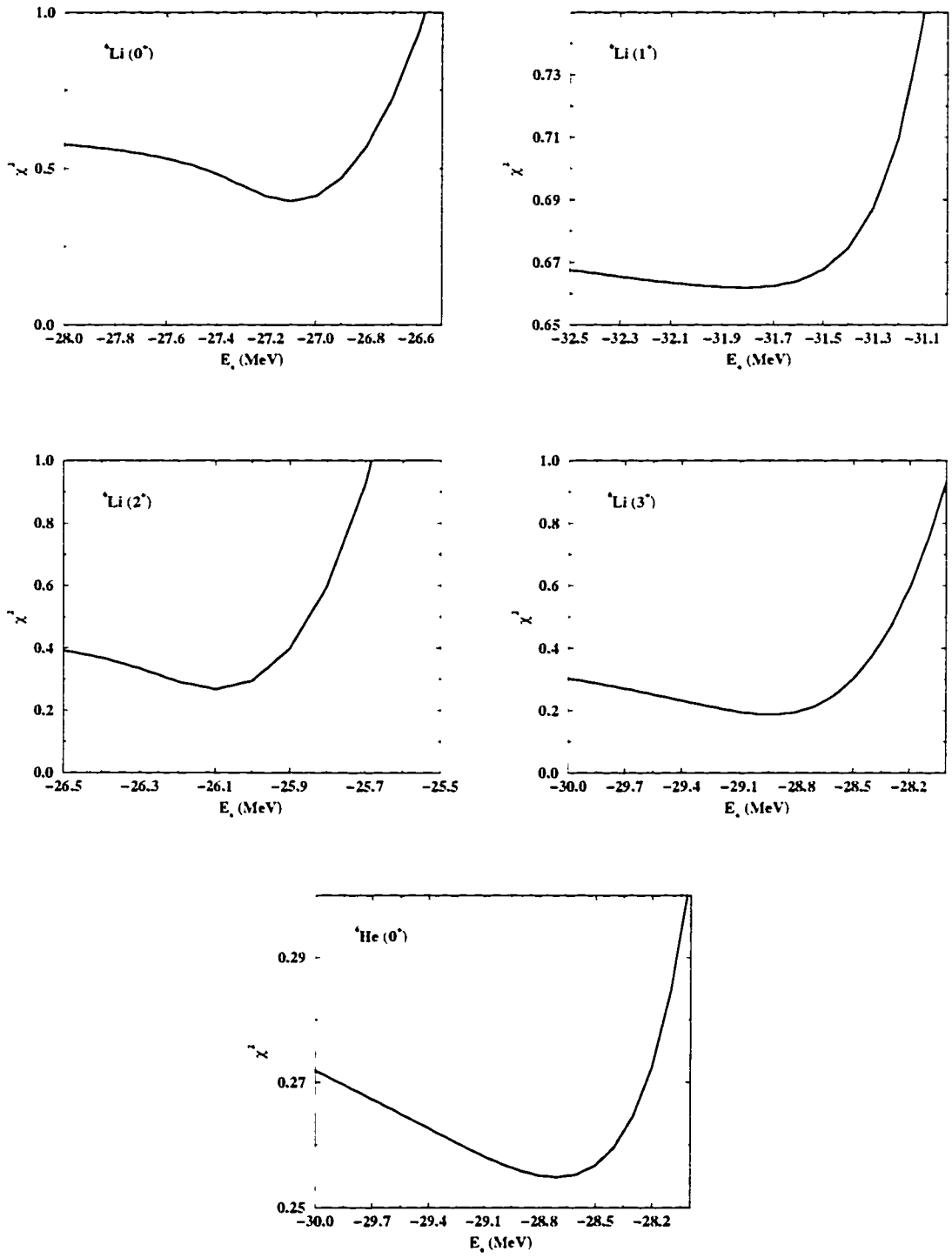


Figure 5.5: χ^2 for $^6Li(0^+)$, $^6Li(1^+)$, $^6Li(2^+)$, $^6Li(3^+)$, and $^6He(0^+)$

Table 5.9: Fit results for 6-body data using "physical" energies. Energies are in MeV

Nucleus	E_o	χ^2	α_1	E_1	α_2	E_2
$^6He(0^+)$	-28.7(8)	0.321	0.090	-15.7	0.070	0.7
$^6Li(0^+)$	-27.4(7)	0.489	0.044	-14.8	2.4×10^{-7}	1.6
$^6Li(1^+)$	-32.1(8)	0.787	0.016	-26.35	0.089	-2.0
$^6Li(2^+)$	-26.0(6)	0.694	1.5×10^{-8}	-14.1	2.7×10^{-7}	2.3
$^6Li(3^+)$	-29.2(7)	0.145	0.052	-16.2	0.024	2.0

state results from Ref. [50]. Since only the energy can be extrapolated as above, all other expectation values are reported as averages over the last 6 data points, from $\tau = 0.04$ to $\tau = 0.06$. We have also included the energy averages, $\bar{E}$, over the last 6 data points, for comparison to the extrapolated values. The $\bar{E}$ energies provide a firm upper bound on the ground state, but the extrapolations will be taken as the converged energies. We will defer our discussion of the spectrum's deviation from experimental spectrum until after we present our 7-body results (Chapter 6).

Since the Hamiltonian used in the propagation is v_8' + Urbana IX, the GFMC wave function is an eigenfunction of this Hamiltonian. We have included the energy expectation value of this Hamiltonian, E_{V8}, in the table. The few MeV difference between the E_{V8} and the $\bar{E}$ expectation values justifies our perturbative treatment of the (v_{18}-v_8') in 6-body nuclei. The ^6He and ^6Li 0^+ states just differ by a rotation in isospin space. Since the propagation Hamiltonian is rotationally invariant in isospin space we should expect that the E_{V8} energies be the same. The values of E_{V8} in Table 5.11 for the ^6He and ^6Li 0^+ states are nearly within error bars of being the same value. It should be noted that slightly different variational wave functions were used as a starting point for these two calculations.

The dominant $\langle v_{14}\rangle$ and $\langle K\rangle$ have similar values in ^4He and ^5He; while in the 6-body states they are closer to the sum of their values in ^4He and ^2H. The $\langle V_{ijk}\rangle$ are similar in ^4He, ^5He, and 6-body states. It is important to note that, though the the $\langle V_{ijk}\rangle$ expectation values make up a small percentage of the total potential energy, they still make a sizeable contribution to the overall energy since there is a large cancelation between the two-body potential and the kinetic energy.

The bulk of the two-body potential energy is provided by the first six terms of the potential

Table 5.10: GFMC expectation values for 3- to 5-body nuclei using Argonne v_{18} + Urbana IX potential. All energies are in MeV.

Nucleus (J^π)	$^3H(\frac{1}{2}^+)$	$^4He(0^+)$	$^5He(\frac{3}{2}^-)$	$^5He(\frac{1}{2}^-)$
Configurations (approx)	40.000	20.000	20.000	20.000
E(Expt.)	-8.48	-28.3	-27.2	-25.8
$\bar{E}$	-8.47(2)	-28.3(1)	-26.5(2)	-25.7(2)
E_{extrap}	-8.47(2)	-28.3(1)	-26.5(2)	-25.7(2)
E_{V8}	-8.70(1)	-30.70(5)		
$\langle K \rangle$	50.0(8)	115.(2)	122.(2)	117.(2)
$\langle v_{ij} \rangle$	-57.6(8)	-139.(2)	-145.(2)	-140.(2)
$\langle v_6 \rangle$	-56.3(8)	-139(2)		
$\langle v_{LS} \rangle$	-0.20(5)	-0.55(14)		
$\langle v_{L^2,(L\cdot S)^2} \rangle$	-1.0(4)	0.2(6)		
$\langle v_\pi \rangle$	-44.1(5)	-106(1)		
$\langle V_{ijk} \rangle$	-1.20(7)	-6.6(4)	-7.0(4)	-6.4(4)
$\langle V_{ijk}^{2\pi} \rangle$	-2.20(6)	-12.6(3)		
$\langle V_{ijk}^{R} \rangle$	1.00(4)	6.0(2)		
$\langle v_{coul}^{pp} \rangle$	0	0.759(4)	0.745(3)	0.751(3)
$\langle v_{em} \rangle$	0.0393(4)	0.120(1)	(0.86)	(0.86)
$\langle v_{CSB} \rangle$	-0.0329(7)	0	0	0
$\langle v_{CIB} \rangle$	0	0	0	0
$\langle K_{CSB} \rangle$	-0.0070(2)	0	0	0
R(n)	1.72(2)	1.42(1)	3.02(3)	3.57(3)
R(p)	1.58(2)	1.42(1)	1.84(2)	1.99(2)
R(p)(Expt.)		1.47		

Table 5.11: GFMC expectation values for 6-body nuclei using Argonne v_{18} + Urbana IX potential. All energies are in MeV.

Nucleus (J)	$^6He(0^+)$	$^6Li(1^+)$	$^6Li(3^+)$	$^6Li(0^+)$	$^6Li(2^+)$
Configurations (approx)	10.000	20.000	10.000	10.000	20.000
E(Expt.)	-29.3	-32.0	-29.8	-28.4	-27.7
$\bar{E}$	-28.0(4)	-31.3(5)	-28.7(5)	-27.2(5)	-26.2(3)
E_{extrap}	-28.7(8)	-31.8(8)	-28.9(7)	-27.1(7)	-26.1(8)
E_{V8}	-30.6(2)	-34.9(3)	-32.5(4)	-30.0(3)	-29.9(1)
$\langle K \rangle$	145(4)	157(4)	159(4)	141(6)	151(3)
$\langle v_{ij} \rangle$	-172(4)	-188(3)	-188(4)	-167(6)	-176(3)
$\langle v_6 \rangle$	-171(4)	-186(3)	-187(4)	-166(6)	-176(3)
$\langle v_{LS} \rangle$	-2.6(5)	-2.4(5)	-4.1(7)	-2.5(4)	-1.7(3)
$\langle v_{L^2\,(L\cdot S)^2} \rangle$	2(1)	1.4(1)	3(1)	1(1)	1.5(1)
$\langle v_\tau \rangle$	-121(2)	-145(3)	-143(3)	-122(4)	-138(2)
$\langle V_{ijk} \rangle$	-7.9(8)	-7.6(9)	-7.7(9)	-7.1(10)	-7.3(6)
$\langle V_{ijk}^{2\tau} \rangle$	-14.8(6)	-15.2(7)	-15.6(7)	-14.6(95)	-14.5(5)
$\langle V_{ijk}^{R} \rangle$	6.9(5)	7.6(5)	8.0(5)	7.3(4)	7.2(3)
$\langle v_{coul}^{pp} \rangle$	0.729(8)	1.60(2)	1.64(3)	1.56(3)	1.58(2)
$\langle v_{em} \rangle$	0.143(3)	0.153(2)	0.152(2)	0.120(4)	0.151(2)
$\langle v_{CSB} \rangle$	-0.048(3)	0	0	0	0
$\langle v_{CIB} \rangle$	0.042(3)	0	0	-0.150(4)	0
$\langle K_{CSB} \rangle$	-0.018(1)	0	0	0	0
R(n)	2.65(7)	2.36(4)	2.32(9)	2.52(8)	2.44(6)
R(p)	1.90(3)	2.36(4)	2.32(9)	2.52(8)	2.44(6)
R(p)(Expt.)		2.43			

Table 5.12: Energy differences in MeV within isospin multiplets

	$^3He - ^3H$	$^6Be - ^6He$	$\frac{1}{2}(^6Be + ^6He) - ^6Li$	$^7Be(\frac{3}{2}) - ^7Li(\frac{3}{2})$
$\langle v_{em} \rangle$	0.677	2.28(4)	0.33	1.54
$\langle K_{csb} \rangle$	0.014	0.036	0.	0.026
$\langle v_{csb} \rangle$	0.066	0.096	0.	0.095
$\langle v_{cib} \rangle$	0.	0.	0.19	0
$\Delta(Calc.)$	0.757(1)	2.41(5)	0.52(5)	1.66(3)
$\Delta(Expt.)$	0.764	2.35	0.34	1.64

as evidenced by $\langle v_6 \rangle$ expectation value. Of the v_6 terms the one-pion exchange portion, $\langle v_\pi \rangle$, appears to contribute the largest amount. This is not to be unexpected considering that the one-pion exchange is the longest range part of the nuclear potential.

The $\langle v_{LS} \rangle$ expectation value contains both **L·S** and **L·S**$(\tau_i \cdot \tau_j)$ terms in the NN interaction. They contribute, along with V_{ijk} [51], to the splitting between $\frac{3}{2}^-$ and $\frac{1}{2}^-$ states of ^5He. However, the calculated splitting of 0.8(3) MeV is much smaller than the observed 1.4 MeV. The magnitude of $\langle v_{LS} \rangle$ is larger in the 6-body 3^+ state than in the 1^+, suggesting that the 1^+ has less contribution from the $(p\frac{3}{2})^2$ configuration in the j-j coupling scheme than the 3^+ state. Conversely, the magnitude of $\langle v_{LS} \rangle$ in the 6-body 2^+ state is smaller than that in the 1^+ state. This is due to the fact that the T=0 2^+ state is a pure $(p\frac{3}{2}p\frac{1}{2})$ state while the 1^+ state is a mixture of states. It is interesting to note that the GFMC $\langle v_{LS} \rangle$ expectation values are all larger in magnitude and, in most cases, of opposite sign from the VMC values. This difference has also been noted between variational calculations which use two-body spin-orbit correlations and those that don't. Such a result confirms the fact that the GFMC propagator builds the **L·S** correlation into the calculation.

In all cases there is very little contribution from the terms quadratic in the angular momentum. The individual expectation values $\langle v_{L^2} \rangle$ and $\langle v_{(L \cdot S)^2} \rangle$ are on the order of 21 MeV in the 6-body nuclei, but have opposite sign. Since our technique has difficulties dealing with non-local terms in the potential, the large cancelation between these terms is comforting.

Tables 5.10 and 5.11 list the charge symmetry breaking, $\langle v_{CSB} \rangle$, and the charge independence breaking, $\langle v_{CIB} \rangle$, expectation values as well as correction to the kinetic energy for the proton-neutron mass difference, $\langle K_{CSB} \rangle$. The present Hamiltonian explains the isovector

energy differences fairly well. Unfortunately, the calculated value of the isotensor difference $\frac{1}{2}(^6Be +^6 He) -^6 Li$ is much larger than observed. The observed difference is essentially explained by the electromagnetic interaction alone. This is very puzzling because most of the contribution of the isotensor v_{ab} to this difference should be from the relative $^1S_0$ two-nucleon state in which the difference between pp and np phase shifts seems to be well established [52, 53].

The non-perturbative contribution of the Coulomb interaction, particularly in ^6Be, neglected here, may reduce the value of this isotensor difference. However, our use of a reduced isospin basis prevents us from using isospin symmetry breaking terms in our propagator. Variational calculations of the ^6Be ground state with a slightly more diffuse core have failed to significantly modify the isotensor mass difference. It has also been suggested that charge-dependence in the three-body interaction could correct the isotensor mass splitting. Wiringa has investigated the contributions of the charge-dependence of the two-pion exchange V_{ijk} to the isotensor mass splitting and has determined that such contributions are negligible.

The last three lines of Table 5.11 give the proton and neutron rms point radii. The Coulomb interaction accounts for most of the isovector $^6Be -^6 He$ mass difference (Table 5.12). Since this difference is correctly predicted, the $R(p)$ of ^6Be, assumed to be equal to the $R(n)$ of ^6He, appears to have a reasonable value.

Figures 5.6 and 5.7 show the point one nucleon and two-nucleon (pair) separation densities for ^6Li (1^+) and ^6He (0^+). The data points show the results of the GFMC calculation, while the solid lines give the densities obtained from Wiringa's optimum wave function. The ^6Li point proton density is normalized to Z=3, while the ^6He proton and neutron densities are normalized to Z=2 and A-Z=4 respectively. Each of the pair separation densities is normalized to the appropriate total number of proton-proton, neutron-neutron or proton-neutron pairs. In all cases, there is very good agreement between the GFMC and the VMC densities. The GFMC pair densities appear to be peaked at a slightly larger separation than the VMC, an observation that may be useful to improve upon the present Ψ_V.

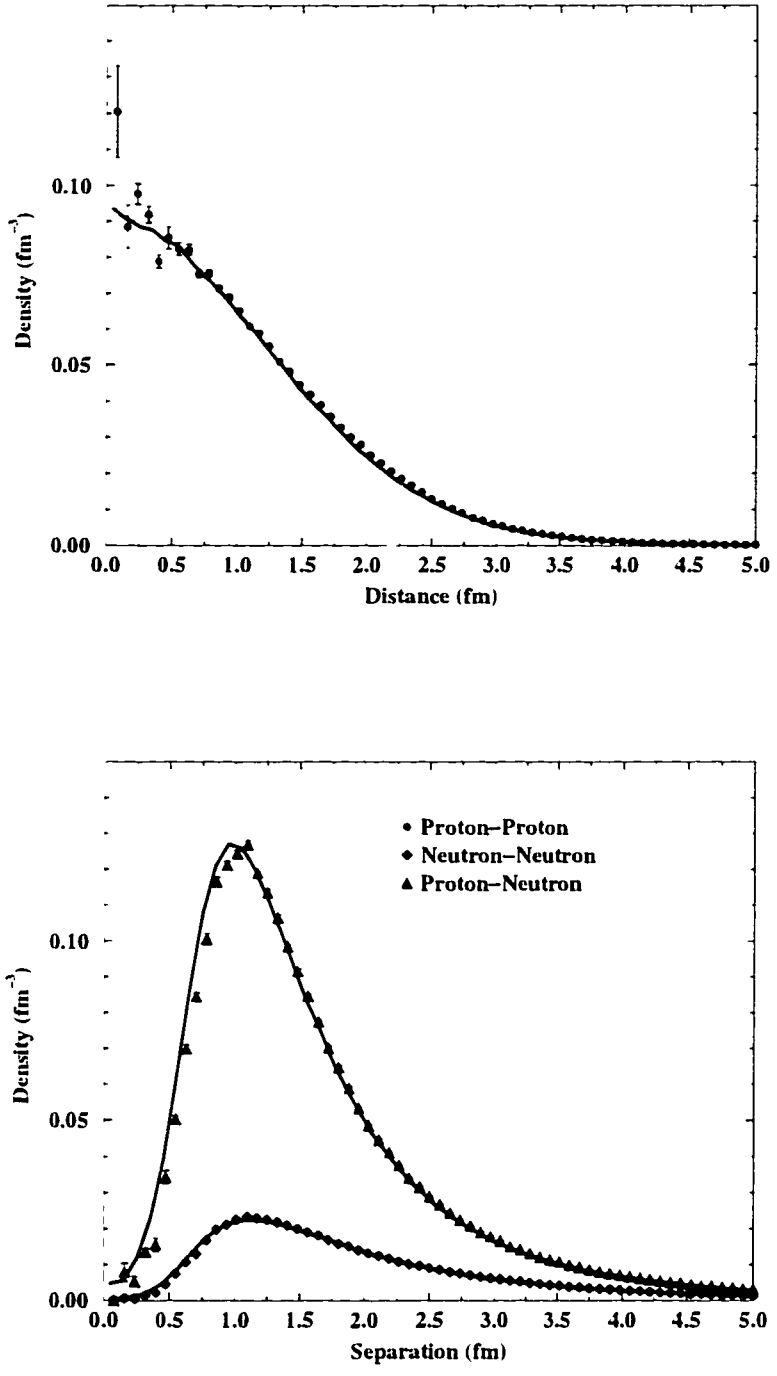


Figure 5.6: Point nucleon and pair densities for ^6Li (1^+) where the solid lines represent VMC results.

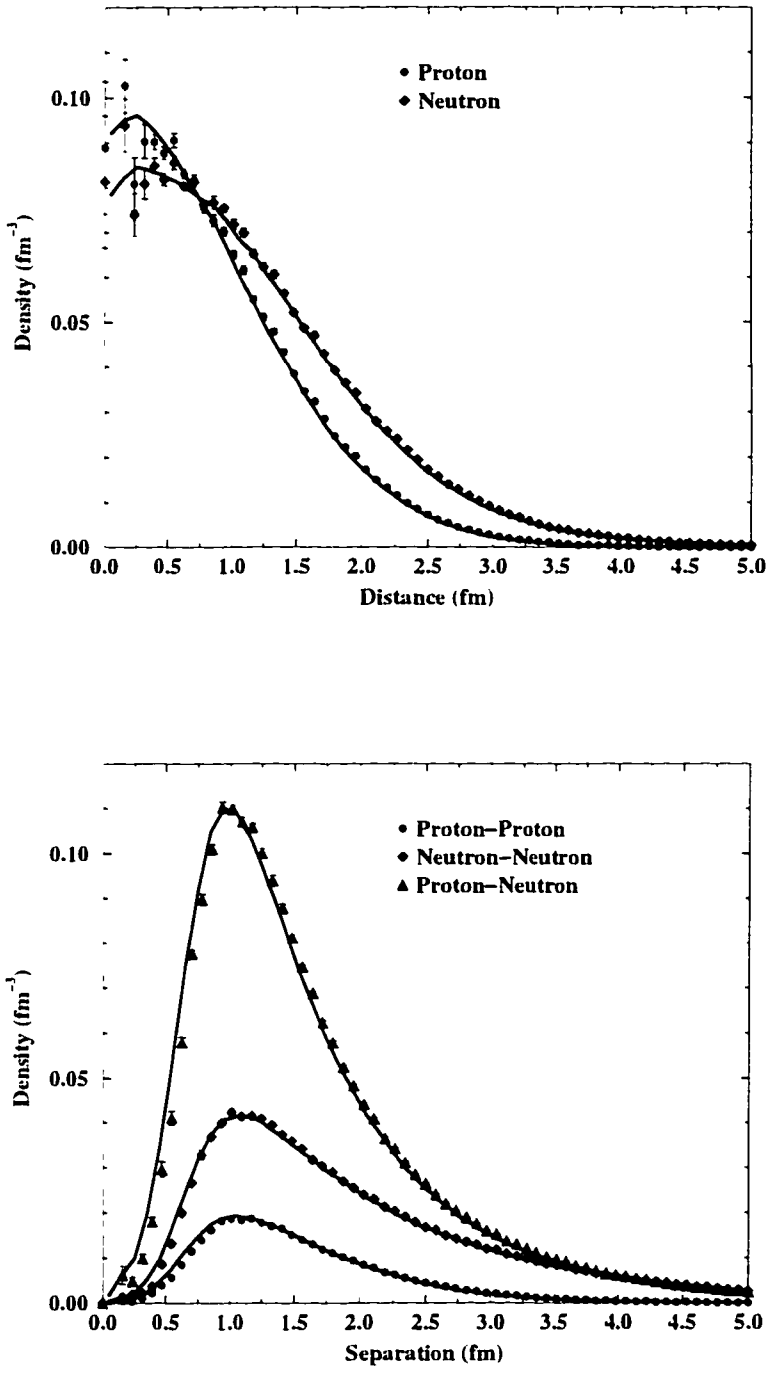


Figure 5.7: Point nucleon and pair densities for ^6He (0^+) where the solid lines represent VMC results.

5.4 Seven-body Results

Though we have only just touched the tip of possible calculations which can be done in the six-body system, it is important to get an idea of the weaknesses in our Hamiltonian. For that reason, we have pushed on to calculate the 7-body spectrum. There are several ^7Li and ^7Be bound states ($\frac{1}{2}^-, \frac{3}{2}^-, \frac{5}{2}^-, \frac{7}{2}^-$) which will provide a valuable testing ground for our Hamiltonian. Furthermore, we will be able to determine the isovector mass splitting ^7Li-^7Be for each one of the bound states.

The 7-body Jastrow wave function is easily generalized from the 6-body Jastrow wave function presented in Eqs. (5.3) and (5.5).

$$
\begin{aligned}
|\Psi_J\rangle = \mathcal{A}[& \prod_{i<j<k\leq 4} f^c_{sss}(\mathbf{r}_{ij}, \mathbf{r}_{ik}, \mathbf{r}_{jk}) \prod_{\substack{i<j\leq 4\\ 5\leq k}} f^c_{ssp}(\mathbf{r}_{ik}, \mathbf{r}_{jk}) \prod_{\substack{i\leq 4\\ 5\leq j<k}} f^c_{spp}(\mathbf{r}_{ij}, \mathbf{r}_{ik}) \\
\times & \prod_{i<j\leq 4} f^c_{ss}(r_{ij}) \prod_{1\leq k\leq 4} f^c_{sp}(r_{k5}) f^c_{sp}(r_{k6}) f^c_{sp}(r_{k7})\, f^c_{pp}(r_{56}) f^c_{pp}(r_{57}) f^c_{pp}(r_{67}) \\
\times & |\Phi(JMTT_z)_{1234:567}\rangle\ ] . \qquad (5.11)
\end{aligned}
$$

where,

$$
\begin{aligned}
&|\Phi(JMTT_z)_{1234:567}\rangle = |\Phi_\alpha(0000)_{1234}\ o_p(R_{\alpha 5}) o_p(R_{\alpha 6}) o_p(R_{\alpha 7}) \\
&\left\{[Y_1^{m_l}(\Omega_{\alpha 5}) Y_1^{m_l'}(\Omega_{\alpha 6}) Y_1^{m_l''}(\Omega_{\alpha 7})]_{LM_L} \times [\chi_5(\frac{1}{2}m_s)\chi_6(\frac{1}{2}m_s')\chi_7(\frac{1}{2}m_s'')]_{SM_S}\right\}_{JM} \\
&\times [\nu_5(\frac{1}{2}t_z)\nu_6(\frac{1}{2}t_z')\nu_7(\frac{1}{2}t_z'')]_{TT_z}\rangle\ . \qquad (5.12)
\end{aligned}
$$

The $\frac{1}{2}^-$ and $\frac{3}{2}^-$ states are constructed by placing the three p-shell nucleons into an anti-symmetric T=$\frac{1}{2}$, S=$\frac{1}{2}$ (triton-like) spin-isospin state with an overall symmetric L=1 angular wave function. Thus the spatial part of the wave function is symmetric while the anti-symmetry is contained in the spin-isospin part. The $\frac{5}{2}^-$ and $\frac{7}{2}^-$ states are also constructed similarly with the L=3 spatially symmetric state. Shell model studies have indicated that the lowest $\frac{1}{2}^-$, $\frac{3}{2}^-$, $\frac{5}{2}^-$, and $\frac{7}{2}^-$ states are almost purely spatially symmetric [54]. With that assumption, the Φ is unique without any parameters apart from those in the radial functions $o_p(R_{\alpha i})$. A listing expectation values from our ^7Li VMC calculations can be found in Table 5.13

There are two $\frac{5}{2}^-$ T=$\frac{1}{2}$ states for ^7Li separated by roughly 0.8 MeV in the energy spectrum. Shell model studies indicate that the lower energy $\frac{5}{2}^-$ state is dominated by the symmetric

Table 5.13: VMC expectation values for 7-body nuclei using Argonne v_{18} + Urbana IX potential. All energies are in MeV.

Nucleus (J)	$^7Li(\frac{3}{2}^-)$	$^7Li(\frac{1}{2}^-)$	$^7Li(\frac{7}{2}^-)$	$^7Li(\frac{5}{2}^-)$
N	10.000	10.000	10.000	10.000
E	-27.7(5)	-28.0(3)	-21.1(4)	-19.4(4)
$\langle K \rangle$	203(3)	202(2)	207(3)	202(3)
$\langle v_{ij} \rangle$	-230(3)	-230(2)	-228(3)	-222(3)
$\langle v_{LS} \rangle$	0.7(1)	1.0(1)	0.1(2)	1.5(1)
$\langle v_\tau \rangle$	-178(2)	-174(2)	-171(2)	-171(2)
$\langle V_{ijk} \rangle$	-9.7(9)	-9.5(5)	-10.1(5)	-9.4(4)
R(n)	2.31(5)	2.31(4)	2.26(5)	2.26(4)
R(p)	2.18(4)	2.18(3)	2.12(4)	2.13(3)

L=3 state while the higher energy state is predominantly composed of states with mixed-symmetry. Since these states are so close in energy, they probably mix with each other. In the VMC, such a mixing is not considered. In principle a GFMC calculation carried out up to $\tau > 1$ MeV^{-1} (the inverse of the energy separating the states) will project out the correct low energy state. The present GFMC with $\tau < 0.06$ MeV^{-1} will not.

We have carried out GFMC calculations for the ^7Li $J^\pi = \frac{1}{2}^-, \frac{3}{2}^-, \frac{5}{2}^-$ and $\frac{7}{2}^-$ states. As in the 6-body system the ^7Be states were found by exchanging protons and neutrons in the isospin symmetry breaking expectation values. Once again, the exact two-body propagator, described in Chapter 4, was used with a time step of $\Delta\tau = 0.0005$ MeV^{-1}. Starting with 10,000 variational configurations the calculation propagated up to $\tau = 0.06$ MeV^{-1} with with transient energy estimates evaluated every 20 time steps. E(τ) for the 7-body states is shown in Figures 5.8 and 5.9.

As one can see, there is a dramatic drop of ~ 8 MeV in E(τ) within the first 0.01 MeV^{-1} with only moderate (< 2 MeV) decreases in energy thereafter. The $\frac{3}{2}^-$ and $\frac{5}{2}^-$ states appear to have converged by $\tau = 0.04$ MeV^{-1} while the $\frac{1}{2}^-$ and $\frac{7}{2}^-$ states have not quite converged by $\tau = 0.06$ MeV^{-1}. This may indicate that the variational wave functions for the later two states contain admixtures of lower lying states than the former two.

Extrapolations were performed for the $\frac{1}{2}^-, \frac{3}{2}^-$ and $\frac{7}{2}^-$ states in the same manner as for the 6-body states. Table 5.14 shows the extracted ground state energies and the parameters for these fits. No fit was performed for the $\frac{5}{2}^-$ state since it appears to have converged.

Table 5.14: Fit results for 7-body data. Energies are in MeV

Nucleus	E_o	χ^2	α_1	E_1
$^7Li(\frac{1}{2})$	-37.3(9)	0.397	0.042	88.7
$^7Li(\frac{3}{2})$	-37.1(9)	0.160	0.025	103.
$^7Li(\frac{7}{2})$	-31.8(8)	0.588	0.054	37.5

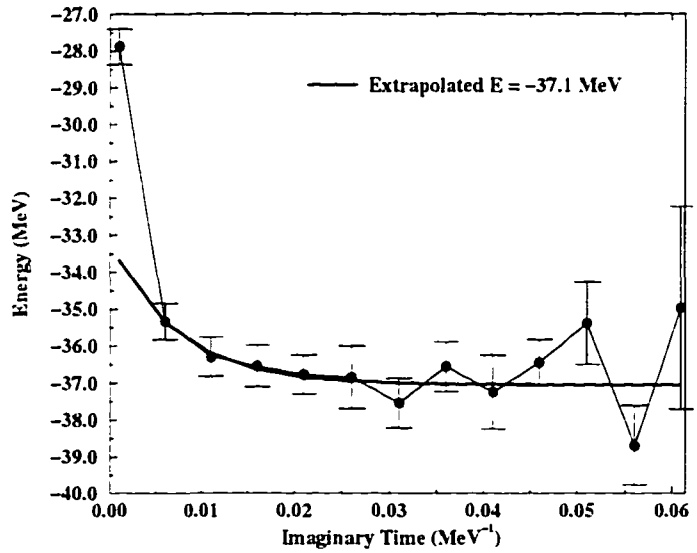
Table 5.15: Fit results for 7-body data using "physical" energies. Energies are in MeV

Nucleus	E_o	χ^2	α_1	E_1	α_2	E_2
$^7Li(\frac{1}{2})$	-37.4(9)	0.563	0.013	-8.7.	0.011	11.3
$^7Li(\frac{3}{2})$	-36.9(9)	0.886	0.013	-29.35	5.3×10^{-8}	10.8
$^7Li(\frac{7}{2})$	-32.4(8)	0.751	0.096	-4.5	1.6×10^{-9}	15.5

Furthermore, fits using more "physical" excitation energies were carried out as they were for the 6-body states. Table 5.15 shows the results of these fits. The $\frac{1}{2}^-$ and $\frac{7}{2}^-$ data was fit using E_1-E_o = 30 MeV. (There is a physical $\frac{7}{2}^-$ excitation at 5.04 MeV, but it proved to be too shallow to obtain a reasonable fit to the data.) For the $\frac{3}{2}^-$ state, we set E_1-E_o to the physical value of 9.85 MeV. The value of E_2-E_o was set to 50 MeV for all of the states. As before, there is very little dependence on the second excited state and there is no statistically significant difference from the earlier fits.

Table 5.4 shows the 7-body expectation values. Since the statistical noise seemed to become significant beyond τ = 0.04 MeV^{-1}, the results were averaged between τ = 0.025 MeV^{-1} and τ = 0.04 MeV^{-1} and treated as upper bounds. Many of the trends seen in the 6-body results are reproduced in the 7-body results. The $\langle v_6 \rangle$ comprises the bulk of the potential energy with a sizeable fraction coming from $\langle v_\pi \rangle$. The contribution of the three-nucleon interaction still makes up roughly 5% of the potential energy and about 25% of the binding energy. As in the 6-body results, the GFMC $\langle v_{LS} \rangle$ is attractive and larger in magnitude than the VMC result. The proton rms radius, R(p), for the $\frac{3}{2}^-$ is 2.27(3) fm, which compares very well with the experimental value of 2.27 fm.

Table 5.16 shows the $^7Be - ^7Li$ mass differences for each of the seven-body states. The



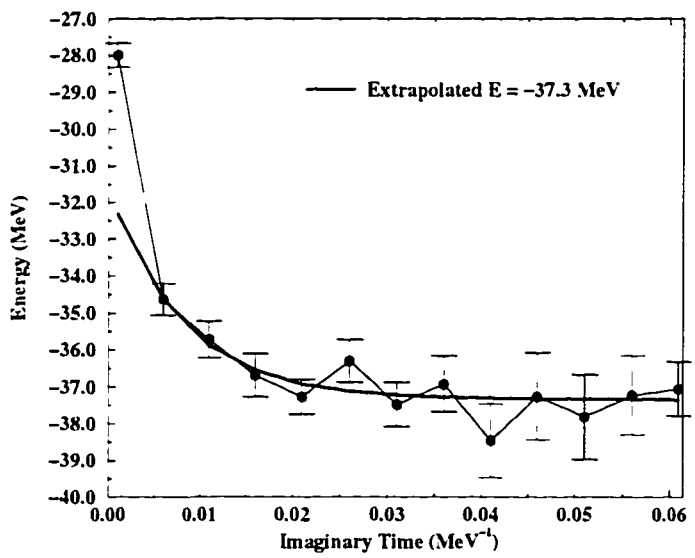


Figure 5.8: E(τ) and fit for ^7Li ($\frac{3}{2}^-$) and ^7Li ($\frac{1}{2}^-$) states using Argonne v_{18} + Urbana IX and $\Delta\tau = 0.0005$ MeV^{-1}

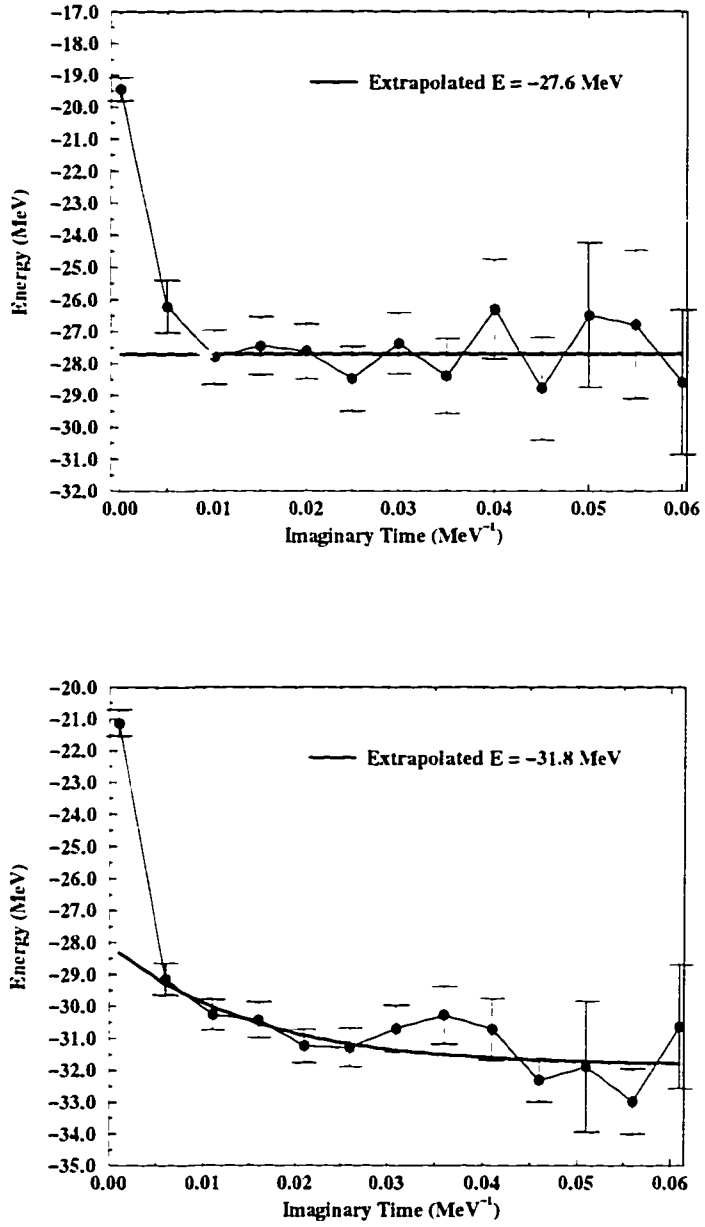


Figure 5.9: E(τ) and fit for ^7Li ($\frac{5}{2}^-$) and ^7Li ($\frac{7}{2}^-$) states using Argonne v_{18} + Urbana IX and $\triangle\tau = 0.0005$ MeV^{-1}

Nucleus (J)	$^7Li(\frac{3}{2}^-)$	$^7Li(\frac{1}{2}^-)$	$^7Li(\frac{7}{2}^-)$	$^7Li(\frac{5}{2}^-)$
E(Expt.)	-39.2	-38.7	-34.5	-32.5
$\bar{E}$	-37.1(5)	-37.3(5)	-30.7(5)	-27.6(7)
E_{extrap}	-37.1(9)	-37.3(9)	-31.8(8)	
E_{V8}	-41.0(4)	-41.4(4)	-35.5(4)	-33.1(5)
$\langle K \rangle$	198(8)	202(4)	198(5)	202(5)
$\langle v_{ij} \rangle$	-235(8)	-239(4)	-229(4)	-230(4)
$\langle v_6 \rangle$	-235(8)	-239(4)	-229(4)	-231(4)
$\langle v_{LS} \rangle$	-3.4(7)	-3.1(7)	-4.6(7)	-3.5(6)
$\langle v_{L^2.(L\cdot S)^2} \rangle$	3(1)	3(2)	5(1)	4(2)
$\langle v_\pi \rangle$	-173(3)	-177(3)	-169(4)	-171(3)
$\langle V_{ijk} \rangle$	-9.8(12)	-10.1(10)	-9.8(10)	-9.9(10)
$\langle V_{ijk}^{2\pi} \rangle$	-19.5(11)	-20.0(8)	-20.1(8)	-19.8(15)
$\langle V_{ijk}^{R} \rangle$	9.7(7)	9.9(6)	10.3(6)	9.9(10)
$\langle v_{coul}^{pp} \rangle$	1.62(2)	1.63(2)	1.71(3)	1.65(3)
$\langle v_{em} \rangle$	0.178(3)	0.187(2)	0.173(3)	0.187(4)
$\langle v_{CSB} \rangle$	-0.047(2)	-0.052(3)	-0.038(3)	-0.047(2)
$\langle v_{CIB} \rangle$	0	0	0	0
$\langle K_{CSB} \rangle$	-0.0130(8)	-0.0136(6)	-0.0130(7)	-0.0136(5)
R(n)	2.42(3)	2.36(4)	2.33(5)	2.34(4)
R(p)	2.27(3)	2.23(4)	2.14(5)	2.19(4)
R(p)(Expt.)	2.27			

Table 5.16: Energy differences in MeV within 7-body isospin doublets

J	$^7Be - ^7Li$	$^7Be - ^7Li$ (Expt.)
$\frac{3}{2}$	1.66(3)	1.64
$\frac{1}{2}$	1.69(5)	1.59
$\frac{1}{2}$	1.61(6)	1.58
$\frac{5}{2}$	1.67(5)	1.69

$^7Be - ^7Li$ mass differences are all within error bars of the experimental values except for the $\frac{1}{2}^-$ state.

Our calculations seem to fare rather poorly for the $\frac{5}{2}^-$ and $\frac{7}{2}^-$ states. This failure certainly needs to be investigated further. It could be that the assumed L=3 spatially symmetric structure of the wave function is incorrect or the Hamiltonian or both. In the following discussion we will not consider these states pending further tests.

5.5 Neutron Drop Results

Properties of neutron matter are vitally important in determining the structure of neutron stars [55], they have a strong bearing on the energies of neutron-rich nuclei, and on the r-process in nucleosynthesis [56]. It is impossible to extrapolate available data on nuclei to the region of neutron matter with sufficient precision using effective interactions. Different effective interactions that fit the energies of laboratory nuclei rather well predict very different equations of state for neutron matter [55]. In contrast, it appears that modern calculations of neutron matter based on realistic models of nuclear forces are much more consistent with each other at densities $\rho \sim 0.16 fm^{-3}$[57], and therefore are presumably more reliable. The two-nucleon interaction in these realistic models is better determined from the scattering data in isospin T=1 states than that in T=0, and the uncertainties coming from three-nucleon forces and relativistic effects are also much smaller in neutron than in nuclear matter. Calculations of uniform neutron matter have provided important constraints on Skyrme-type effective interactions used to study neutron-rich systems. They do not, however, provide information on the strength of the spin-orbit interaction, nor on other terms sensitive to density gradients, both of which may affect significantly the predicted properties of drip-line nuclei and of neutron-star matter.

Ab initio calculations of finite nuclei, based on realistic models of nuclear forces, can provide the necessary additional information. Using GFMC, VMC and Cluster VMC (CVMC) techniques, the ground state properties of 7 and 8 neutron drop clusters have been studied for the Argonne v_{18} + Urbana IX potential with a weak external well. Neutron matter is not bound, therefore an external well (V_{ex}) is necessary to hold the neutrons together. We have used a Woods-Saxon well with $V_o = -20$ MeV, $R = 3$ fm, and $a = 0.65$ fm, chosen such that with it alone only the single-neutron 1s state is bound at -5.73 MeV, while the 1p and higher states are unbound. The investigated states of seven and eight neutrons are thus bound by both the well and the interaction between neutrons. We denote them by ^{8}n($J^\pi = 0^+$) and ^{7}n($J^\pi = 1/2^-$ and $3/2^-$).

The variational wave function used for the neutron clusters is much simpler than than used for nuclei because all of the nucleons are neutrons requiring only one isospin state. The isospin dependence of the interaction can be analytically calculated and eliminated from the numerical computations. The variational wave function used is as follows:

$$|\Psi_V\rangle = \left[S \prod_{i<j}(1 + U_{ij})\right] \left[\prod_{i<j} f_c(r_{ij})\right] |\Phi\rangle \,. \tag{5.13}$$

$$U_{ij} = u_\sigma(r_{ij})\, \sigma_i \cdot \sigma_j + u_t(r_{ij}) S_{ij} \,. \tag{5.14}$$

where $|\Phi\rangle$ is an antisymmetric shell model wave function. The three-body correlations commonly used in nuclear Ψ_V are omitted because they have little effect on low-density neutron systems.

The single particle shell model states are determined variationally using a Woods-Saxon well (Eq. (5.6)) where the variational parameters are $V_o = -38.2751$ MeV, $R_v = 3$ fm, and $a_v = 0.6$ fm. Fig. (5.10) shows a plot of the external well, the variational well and the 1s and 1p shell model wave functions.

The 8×8 (or 7×7) determinant

$$\Phi = \begin{vmatrix} \phi_S\uparrow & \phi_S\downarrow & \phi_{Px}\uparrow & \phi_{Px}\downarrow & \phi_{Py}\uparrow & \phi_{Py}\downarrow & \phi_{Pz}\uparrow & \phi_{Pz}\downarrow \\ \phi_S\uparrow & \phi_S\downarrow & \phi_{Px}\uparrow & \phi_{Px}\downarrow & \phi_{Py}\uparrow & \phi_{Py}\downarrow & \phi_{Pz}\uparrow & \phi_{Pz}\downarrow \\ \phi_S\uparrow & \phi_S\downarrow & \phi_{Px}\uparrow & \phi_{Px}\downarrow & \phi_{Py}\uparrow & \phi_{Py}\downarrow & \phi_{Pz}\uparrow & \phi_{Pz}\downarrow \\ \phi_S\uparrow & \phi_S\downarrow & \phi_{Px}\uparrow & \phi_{Px}\downarrow & \phi_{Py}\uparrow & \phi_{Py}\downarrow & \phi_{Pz}\uparrow & \phi_{Pz}\downarrow \\ \phi_S\uparrow & \phi_S\downarrow & \phi_{Px}\uparrow & \phi_{Px}\downarrow & \phi_{Py}\uparrow & \phi_{Py}\downarrow & \phi_{Pz}\uparrow & \phi_{Pz}\downarrow \\ \phi_S\uparrow & \phi_S\downarrow & \phi_{Px}\uparrow & \phi_{Px}\downarrow & \phi_{Py}\uparrow & \phi_{Py}\downarrow & \phi_{Pz}\uparrow & \phi_{Pz}\downarrow \\ \phi_S\uparrow & \phi_S\downarrow & \phi_{Px}\uparrow & \phi_{Px}\downarrow & \phi_{Py}\uparrow & \phi_{Py}\downarrow & \phi_{Pz}\uparrow & \phi_{Pz}\downarrow \\ \phi_S\uparrow & \phi_S\downarrow & \phi_{Px}\uparrow & \phi_{Px}\downarrow & \phi_{Py}\uparrow & \phi_{Py}\downarrow & \phi_{Pz}\uparrow & \phi_{Pz}\downarrow \end{vmatrix} \tag{5.15}$$

needed to construct the anti-symmetric shell model wave function can be reduced to a sum of products of two 4×4 (or one 4×4 and one 3×3) matrices over all particle exchanges. This is accomplished by decoupling the spin degrees of freedom from the spatial determinants. Thus, Φ, for the 8-neutron system is given by

$$\Phi = \sum_{permutations} \begin{vmatrix} o_S(r_1) & o_{P_x}(r_2) & o_{P_y}(r_3) & o_{P_z}(r_4) \\ o_S(r_1) & o_{P_x}(r_2) & o_{P_y}(r_3) & o_{P_z}(r_4) \\ o_S(r_1) & o_{P_x}(r_2) & o_{P_y}(r_3) & o_{P_z}(r_4) \\ o_S(r_1) & o_{P_x}(r_2) & o_{P_y}(r_3) & o_{P_z}(r_4) \end{vmatrix} \times$$

$$\begin{vmatrix} o_S(r_5) & o_{P_x}(r_6) & o_{P_y}(r_7) & o_{P_z}(r_8) \\ o_S(r_5) & o_{P_x}(r_6) & o_{P_y}(r_7) & o_{P_z}(r_8) \\ o_S(r_5) & o_{P_x}(r_6) & o_{P_y}(r_7) & o_{P_z}(r_8) \\ o_S(r_5) & o_{P_x}(r_6) & o_{P_y}(r_7) & o_{P_z}(r_8) \end{vmatrix} |\uparrow_1\uparrow_2\uparrow_3\uparrow_4\downarrow_5\downarrow_6\downarrow_7\downarrow_8\rangle \tag{5.16}$$

For 8 neutrons, there are 70 independent exchanges (16 one particle, 36 two particle, 16 three particle, 1 four particle, and the original arrangement) and for the 7 neutron systems, there are 35 independent exchanges (12 one particle, 18 two particle, 4 three particle, and the original arrangement). Note that the expression of Φ in terms of single particle states is not translationally invariant as in previous calculations of nuclear ground states. The external well breaks the translational invariance.

The ^{7}n($J^\pi = 3/2^-$) state is constructed by removing a $m_s = -\frac{1}{2}$ electron from the L=1, M_L=-1 state of a closed shell configuration. The ^{7}n($J^\pi = 1/2^-$) ground state is a superposition of two states, one where a $m_s = -\frac{1}{2}$ electron is removed from a L=1, M_L=0 state and one where a $m_s = \frac{1}{2}$ electron is removed from a L=1, M_L=-1.

The correlation functions f_c, u_σ and u_t are calculated for neutron matter as in [58]. The optimum parameters are: k_f = 1.1 MeV^{-1}, d_c=d_t=3.1 fm, β_c = 1.5, β_t = 2.0 and α = 0.6.

The GFMC calculations are carried out as described in chapter 4 with a simpler Hamiltonian:

$$H = -\sum_i \frac{\hbar^2}{2m}\nabla_i^2 + \sum_i V_{ex}(i) + \sum_{i<j} v_8'(ij) + \sum_{i<j<k} V_{ijk}\,. \tag{5.17}$$

The small difference between the full v_{18} and v_8' is treated as a first-order perturbation, whose contribution to the calculated energies is < 0.2 MeV.

The calculated transient energies, E(τ), are shown in Fig. (5.11) The E($\tau \to \infty$) converges to the lowest eigenvalue of the chosen J^π. In Fermi systems, the statistical error in E(τ) increases with τ as configurations diffuse across nodal surfaces of the wave function[27]. Due to the large number of nodal surfaces in the wave functions of ^{7}n and ^{8}n states, it is

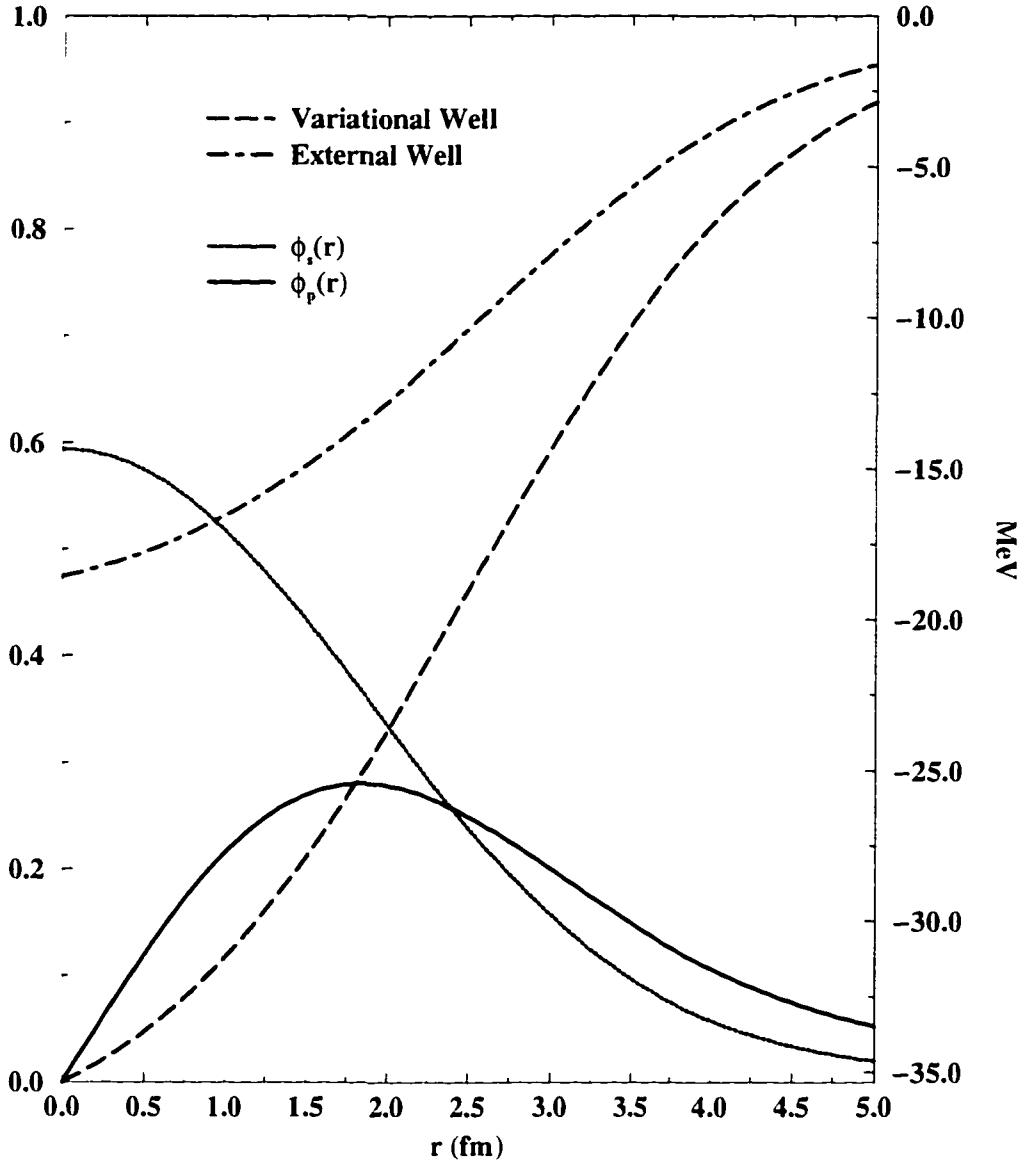


Figure 5.10: Single particle wave functions and Woods Saxon wells for 8 neutron variational wave function

difficult to study their transient energy for values of $\tau > 0.04$ MeV^{-1}. The average values of E(τ) for $\tau = 0.032$, 0.036 and 0.04 MeV^{-1}, denoted by $\bar{E}$, are shown by horizontal lines in Fig. (5.11) The E(τ) of the ^{8}n(0^+) and ^{7}n($1/2^-$) states do not have much τ-dependence for $\tau > 0.015$ MeV^{-1}, suggesting that their $\bar{E}$ can be identified with the eigenvalues. In contrast, the E(τ) of the ^{7}n($3/2^-$) state has more τ-dependence. Consequently, the eigenvalue of the lowest ^{7}n($3/2^-$) state could be a little below its $\bar{E}$ value. However, we will neglect that difference and regard it as our best estimate of the eigenvalue. The GFMC estimate of the density distribution of neutrons in the ^{8}n(0^+) drop is shown in Fig. (5.11). These results can be used to test the accuracy of the CVMC method and to further constrain the Skyrme type energy-density functionals used to study neutron-rich nuclei and neutron star crusts as discussed below.

The CVMC method and its modification to calculate spin-orbit splittings (SOS) are described in Refs. [59, 51]. The present CVMC calculations, carried out by Steven Pieper, are more accurate; they include contributions of all correlations and interactions up to five-body clusters. In contrast, in [59, 51] contributions of only static correlations and interactions were calculated up to four-body clusters and the momentum-dependent terms were evaluated only at the two-body level. With the simpler Ψ_V given by Eq. (5.13), the 1- to 5-body cluster contributions to the energy of ^{8}n(0^+) state are respectively 12.9, -54.5, 11.1, -3.8, and 1.1 MeV, which sum to -33.3(2) MeV. The E($\tau = 0$) is just the variational energy calculated to all orders without cluster expansion. Its value of -33.7(1) MeV is very close to the CVMC result retaining up to five-body clusters. Even in this rather low-density system, the cluster expansion has a slow convergence and it was necessary to include five-body cluster contributions to reduce the truncation error to < 2%.

The simple Ψ_V (Eq. 5.13) is not very accurate; the energy obtained with it is ~ 4 MeV (or $\sim 11\%$) too large. In CVMC, the more general variational wave functions :

$$|\Psi_V'\rangle = \left[1 + \sum_{i<j<k} U_{ijk}\right] \left[S \prod_{i<j}(1 + U_{ij})\right] \left[1 + \sum_{i<j} u_{L\cdot S}(r_{ij}) L_{ij} \cdot (\sigma_i + \sigma_j)\right] \left[\prod_{i<j} f_c(r_{ij})\right] |\Phi\rangle \tag{5.18}$$

were used where the three-body correlations, U_{ijk}, are of the kind used in Refs. [59, 51], and as before the U_{ij} contains spin and tensor terms. The energies obtained with the Ψ_V' variational wave functions are respectively -35.6(1), -31.2(1), -29.7(1) MeV for the ^{8}n(0^+), ^{7}n($\frac{1}{2}^-$) and ^{7}n($\frac{3}{2}^-$) states. They are only ~ 4 % above the GFMC energies -37.6(3), -32.3(2)

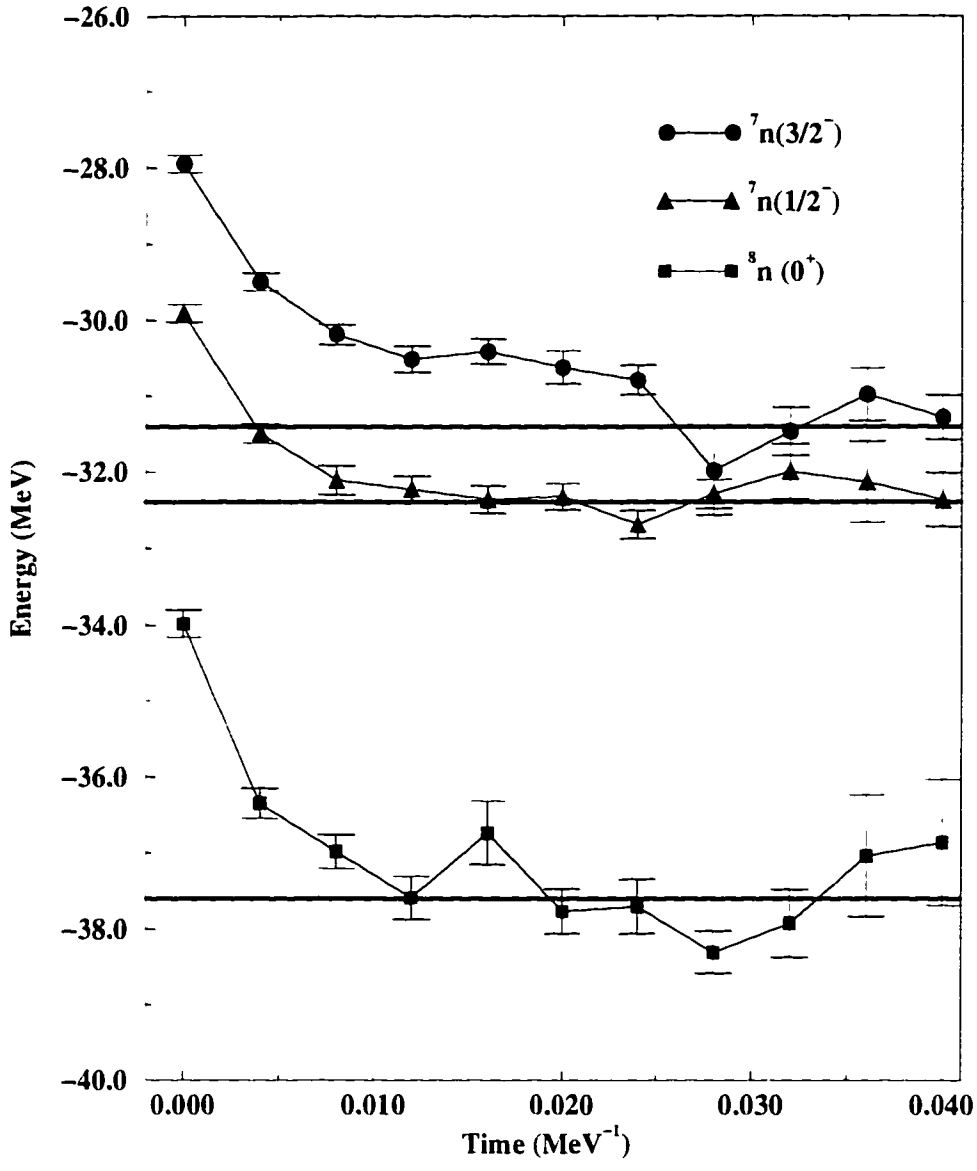


Figure 5.11: The transient energy $E(\tau)$ for the GFMC calculations of $^8n(0^+)$ and $^7n(1/2^-$ and $3/2^-)$ as a function of the imaginary time τ.

and -31.2(2) MeV. The CVMC calculations require about a factor of 25 less computer time than the GFMC, even allowing for the variational search, and it is possible to reduce the statistical error to a fraction of one percent. Much of the improvement in Ψ'_V comes from the spin-orbit correlations omitted in the simpler Ψ_V. In the present GFMC calculations, the spin-orbit correlations are built in exactly via the propagation in imaginary time.

The contributions of the two-pion exchange part of the three-nucleon interaction , $V_{ijk}^{2\pi}$, to the GFMC energies of ^{8}n(0^+), ^{7}n($\frac{1}{2}^-$ and $\frac{3}{2}^-$) are respectively, 0.23, 0.13, and 0.13 MeV while those of the phenomenological V_{ijk}^R are 0.65, 0.42, and 0.59 with a sampling error of $\sim$ 0.08 MeV. In laboratory nuclei, these contributions are much larger, eg. the total $\langle V_{ijk}\rangle$ is ~ -6.2 MeV in ^6Li (Table 5.11). Their small values here are presumably due to the Pauli principle forbidding three neutrons to come close and the absence of the n-p $^3S_1 - {}^3D_1$ coupled state through which the $V_{ijk}^{2\pi}$ gives a large negative contribution.

Fragmentation of the $p_{3/2}$ strength in ^{15}N results in the SOS of $p_{1/2}$ and $p_{3/2}$ quasi-hole states being $\sim$0.6 MeV larger than the observed splitting between the lowest $3/2^-$ and $1/2^-$ states in ^{15}N [51, 60]. If the $3/2^-$ hole strength in ^{7}n is similarly fragmented, the difference in the energies of lowest ^{7}n($1/2^-$) and ^{7}n($3/2^-$) states could be smaller than the SOS in ^{7}n. If W denotes the energy width of the fragmentation, the GFMC transient energies, E(τ), for $\tau \geq 1/W$ will include fragmentation effects. Since W is expected to be only a few MeV, it is very unlikely that the present $\bar{E}$ evaluated up to $\tau \sim 0.04$ has any fragmentation effects. The ^{7}n variational wave functions are constructed by removing an appropriate state from the $|\Phi\rangle$ and thus correspond to quasi-hole states. Presumably, they too do not contain any fragmentation effects. Hence, we identify the difference between the calculated energies of the ^{7}n($1/2^-$) and ($3/2^-$) states as the spin-orbit splitting. Its value is 1.1$\pm$0.3 and 1.4$\pm$0.1 MeV in the GFMC and CVMC calculations respectively.

Table 5.17 shows a comparison of the GFMC and CVMC results with that of four energy-density functionals (EDF) used for astrophysical investigations involving dense matter: (i) Skyrme 1′ - Vautherin-Brink Skyrme model 1 [61] modified [62] to fit the neutron-matter E(ρ) of Ref. [63]; (ii) SkM - Skyrme model M [64]; (iii) FPS and (iv) FPS-21 are generalized Skyrme models fitted approximately [23] and accurately [55] to the nuclear- and neutron-matter energies of Ref. [65]. These EDF's reproduce the ground-state energies of stable closed-shell nuclei rather accurately. Their ability to fit the binding energies of magic (ie. closed shell) nuclei is measured by the rms value of $\Delta E_i/E_i$, where ΔE_i is the difference

Table 5.17: Comparison of microscopic and Skyrme-model energies.

	^{8}n(0^+) MeV	^{7}n SOS MeV	Magic Nuclei $\lvert\Delta E/E\rvert_{rms}$%	^{15}N SOS[a] MeV	$W_{\ell s}$ MeV^5
GFMC	-37.6(3)	1.1(3)			
CVMC	-35.5(1)	1.4(1)	-	6.1[b]	
SkM	-47.4	3.0	1.1	6.3	130.
FPS-21	-42.2	3.0	1.1	6.7	110.
Skyrme 1′	-38.7	2.9	1.8	6.9	120.
FPS	-32.5	3.5	1.2	6.7	110.

between calculated and observed binding energy of a nucleus in the data set. These values are also listed in Table 5.17. This table also contains their predictions for the energy of the ^{8}n ground state and the SOS in ^{7}n and ^{15}N. The results for the density distributions $\rho(r)$ of the ^{8}n(0^+) state are compared with the GFMC and CVMC ρ(r) in Fig. (5.12)

It has been suggested [66] that the departures of the EDF energies for ^{8}n and ^{7}n from the microscopic calculations can be compensated by adding a gradient term to the FPS21 EDF, $\frac{1}{2}\alpha(\rho_n+\rho_p)^\beta(\nabla\rho_n-\nabla\rho_p)^2$, where α and β are a parameters to be fit. Such a term has no effect in uniform neutron matter to which FPS21 is fitted. It also has little effect on energies of magic nuclei in which $(\nabla\rho_n-\nabla\rho_p)$ is small. However, it can be important in on the surface of neutron clusters and can correct for the over binding of ^{8}n(0^+), and also reduce the central neutron densities. Good fits to both of these quantities cannot be made simultaneously, however. For the two sets of coefficients $(\beta,\alpha)=(0,150$ MeV fm^8) and $(2.7\times10^4$MeV fm$^{14})$ the ground state energies of ^{8}n are -39.6MeV and -40.2MeV respectively, and the neutron density distributions are shown as curves A and B in Fig. (5.12).

The present CVMC and GFMC results clearly indicate that the SOS predicted by the unadjusted Skyrme models for ^{7}n is too large, while it is good for ^{15}N. Relativistic mean-field models also predict a weaker spin-orbit potential in neutron-rich nuclei [67]. The neutron spin-orbit potential in these Skyrme models is of the form

$$V_{\ell s}^{(n)}(r)=W_{\ell s}\frac{1}{r}\frac{d}{dr}\left(\rho(r)+\rho_n(r)\right). \tag{5.19}$$

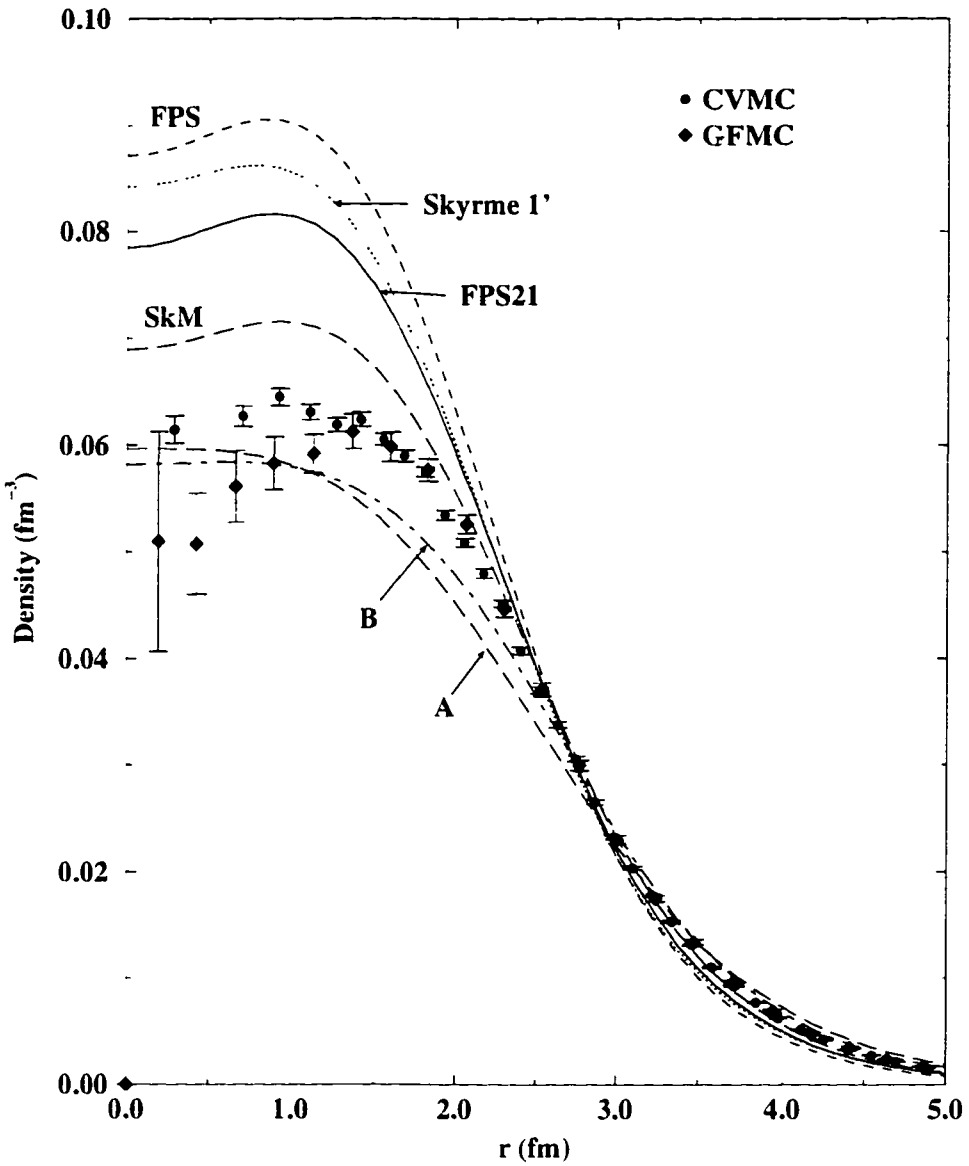


Figure 5.12: Neutron density distribution for ^{8}n, according to methods described in the text. The curves labeled A and B come from modified versions of the effective interaction FPS21.

obtained by Vautherin and Brink[61] assuming that it originates from two-nucleon interactions in the triplet-P state. Here $\rho(r) = \rho_n(r) + \rho_p(r)$ and $W_{\ell s}$ is a constant determined from the SOS in laboratory nuclei and listed in Table 5.17. The form of this potential, proportional to a radial derivative of the densities, indicates that apart from other dependences, the SOS obtained with a given EDF will depend on the nucleon central densities given by that model. The two modified versions of FPS21 just described, for the same value $W_{\ell s}$ = 110MeV used in Table I, each give a SOS in ^{7}n of 2.2 MeV. The reduction from the value of 3 MeV for unmodified FPS21 occurs because of the reduced central densities induced by the extra gradient term. The SOS in these modified models is still about double the value predicted by CVMC and GFMC, however.

As discussed in Ref. [51], more than half the SOS in ^{15}N comes from three-nucleon contributions involving either two neutrons and a proton or vice-versa. In contrast, the three-body interaction and clusters give a very small contribution to the SOS in ^{7}n in CVMC calculations. This suggests adding terms to the EDF that will produce a spin-orbit potential:

$$V_{\ell s}^{(n)}(r) = W_{\ell s}^{(2)} \frac{1}{r} \frac{d}{dr}(\rho(r) + \rho_n(r)) + W_{1,\ell s}^{(3)} \frac{1}{r} \frac{d}{dr}(\rho_n(r)\rho_p(r)) + W_{2,\ell s}^{(3)} \frac{1}{r} \frac{d}{dr}(\rho_p(r))^2 \qquad (5.20)$$

having separate two- and three-body contributions. The $(\rho_n(r))^2$ term is omitted because three-neutron clusters seem to give negligible $V_{\ell s}^{(n)}(r)$. In neutron drops only the two-body $W_{\ell s}^{(2)}$ contributes, while in N $\sim$ Z nuclei, like ^{15}N, $\rho_n(r)\rho_p(r) \sim \rho_p(r)^2$ and the sum $W_{\ell s}^{(3)} = W_{1,\ell s}^{(3)} + W_{2,\ell s}^{(3)}$ is the only relevant new parameter.

With this modification to the spin-orbit interaction, and the FPS21 parameterization for the central interaction (including the gradient term for the neutron drops but not for ^{15}N), we can fit the spin-orbit splittings of ^{15}N and ^{7}n exactly, using the parameter values $W_{\ell s}^{(2)}$ = 61 MeV fm^5 and $W_{\ell s}^{(3)}$ = 745 MeV fm^8. The spin-orbit splittings in the eight closed-shell nuclei mentioned earlier are modified only slightly, a not unexpected result in view of their relatively small values of neutron excess

Chapter 6

Conclusions

6.1 Tying It All Together

At this point we can step back and look at our results to see what can be learned about our Hamiltonian. Table 6.1 shows the experimental and GFMC energy expectation values for all of the nuclei studied in this work along with their deviation from the experiment energies. Note that there are only three states, ^6Li 1^+, ^6He 0^+, ^7Li $\frac{7}{2}^-$, (and their isospin flipped states) where the extrapolated energy values are outside the error bars of the upper bound average $\bar{E}$. In these systems we will take the GFMC energy to be the extrapolated value. In all other systems, the value of $\bar{E}$ will be considered to be the converged energy. Figure 6.1 shows the 3- to 7-body spectrum for the data in Table 6.1.

A comparison of the calculated binding energies to the experimental values points to problems which are likely due to deficiencies in the Hamiltonian. The first sign of trouble is the failure to reproduce the spin-orbit splitting in ^5He, specifically the binding energy of the $\frac{3}{2}^-$ state. This, coupled with the systematic underbinding of the 3^+, 0^+, and 2^+ ^6Li states, indicates that perhaps the interaction needs to be more attractive in $p\frac{3}{2}$ states. In the 7-body spectrum, the underbinding is more dramatic. However, we can only be confident of our results for the $\frac{1}{2}^-$ and $\frac{3}{2}^-$ states as they are well separated from their excited states. Even these states are underbound by $\sim$ 2 MeV.

It possible that the errors in the spectrum of states with A$\geq$5 are caused by our simplistic approach to three-nucleon interactions. It should be pointed out that, previous to these calculations, we had only two data points, ^3H and ^4He, with which to tune our three-nucleon interaction. Since both of the ^3H and ^4He binding energies can be fit with the single parameter $A_{2\pi}$, it was not possible to explore more sophisticated models of three-

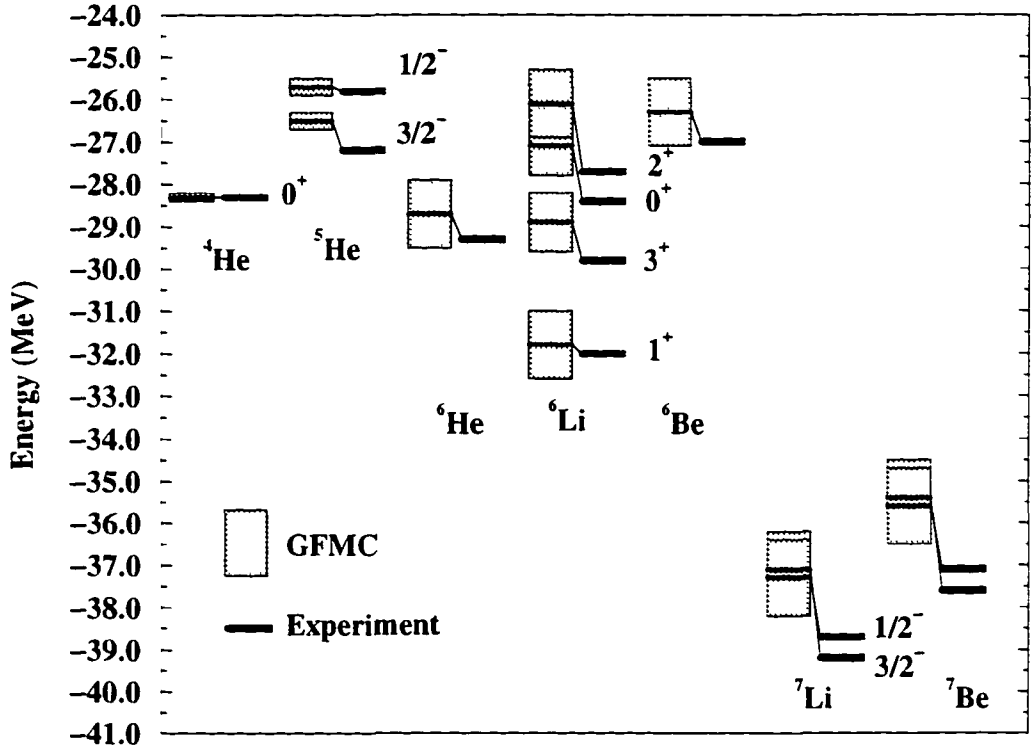


Figure 6.1: The GFMC energy spectrum for 4- to 7-body nuclei compared to experiment.

Table 6.1: Ground state energies calculated using the Argonne v_{18} + Urbana IX interactions

Nucleus	Expt.	$\bar{E}$	Extrapolated	Error
$^3H(1/2^+)$	-8.48	-8.47(2)		-0.01(2)
$^3He(1/2^+)$	-7.72	-7.72(2)		0.00(2)
$^4He(0^+)$	-28.3	-28.3(1)		0.0(1)
$^5He(3/2^-)$	-27.2	-26.5(2)		-0.7(5)
$^5He(1/2^-)$	-25.8	-25.7(2)		-0.1(2)
$^6He(0^+)$	-29.3	-28.0(4)	-28.7(8)	-0.6(8)
$^6Li(0^+)$	-28.4	-27.2(5)	-27.1(7)	-1.2(5)
$^6Li(1^+)$	-32.0	-31.3(5)	-31.8(8)	-0.2(8)
$^6Li(2^+)$	-27.7	-26.2(3)	-26.1(8)	-1.5(3)
$^6Li(3^+)$	-29.8	-28.7(5)	-28.9(7)	-1.1(5)
$^7Li(3/2^-)$	-39.2	-37.1(5)	-37.1(9)	-2.1(5)
$^7Li(1/2^-)$	-38.7	-37.3(5)	-37.3(9)	-1.4(5)
$^6Be(0^+)$	-27.0	-25.6(4)	-26.3(8)	-0.7(8)
$^7Be(3/2^-)$	-37.6	-35.4(6)	-35.4(9)	-2.2(6)
$^7Be(1/2^-)$	-37.1	-35.6(6)	-35.6(9)	-1.5(6)

nucleon interactions. With this in mind, it is pleasing to find that the errors in the ab-inito 5- through 7-body calculations are only on the order of 10% to 20% of the three-nucleon interaction. It is hoped that this spectrum will provide data points to explore more detailed models of three-body forces.

One might question what flexibility in the current model can be exploited to reproduce the experimental spectrum. Disregarding Fujita-Miazawa's estimate for $A_{2\pi}$ and ignoring the estimate of U_o from the saturation density of nuclear matter, we can try to determine the values of $A_{2\pi}$ and U_o by performing a least squares fit to the observed spectrum. Using the values of $V^{2\pi}_{ijk}$ and V^{R}_{ijk} from Tables 5.10, 5.11, and 5.4, we varied $A_{2\pi}$ and U_o to minimize the χ^2 for the deviation of the new energies from the experimental energies. The only constraint on our fit has been that U_o remain positive so that there is a repulsive term to balance the attraction. Table 6.2 shows the results of our fit. The optimum parameters are setting U_o to zero and a decrease in the strength of $A_{2\pi}$ by roughly 70%. This is not surprising since the contribution from the repulsive part of the potential is close to 50% of the attractive part in almost all of the nuclei considered. The χ^2 for this fit is 3.29. For comparison, a fit was conducted to search for the optimum $A_{2\pi}$ keeping U_o fixed at its current value of 0.0048

Table 6.2: Fits to the experimental spectrum. All energies are in MeV

Nucleus	Expt.	GFMC Error	Fit Error (U_o=0.005)	Fit Error (U_o=0.0)
$^3H(1/2^+)$	-8.48	-0.01(2)	0.04(2)	0.00(2)
$^4He(0^+)$	-28.3	0.0(1)	0.31(1)	0.3(1)
$^5He(3/2^-)$	-27.2	-0.7(5)	-0.4(1)	-0.3(1)
$^5He(1/2^-)$	-25.8	-0.1(2)	0.2(1)	0.2(1)
$^6He(0^+)$	-29.3	-0.6(8)	-0.2(8)	-0.4(8)
$^6Li(0^+)$	-28.4	-1.2(5)	-0.8(5)	-0.5(5)
$^6Li(1^+)$	-32.0	-0.2(8)	0.2(8)	0.5(8)
$^6Li(2^+)$	-27.7	-1.5(3)	-1.1(3)	-0.8(3)
$^6Li(3^+)$	-29.8	-1.1(5)	-0.7(5)	-0.1(5)
$^7Li(1/2^-)$	-38.7	-1.4(5)	-0.9(6)	-1.2(6)
$^7Li(3/2^-)$	-39.2	-2.1(5)	-1.6(6)	-0.7(6)
$^7Li(5/2^-)$	-32.5	-4.9(7)	-4.4(7)	-3.9(7)
$^7Li(7/2^-)$	-34.5	-2.7(8)	-2.2(8)	-1.5(8)

MeV; the resulting $A_{2\pi}$ was -0.03 MeV with χ^2=5.66. The results of this fit are also included in Table 6.2. As one can see these fits are not very different. In both fits, ^4He is overbound and the A = 5 to 7 p-shell nuclei are underbound. We thus conclude that additional terms in V_{ijk} must be considered to reproduce nuclear binding energies with high precision. Also, the present calculations cannot determine U_o and $A_{2\pi}$ separately.

Attempts are being made to explore a parameterization of the three-nucleon interaction which improves the errors in the A> 5 spectrum while maintaining the ^3H and ^4He energies. Such interactions should be selectively attractive in the p-waves. Thus far such attempts have fixed the ^5He spin-orbit splitting, but the improvement to the 6- and 7-body spectrum has been marginal.

A much more difficult source of trouble might be our perturbative treatment of the terms quadratic in the angular momentum. Although the effect of such a treatment is small in ^3H and ^4He [41], it is possible that for A≥5, where the wave functions have a much larger p-wave component, the corrections may become significant. As we have discussed in Chapter 4, a suitable technique for including these terms in the propagator and the sampling has yet to be developed. Another possible candidate for the cause of these problems are relativistic effects. Though it has been found that the relativistic corrections scale like the three nucleon

interaction in ^3H and ^4He[2], the effect of these corrections in p-wave nuclei has yet to be assessed.

One of the main goals of the Argonne v_{18} potential was to accurately reproduce isospin symmetry breaking effects in nuclear ground states. We have found in this work that the potential successfully reproduces the isovector mass splittings $^3He -^3 H$, $^6Be -^6 He$, and $^7Be -^7 Li$. The isotensor splitting, $\frac{1}{2}(^6Be +^6 He) -^6 Li$, was found to be too large. It is unclear at this point if this is due to a failure of the potential or our treatment of ^6Be as a bound state when it is actually a narrow resonance.

6.2 Fermion Sign Problem

Frequently, the quality of our results was limited by the onset of the "sign problem". We monitor the "sign problem" by keeping track of the normalization (Eq. 4.8) as a function of imaginary time. When the normalization drops below 0.8 that means that roughly 10% of the configurations are carrying a negative weight. Imperially, we limit our evolution in imaginary time to the range where the normalizations are above 0.6 (about 20% carrying negative weight.) After that point, the statistical noise tends to become significant. Figure 6.2 shows a plot of the normalization (Eq. 4.8) as a function of τ for several of the systems studied in this work. As the number of nucleons increases the normalization drops faster as a function of τ. This is expected since, by increasing the number of nucleons, we have increased the number of nodal surfaces which can be crossed. Furthermore, we have found that the "sign problem" in the neutron drop system is aggravated by increasing the well depth (increasing the density). By increasing the density, the nodes are closer and more likely to be crossed. From the graph, we can see that increasing the number of nucleons limits the propagation time and thus our ability to obtain converged results. Clearly this problem will limit the size of the system we will be able to study with our current technique.

6.3 Musings on the Future

There is a great deal that can be learned about our Hamiltonian and nuclear structure from our results and there is a great deal left to be done.

Wiringa and Schiavilla have recently completed elastic and inelastic form factor calculations for ^6Li using variational wave functions. Using consistent charge and current operators

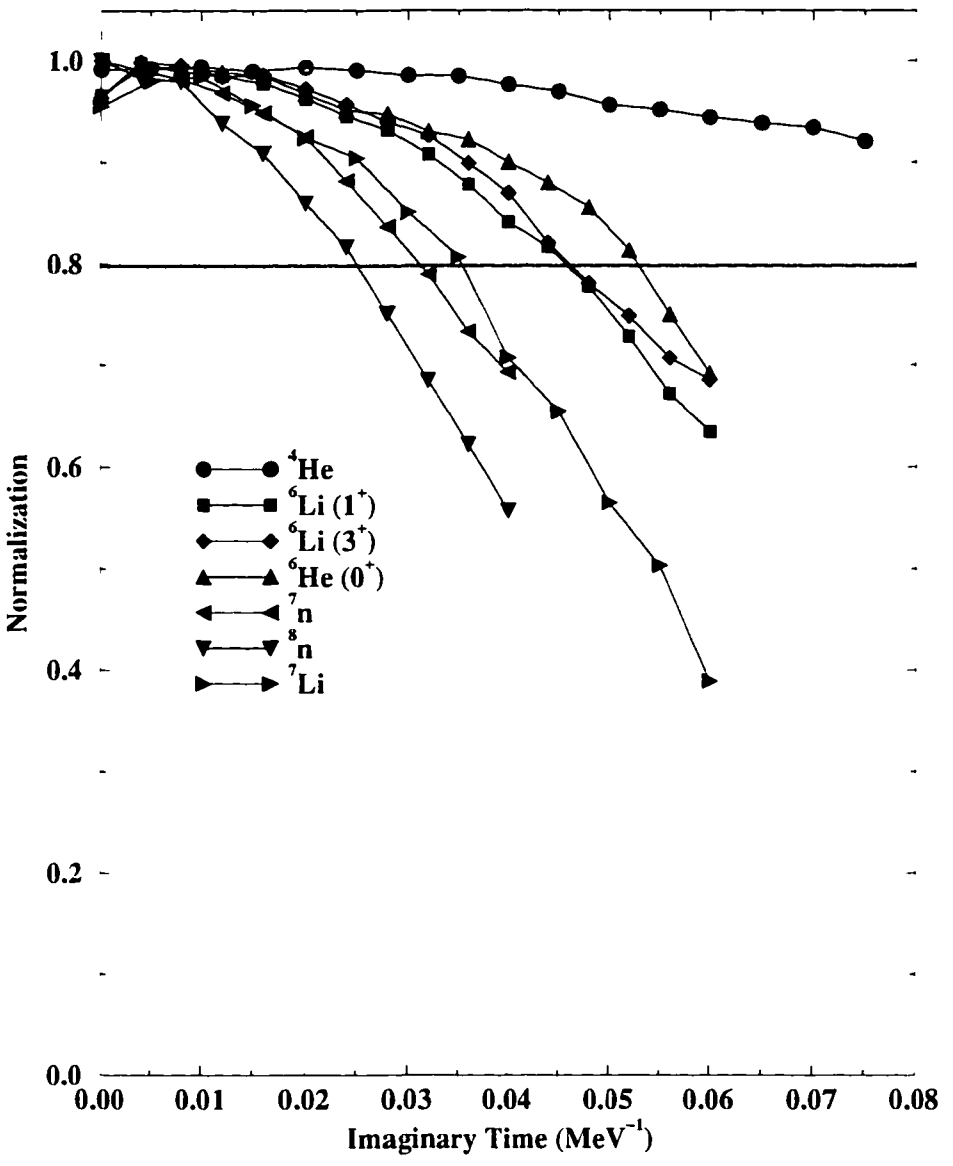


Figure 6.2: Normalization as a function of τ for systems studied in this work

with one- and two-body terms they have been able to evaluate the elastic longitudinal and transverse form factors as well as the tensor polarization. They have also calculated the transition form factors for the $(3^+:0)$, $(0^+:1)$, $(2^+:0)$, and $(2^+:1)$ 6-body states. Currently, we are trying to adapt these calculations for use with the GFMC wave functions.

Recently, interesting short range toroidal and dumbbell structures have been found in the two-nucleon density distributions in states with isospin T=0 and S=1 with spin projection $M_S=0$ and $M_S=\pm 1$. Work is under way to explore these structures in the deuteron and in ^3H, ^4He, ^6Li, ^7Li, and ^{16}O using variational techniques. They have provided insights into the deuteron structure and the D-wave components in ^3H, ^4He and ^6Li as well as revealing a possible connection with QCD. It would interesting to investigate these structures more accurately using the GFMC.

There are also several reactions which we hope to be able to study using the results of the GFMC calculations conducted in this work. For example, we will be able to examine the ^6He β-decay and the $\alpha - d$ radiative capture to ^6Li as well as the electron scattering response. In the seven-body system we hope to be able to study the ^4He $(^3He,\gamma)$ ^7Be radiative capture which is of interest in solar physics.

As we have said before, the Hamiltonian used in this work neglects relativistic corrections which are thought to be of the order of the three-nucleon interaction. A relativistic nuclear Hamiltonian has recently been developed [3, 4] and calculations are under way of relativistic corrections to 2- 4-body nuclei using variational techniques. These corrections will first be treated as a perturbation to our GFMC results, but ultimately a relativistic formulation of the GFMC techniques presented here will be required. It is clear that the exact two-body propagator discussed in Chapter 4 has the flexibility to tackle the expected non-local terms of a relativistic potential, however, the momentum dependent relativistic corrections will present the same sampling problems as the momentum dependent terms in our potential (Chapter 4).

It is natural at this point to question the future directions of GFMC in the study of few-body nuclear ground states. There are, without a doubt, many interesting calculations just one or two nucleons away from our current states. The eight-body nuclear ground states, ^8He, ^8Be, and ^8B provide very appetizing targets for the next generation of GFMC calculations. Investigations of ^8He will provide us with valuable information regarding the isospin dependence of the three-nucleon potential. There is also the important solar physics

reaction ^7Be (p,γ) ^8B and the simplest fission reaction $^8Be \rightarrow {}^4He + {}^4He$. There is the ^9n spin-orbit splitting and the ^{10}n ground state which will provide data on the strength of the two-neutron pairing interaction. Put a proton into the ^8n calculation and we have ^9H, which will serve as an interesting starting point for studies of halo nuclei.

There are several real obstacles which must be overcome if these future calculations are to become a reality. As we have seen in this work, the gap between the quality of the variational calculations and the GFMC calculations is widening as we increase the number of nucleons. The source of this difference must be understood if VMC is to continue to provide suitable starting points for GFMC calculations. We have also noted the increase of the sign problem as the number of nucleons is increased (Figure 6.2). Future calculations of larger nuclear systems will have to deal with this problem either by incorporating some of the techniques mentioned in Chapter 4 or, since it appears the first few time steps account for a large measure of the GFMC improvement over the variational, we could settle for just obtaining upperbounds on the ground state.

We have mentioned before that the computational complexity of these calculations grows rapidly with the number of nucleons. Clearly our appetite will have to be limited, but the question is how far can we go based on the current (and foreseeable) state of the art in high performance computing. A 10,000 node hour computation per state (10 calculations of 1000 configurations each) on the IBM SP2 at Cornell is within our current reach. Based on Table 5.1, a rough estimate of the size of the nuclear system which corresponds to this can be made. If the computation time scales like the size of the wave function (N) times the number of pairs (P), then the ratio of computer time to NP for ^7Li is 0.004. This means that it would take roughly 10,000 node hours for an 8-body system with a wave function of size 8900 elements. This type of extrapolation is very optimistic; it is likely that cache size limitations will cause severe performance penalties for N less than this value. However, even if we take 75% of this size, all of the neutron drop states mentioned above are within range. The ^8He, ^8Be, and ^8B states have wave function of size 5120, 3584, and 7168 respectively. This puts ^8He and ^8Be within the reach of the current state of the art. According to this analysis, a ^8B GFMC calculation will likely be at the very limits of what can be accomplished using the IBM SP2. On the other hand, with microprocessor performance doubling every three to four years, we can expect that GFMC calculations of the next generation of few-body nuclear ground states will be completed before the turn of the century.

Appendix: Reduced Isospin States

^{3}H T=$\frac{1}{2}$ T$_z$ = $\frac{1}{2}$

$$\frac{1}{\sqrt{2}}(|pnn\rangle - |npn\rangle)$$

$$\frac{1}{\sqrt{6}}(|pnn\rangle + |npn\rangle - 2|nnp\rangle)$$

^{4}He T=0

$$\frac{1}{2}(|pnpn\rangle - |nppn\rangle - |pnnp\rangle + |npnp\rangle)$$

$$\frac{1}{\sqrt{12}}(2|ppnn\rangle - |pnpn\rangle - |nppn\rangle - |pnnp\rangle - |npnp\rangle + 2|nnpp\rangle)$$

^{5}He T=$\frac{1}{2}$ T$_z$ = $\frac{1}{2}$

$$\frac{1}{2}(|pnpnn\rangle - |nppnn\rangle - |pnnpn\rangle + |npnpn\rangle)$$

$$\frac{1}{\sqrt{12}}(2|ppnnn\rangle - |pnpnn\rangle - |nppnn\rangle - |pnnpn\rangle - |npnpn\rangle + 2|nnppn\rangle)$$

$$\frac{1}{\sqrt{12}}(|pnpnn\rangle - |nppnn\rangle + |pnnpn\rangle - |npnpn\rangle - 2|pnnnp\rangle + 2|npnnp\rangle)$$

$$\frac{1}{6}(2|ppnnn\rangle - |pnpnn\rangle - |nppnn\rangle + |pnnpn\rangle + |npnpn\rangle - 2|nnppn\rangle - 2|pnnnp\rangle$$
$$-2|npnnp\rangle + 4|nnpnp\rangle)$$

$$\frac{1}{\sqrt{18}}(|ppnnn\rangle + |pnpnn\rangle + |nppnn\rangle - |pnnpn\rangle - |npnpn\rangle - |nnppn\rangle - |pnnnp\rangle$$
$$-|npnnp\rangle - |nnpnp\rangle + 3|nnnpp\rangle)$$

^{6}Li(1^+,2^+,3^+) T=0

$$\frac{1}{\sqrt{8}}(|pnpnpn\rangle - |nppnpn\rangle - |pnnppn\rangle + |npnppn\rangle - |pnpnnp\rangle + |nppnnp\rangle$$
$$+|pnnpnp\rangle - |npnpnp\rangle)$$

$$\frac{1}{\sqrt{24}}(2|ppnnpn\rangle - |pnpnpn\rangle - |nppnpn\rangle - |pnnppn\rangle - |npnppn\rangle + 2|nnpppn\rangle$$
$$-2|ppnnnp\rangle + |pnpnnp\rangle + |nppnnp\rangle + |pnnpnp\rangle + |npnpnp\rangle - 2|nnppnp\rangle)$$

$$\frac{1}{\sqrt{24}}(2|pnppnn\rangle - 2|nppnn\rangle - |pnpnpn\rangle + |nppnpn\rangle - |pnnppn\rangle + |npnppn\rangle$$
$$-|pnpnnp\rangle + |nppnnp\rangle - |pnnpnp\rangle + |npnpnp\rangle + 2|pnnnpp\rangle - 2|npnnpp\rangle)$$

$$\frac{1}{\sqrt{72}}(4|ppnpnn\rangle - 2|pnppnn\rangle - 2|nppnn\rangle - 2|ppnnpn\rangle + |pnpnpn\rangle + |nppnpn\rangle$$
$$-|pnnppn\rangle - |npnppn\rangle + 2|nnpppn\rangle - 2|ppnnnp\rangle + |pnpnnp\rangle + |nppnnp\rangle$$
$$-|pnnpnp\rangle - |npnpnp\rangle + 2|nnppnp\rangle + 2|pnnnpp\rangle + 2|npnnpp\rangle - 4|nnpnpp\rangle)$$

$$\frac{1}{6}(3|pppnnn\rangle - |ppnpnn\rangle - |pnppnn\rangle - |nppnn\rangle - |ppnnpn\rangle - |pnpnpn\rangle$$
$$-|nppnpn\rangle + |pnnppn\rangle + |npnppn\rangle + |nnpppn\rangle - |ppnnnp\rangle - |pnpnnp\rangle$$
$$-|nppnnp\rangle + |pnnpnp\rangle + |npnpnp\rangle + |nnppnp\rangle + |pnnnpp\rangle + |npnnpp\rangle$$
$$+|nnpnpp\rangle - 3|nnnppp\rangle)$$

^{6}He(0^+) T=1 T_z=-1

$$\frac{1}{2}(|pnpnnn\rangle - |nppnnn\rangle - |pnnpnn\rangle + |npnpnn\rangle)$$

$$\frac{1}{\sqrt{12}}(2|ppnnnn\rangle - |pnpnnn\rangle - |nppnnn\rangle - |pnnpnn\rangle - |npnpnn\rangle + 2|nnppnn\rangle)$$

$$\frac{1}{\sqrt{12}}(|pnpnnn\rangle - |nppnnn\rangle + |pnnpnn\rangle - |npnpnn\rangle - 2|pnnnpn\rangle + 2|npnnpn\rangle)$$

$$\frac{1}{6}(2|ppnnnn\rangle - |pnpnnn\rangle - |nppnnn\rangle + |pnnpnn\rangle + |npnpnn\rangle - 2|nnppnn\rangle$$
$$-2|pnnnpn\rangle - 2|npnnpn\rangle + 4|nnpnpn\rangle)$$

$$\frac{1}{18}(|ppnnnn\rangle + |pnpnnn\rangle + |nppnnn\rangle - |pnnpnn\rangle - |npnpnn\rangle - |nnppnn\rangle - |pnnnpn\rangle$$
$$-|npnnpn\rangle - |nnpnpn\rangle + 3|nnnppn\rangle)$$

$$\frac{1}{24}(|pnpnnn\rangle - |nppnnn\rangle + |pnnpnn\rangle - |npnpnn\rangle + |pnnnpn\rangle - |npnnpn\rangle$$
$$-3|pnnnnp\rangle + 3|npnnnp\rangle)$$

$$\frac{1}{72}(2|ppnnnn\rangle - |pnpnnn\rangle - |nppnnn\rangle + |pnnpnn\rangle + |npnpnn\rangle - 2|nnppnn\rangle + |pnnnpn\rangle$$
$$+|npnnpn\rangle - 2|nnpnpn\rangle - 3|pnnnnp\rangle - 3|npnnnp\rangle + 6|nnpnnp\rangle)$$

$$\frac{1}{12}(2|ppnnnn\rangle + 2|pnpnnn\rangle + 2|nppnnn\rangle - 2|pnnpnn\rangle - 2|npnpnn\rangle - 2|nnppnn\rangle$$
$$+|pnnnpn\rangle + |npnnpn\rangle + |nnpnpn\rangle - 3|nnnppn\rangle - 3|pnnnnp\rangle - 3|npnnnp\rangle$$
$$-3|nnpnnp\rangle + 9|nnnpnp\rangle)$$

$$\frac{1}{\sqrt{240}}(2|ppnnnn\rangle + 2|pnpnnn\rangle + 2|nppnnn\rangle + 2|pnnpnn\rangle + 2|npnpnn\rangle + 2|nnppnn\rangle$$
$$-3|pnnnpn\rangle - 3|npnnpn\rangle - 3|nnpnpn\rangle - 3|nnnppn\rangle - 3|pnnnnp\rangle$$
$$-3|npnnnp\rangle - 3|nnpnnp\rangle - 3|nnnpnp\rangle + 12|nnnnpp\rangle)$$

^{6}Li (0^+) T=1 T_z=0

$$\frac{1}{\sqrt{8}}(|pnpnpn\rangle - |nppnpn\rangle - |pnnppn\rangle + |npnppn\rangle + |pnpnnp\rangle - |nppnnp\rangle$$
$$-|pnnpnp\rangle + |npnpnp\rangle)$$

$$\frac{1}{\sqrt{24}}(2|ppnnpn\rangle - |pnpnpn\rangle - |nppnpn\rangle - |pnnppn\rangle - |npnppn\rangle + 2|nnpppn\rangle$$
$$+2|ppnnnp\rangle - |pnpnnp\rangle - |nppnnp\rangle - |pnnpnp\rangle - |npnpnp\rangle + 2|nnppnp\rangle)$$

$$\frac{1}{\sqrt{24}}(2|pnppnn\rangle - 2|nppnnn\rangle - |pnpnpn\rangle + |nppnpn\rangle - |pnnppn\rangle + |npnppn\rangle$$
$$+|pnpnnp\rangle - |nppnnp\rangle + |pnnpnp\rangle - |npnpnp\rangle - 2|pnnnpp\rangle + 2|npnnpp\rangle)$$

$$\frac{1}{\sqrt{72}}4|ppnpnn\rangle - 2|pnppnn\rangle - 2|nppnnn\rangle - 2|ppnnpn\rangle + |pnpnpn\rangle + |nppnpn\rangle$$
$$-|pnnppn\rangle - |npnppn\rangle + 2|nnpppn\rangle + 2|ppnnnp\rangle - |pnpnnp\rangle - |nppnnp\rangle$$
$$+|pnnpnp\rangle + |npnpnp\rangle - 2|nnppnp\rangle - 2|pnnnpp\rangle - 2|npnnpp\rangle + 4|nnpnpp\rangle)$$

$$\frac{1}{6}(3|pppnnn\rangle - |ppnpnn\rangle - |pnppnn\rangle - |nppnnn\rangle - |ppnnpn\rangle - |pnpnpn\rangle$$
$$-|nppnpn\rangle + |pnnppn\rangle + |npnppn\rangle + |nnpppn\rangle + |ppnnnp\rangle + |pnpnnp\rangle$$

$$+|nppnnp\rangle - |pnnpnp\rangle - |npnpnp\rangle - |nnppnp\rangle - |pnnnpp\rangle - |npnnpp\rangle$$
$$-|nnpnpp\rangle + 3|nnnppp\rangle)$$

$$\frac{1}{\sqrt{12}}(|pnppnn\rangle - |npppnn\rangle + |pnpnpn\rangle - |nppnpn\rangle + |pnnppn\rangle - |npnppn\rangle$$
$$-|pnpnnp\rangle + |nppnnp\rangle - |pnnpnp\rangle + |npnpnp\rangle - |pnnnpp\rangle + |npnnpp\rangle)$$

$$\frac{1}{6}(2|ppnpnn\rangle - |pnppnn\rangle - |npppnn\rangle + 2|ppnnpn\rangle - |pnpnpn\rangle - |nppnpn\rangle$$
$$+|pnnppn\rangle + |npnppn\rangle - 2|nnpppn\rangle - 2|ppnnnp\rangle + |pnpnnp\rangle + |nppnnp\rangle$$
$$-|pnnpnp\rangle - |npnpnp\rangle + 2|nnppnp\rangle - |pnnnpp\rangle - |npnnpp\rangle + 2|nnpnpp\rangle)$$

$$\frac{1}{\sqrt{72}}(3|pppnnn\rangle - |ppnpnn\rangle - |pnppnn\rangle - |npppnn\rangle + 2|ppnnpn\rangle + 2|pnpnpn\rangle$$
$$+2|nppnpn\rangle - 2|pnnppn\rangle - 2|npnppn\rangle - 2|nnpppn\rangle - 2|ppnnnp\rangle$$
$$-2|pnpnnp\rangle - 2|nppnnp\rangle + 2|pnnpnp\rangle + 2|npnpnp\rangle + 2|nnppnp\rangle$$
$$-|pnnnpp\rangle - |npnnpp\rangle - |nnpnpp\rangle + 3|nnnppp\rangle)$$

$$\frac{1}{\sqrt{120}}(3|pppnnn\rangle + 3|ppnpnn\rangle + 3|pnppnn\rangle + 3|npppnn\rangle - 2|ppnnpn\rangle - 2|pnpnpn\rangle$$
$$-2|nppnpn\rangle - 2|pnnppn\rangle - 2|npnppn\rangle - 2|nnpppn\rangle - 2|ppnnnp\rangle$$
$$-2|pnpnnp\rangle - 2|nppnnp\rangle - 2|pnnpnp\rangle - 2|npnpnp\rangle - 2|nnppnp\rangle$$
$$+3|pnnnpp\rangle + 3|npnnpp\rangle + 3|nnpnpp\rangle + 3|nnnppp\rangle)$$

^{7}Li ($\frac{3}{2}^-, \frac{1}{2}^-, \frac{7}{2}^-, \frac{5}{2}^-$) T=$\frac{1}{2}$, T$_z$=$\frac{1}{2}$

$$\frac{1}{\sqrt{8}}(|pnpnpnn\rangle - |nppnpnn\rangle - |pnnppnn\rangle + |npnppnn\rangle - |pnpnnpn\rangle + |nppnnpn\rangle$$
$$+|pnnpnpn\rangle - |npnpnpn\rangle)$$

$$\frac{1}{\sqrt{24}}(2|ppnnpnn\rangle - |pnpnpnn\rangle - |nppnpnn\rangle - |pnnppnn\rangle - |npnppnn\rangle$$
$$+2|nnpppnn\rangle - 2|ppnnnpn\rangle + |pnpnnpn\rangle + |nppnnpn\rangle + |pnnpnpn\rangle$$
$$+|npnpnpn\rangle - 2|nnppnpn\rangle)$$

$$\frac{1}{\sqrt{24}}(2|pnppnnn\rangle - 2|npppnnn\rangle - |pnpnpnn\rangle + |nppnpnn\rangle - |pnnppnn\rangle$$
$$+|npnppnn\rangle - |pnpnnpn\rangle + |nppnnpn\rangle - |pnnpnpn\rangle + |npnpnpn\rangle$$
$$+2|pnnnppn\rangle - 2|npnnppn\rangle)$$

$$\frac{1}{\sqrt{72}}(4|ppnpnnn\rangle - 2|pnppnnn\rangle - 2|npppnnn\rangle - 2|ppnnpnn\rangle + |pnpnpnn\rangle$$
$$+|nppnpnn\rangle - |pnnppnn\rangle - |npnppnn\rangle + 2|nnpppnn\rangle - 2|ppnnnpn\rangle$$
$$+|pnpnnpn\rangle + |nppnnpn\rangle - |pnnpnpn\rangle - |npnpnpn\rangle + 2|nnppnpn\rangle$$
$$+2|pnnnppn\rangle + 2|npnnppn\rangle - 4|nnpnppn\rangle)$$

$$\frac{1}{6}(3|pppnnnn\rangle - |ppnpnnn\rangle - |pnppnnn\rangle - |npppnnn\rangle - |ppnnpnn\rangle$$
$$-|pnpnpnn\rangle - |nppnpnn\rangle + |pnnppnn\rangle + |npnppnn\rangle + |nnpppnn\rangle$$
$$-|ppnnnpn\rangle - |pnpnnpn\rangle - |nppnnpn\rangle + |pnnpnpn\rangle + |npnpnpn\rangle$$

$$+|nnppnpn\rangle + |pnnnppn\rangle + |npnnppn\rangle + |nnpnppn\rangle - 3|nnnpppn\rangle)$$

$$\frac{1}{\sqrt{24}}(|pnpnpnn\rangle - |nppnpnn\rangle - |pnnppnn\rangle + |npnppnn\rangle + |pnpnnpn\rangle$$
$$-|nppnnpn\rangle - |pnnpnpn\rangle + |npnpnpn\rangle - 2|pnpnnnp\rangle + 2|nppnnnp\rangle$$
$$+2|pnnpnnp\rangle - 2|npnpnnp\rangle)$$

$$\frac{1}{\sqrt{72}}(2|ppnnpnn\rangle - |pnpnpnn\rangle - |nppnpnn\rangle - |pnnppnn\rangle - |npnppnn\rangle$$
$$+2|nnpppnn\rangle + 2|ppnnnpn\rangle - |pnpnnpn\rangle - |nppnnpn\rangle - |pnnpnpn\rangle$$
$$-|npnpnpn\rangle + 2|nnppnpn\rangle - 4|ppnnnnp\rangle + 2|pnpnnnp\rangle + 2|nppnnnp\rangle$$
$$+2|pnnpnnp\rangle + 2|npnpnnp\rangle - 4|nnppnnp\rangle)$$

$$\frac{1}{\sqrt{72}}(2|pnppnnn\rangle - 2|npppnnn\rangle - |pnpnpnn\rangle + |nppnpnn\rangle - |pnnppnn\rangle$$
$$+|npnppnn\rangle + |pnpnnpn\rangle - |nppnnpn\rangle + |pnnpnpn\rangle - |npnpnpn\rangle$$
$$-2|pnnnppn\rangle + 2|npnnppn\rangle - 2|pnpnnnp\rangle + 2|nppnnnp\rangle$$
$$-2|pnnpnnp\rangle + 2|npnpnnp\rangle + 4|pnnnpnp\rangle - 4|npnnpnp\rangle)$$

$$\frac{1}{\sqrt{216}}(4|ppnpnnn\rangle - 2|pnppnnn\rangle - 2|npppnnn\rangle - 2|ppnnpnn\rangle + |pnpnpnn\rangle$$
$$+|nppnpnn\rangle - |pnnppnn\rangle - |npnppnn\rangle + 2|nnpppnn\rangle + 2|ppnnnpn\rangle$$
$$-|pnpnnpn\rangle - |nppnnpn\rangle + |pnnpnpn\rangle + |npnpnpn\rangle - 2|nnppnpn\rangle$$
$$-2|pnnnppn\rangle - 2|npnnppn\rangle + 4|nnpnppn\rangle - 4|ppnnnnp\rangle + 2|pnpnnnp\rangle$$
$$+2|nppnnnp\rangle - 2|pnnpnnp\rangle - 2|npnpnnp\rangle + 4|nnppnnp\rangle$$
$$+4|pnnnpnp\rangle + 4|npnnpnp\rangle - 8|nnpnpnp\rangle)$$

$$\frac{1}{\sqrt{108}}(3|pppnnnn\rangle - |ppnpnnn\rangle - |pnppnnn\rangle - |npppnnn\rangle - |ppnnpnn\rangle$$
$$-|pnpnpnn\rangle - |nppnpnn\rangle + |pnnppnn\rangle + |npnppnn\rangle + |nnpppnn\rangle$$
$$+|ppnnnpn\rangle + |pnpnnpn\rangle + |nppnnpn\rangle - |pnnpnpn\rangle - |npnpnpn\rangle$$
$$-|nnppnpn\rangle - |pnnnppn\rangle - |npnnppn\rangle - |nnpnppn\rangle + 3|nnnpppn\rangle$$
$$-2|ppnnnnp\rangle - 2|pnpnnnp\rangle - 2|nppnnnp\rangle + 2|pnnpnnp\rangle + 2|npnpnnp\rangle$$
$$+2|nnppnnp\rangle + 2|pnnnpnp\rangle + 2|npnnpnp\rangle + 2|nnpnpnp\rangle - 6|nnnppnp\rangle)$$

$$\frac{1}{6}(|pnppnnn\rangle - |npppnnn\rangle + |pnpnpnn\rangle - |nppnpnn\rangle + |pnnppnn\rangle$$
$$-|npnppnn\rangle - |pnpnnpn\rangle + |nppnnpn\rangle - |pnnpnpn\rangle + |npnpnpn\rangle$$
$$-|pnnnppn\rangle + |npnnppn\rangle - |pnpnnnp\rangle + |nppnnnp\rangle - |pnnpnnp\rangle$$
$$+|npnpnnp\rangle - |pnnnpnp\rangle + |npnnpnp\rangle + 3|pnnnnpp\rangle - 3|npnnnpp\rangle)$$

$$\frac{1}{\sqrt{108}}(2|ppnpnnn\rangle - |pnppnnn\rangle - |npppnnn\rangle + 2|ppnnpnn\rangle - |pnpnpnn\rangle$$
$$-|nppnpnn\rangle + |pnnppnn\rangle + |npnppnn\rangle - 2|nnpppnn\rangle - 2|ppnnnpn\rangle$$
$$+|pnpnnpn\rangle + |nppnnpn\rangle - |pnnpnpn\rangle - |npnpnpn\rangle + 2|nnppnpn\rangle$$
$$-|pnnnppn\rangle - |npnnppn\rangle + 2|nnpnppn\rangle - 2|ppnnnnp\rangle + |pnpnnnp\rangle$$

$$+|nppnnnp\rangle - |pnnpnnp\rangle - |npnpnnp\rangle + 2|nnppnnp\rangle - |pnnnpnp\rangle$$
$$-|npnnpnp\rangle + 2|nnpnpnp\rangle + 3|pnnnnpp\rangle + 3|npnnnpp\rangle - 6|nnpnnpp\rangle)$$

$$\frac{1}{\sqrt{216}}(3|pppnnnn\rangle - |ppnpnnn\rangle - |pnppnnn\rangle - |npppnnn\rangle + 2|ppnnpnn\rangle$$
$$+2|pnpnpnn\rangle + 2|nppnpnn\rangle - 2|pnnppnn\rangle - 2|npnppnn\rangle - 2|nnpppnn\rangle$$
$$-2|ppnnnpn\rangle - 2|pnpnnpn\rangle - 2|nppnnpn\rangle + 2|pnnpnpn\rangle + 2|npnpnpn\rangle$$
$$+2|nnppnpn\rangle - |pnnnppn\rangle - |npnnppn\rangle - |nnpnppn\rangle + 3|nnnpppn\rangle$$
$$-2|ppnnnnp\rangle - 2|pnpnnnp\rangle - 2|nppnnnp\rangle + 2|pnnpnnp\rangle + 2|npnpnnp\rangle$$
$$+2|nnppnnp\rangle - |pnnnpnp\rangle - |npnnpnp\rangle - |nnpnpnp\rangle + 3|nnnppnp\rangle$$
$$+3|pnnnnpp\rangle + 3|npnnnpp\rangle + 3|nnpnnpp\rangle - 9|nnnpnpp\rangle)$$

$$\frac{1}{\sqrt{360}}(3|pppnnnn\rangle + 3|ppnpnnn\rangle + 3|pnppnnn\rangle + 3|npppnnn\rangle - 2|ppnnpnn\rangle$$
$$-2|pnpnpnn\rangle - 2|nppnpnn\rangle - 2|pnnppnn\rangle - 2|npnppnn\rangle - 2|nnpppnn\rangle$$
$$-2|ppnnnpn\rangle - 2|pnpnnpn\rangle - 2|nppnnpn\rangle - 2|pnnpnpn\rangle - 2|npnpnpn\rangle$$
$$-2|nnppnpn\rangle + 3|pnnnppn\rangle + 3|npnnppn\rangle + 3|nnpnppn\rangle + 3|nnnpppn\rangle$$
$$-2|ppnnnnp\rangle - 2|pnpnnnp\rangle - 2|nppnnnp\rangle - 2|pnnpnnp\rangle - 2|npnpnnp\rangle$$
$$-2|nnppnnp\rangle + 3|pnnnpnp\rangle + 3|npnnpnp\rangle + 3|nnpnpnp\rangle + 3|nnnppnp\rangle$$
$$+3|pnnnnpp\rangle + 3|npnnnpp\rangle + 3|nnpnnpp\rangle + 3|nnnpnpp\rangle - 12|nnnnppp\rangle)$$

Bibliography

[1] G. E. Brown, W. Weise, G. Baym, and J. Speth, Comments Nucl. Part. Phys. **XVII**, 39 (1987).

[2] J. Carlson, R. Schiavilla, and V. R. Pandharipande, Phys. Rev. C **47**, 484 (1993).

[3] J. L. Forest, V. R. Pandharipande, J. Carlson, and R. Schiavilla, Phys. Rev. C **52**, 576 (1995)

[4] J. L. Forest, V. R. Pandharipande, and J. L. Friar, Phys. Rev. C **52**, 568 (1995)

[5] R. Machleidt, Adv. Nucl. Phys. **19**, 189 (1989).

[6] M. Lacombe, B. Loiseau, J.M. Richard, R. Vinh Mau, J. Côté, P. Pirès, and R. de Tourreil, Phys. Rev. C **21**, 861 (1980).

[7] R. B. Wiringa, R. A. Smith, and T. L. Ainsworth, Phys. Rev. C **29**, 1207 (1984).

[8] R. B. Wiringa, V. G. J. Stoks, and R. Schiavilla, Phys. Rev. C **51**, 38 (1995).

[9] I.E. Lagaris and V.R. Pandharipande, Nucl. Phys. **A359**, 331 (1981).

[10] J. L. Friar, in *Modern Topics in Electron Scattering*, edited by B.Frois and I. Sick, (World Scientific, 1991), p.126

[11] W. Glöckle and H. Kamada, Phys. Rev. Lett. **71**, 971 (1993).

[12] J. Carlson Nucl. Phys. **A522**, 185c (1991)

[13] C. R. Chen, G. L. Payne, J. L. Friar, and B. F. Gibson, Phys. Rev. C **31**, 2266 (1985).

[14] R. B. Wiringa, Phys. Rev. C **43**, 1585 (1991).

[15] J. Lomnitz-Adler and V. R. Pandharipande, Nucl. Phys. **A342**, 404 (1980)

[16] J. Lomnitz-Adler and V. R. Pandharipande, Nucl. Phys. **A361**, 399 (1981)

[17] A. Arriaga, V. R. Pandharipande, R. B. Wiringa, Phys. Rev. C.

[18] M. H. Kalos and P. Whitlock *Monte Carlo Methods Volume 1: Basics* Wiley, New York (1986).

[19] N. Metropolis, et al., J Chem Phys. **21**, 1087 (1953).

[20] R. Schiavilla, V. R. Pandharipande, and A. Fabrocini, Phys. Rev. C **40**, 1484 (1989)

[21] M. H. Kalos, Michael A. Lee, P.A. Whitlock, G. V. Chester Phys. Rev. B **24** 115 (1981).

[22] D. M. Ceperly, Reviews of Modern Physics, **67**, 279 (1995).

[23] R. M. Panoff and J. Carlson, Phys Rev. Lett. **62** 1130 (1989).

[24] Y. Alhassid, G. Maddison, K. Langanke, K. Chow, and S.E. Koonin, Zeitschrift für Physik A **321** 677-685 (1985)

[25] G.H. Lang, C.W. Johnson, S.E. Koonin and W.E. Ormand Phys. Rev. C **48** 1518 (1993)

[26] D. J. Dean, et al., Phys. Rev. Lett. **72**, 4066 (1994).

[27] D. M. Ceperley and B. J. Alder, Phys. Rev. Lett. **45** 566 (1980);

[28] J. C. Grossman and L. Mitas, Phys. Rev. Lett. **74**, 1323 (1995).

[29] M. H. Kalos, Phys. Rev. **128** 4 1791 (1994).

[30] J. Carlson, Phys. Rev. C **36**, 2026 (1987)

[31] J. Carlson, Phys. Rev. C **38**, 1879 (1988)

[32] M. A. Lee, K. E. Schmidt, M. H. Kalos, and G. V. Chester, Phys. Rev. Lett. **46** 728 (1981).

[33] M. H. Kalos, J. Stat. Phys., **63** 1269 (1991)

[34] Shiwei Zhang and M. H. Kalos, Phys. Rev. Lett., **67**, 3074 (1991)

[35] Zhiping Liu, Shiwei Zhang, and M. H. Kalos, Phys. Rev. E, **50**, 3220 (1994)

[36] K. E. Schmidt and Michael A. Lee. Phys. Rev. E. **51**. 5495 (1995)

[37] W. Glöckle. H. Witala. and Th. Cornelius. Nucl. Phys. **A508**. 115C (1990).

[38] A. Kievsky. M. Viviani. and S. Rosati. Nucl. Phys. **A551**. 241 (1993); **A577**. 511 (1994).

[39] J. Carlson. Phys. Rev. C **38**. 1879 (1988)

[40] J. Carlson and R. Schiavilla. in Few Body Systems Suppl. **7**. edited by B. L. G. Bakker and R. van Dantzig (Springer-Verlag. Wien. 1994). p.349.

[41] H. Kamada and W. Glöckle. private communication (1994).

[42] J. Carlson. V. R. Pandharipande. and R. B. Wiringa. Nucl. Phys. **A401**. 59 (1983).

[43] R. Schiavilla. V. R. Pandharipande. and R. B. Wiringa. Nucl. Phys **A449**. 219 (1986)

[44] V. R. Pandharipande. Proceedings of a NATO Advanced Studies Workshop on Hadrons and Hadronic Matter. Cargese. France. (Plenum. 1989). p. 293-341

[45] Y. Wu. S. Ishikawa. and T. Sasakawa. Phys. Rev. Lett. **64**. 1875 (1990); **66**. 242 (1991).

[46] H. De Vries. C. W. De Jager. and C. De Vries. At. Data Nucl. Data Tables **36**. 495 (1987).

[47] R. A. Arndt and L. D. Roper. Nucl. Phys. **A209**. 447 (1973).

[48] J.Carlson. V.R. Pandharipande and R.B. Wiringa. Nucl. Phys. **A424**. 47 (1984).

[49] F. Ajzenberg-Selove. Nucl. Phys. **A413**. 1 (1984).

[50] B. S. Pudliner. V. R. Pandharipande. J. Carlson. and R. B. Wiringa. Phys. Rev. Lett. **74** 4396 (1995).

[51] S. C. Pieper. and V. R. Pandharipande. Phys. Rev. Lett. **70** 2541 (1993).

[52] J. R. Bergervoet. P. C. van Campen. R. A. M. Klomp. J.-L. de Kok. T. A. Rijken. V. G. J. Stoks. and J. J. de Swart. Phys. Rev. C **41**. 1435 (1990).

[53] V. G. J. Stoks. R. A. M. Klomp. M. C. M. Rentmeester. and J. J. de Swart. Phys. Rev. C **48**. 792 (1993).

[54] S. Cohen and D. Kurath, Nucl. Phys **73** 1 (1965); D. Kurath private communication (1996).

[55] C. J. Pethick, D. G. Ravenhall, and C. P. Lorenz, Nucl. Phys. **A584** 675 (1995).

[56] K.-L. Kratz, J.-P. Bitouzet, F.-K. Thielemann, P. Möller, and B. Pfeiffer, Ap. J. **403** 216 (1993). See also J. J. Cowan, F.-K. Thielemann, and J. W. Truran, Phys. Rep. **208** 267 (1991).

[57] For a recent review of these topics, see C. J. Pethick and D. G. Ravenhall, Annu. Rev. Nuc. Part. Sci. **45** 1 (1995).

[58] R.B. Wiringa and V.R. Pandharipande, Rev. Mod. Phys. **51** (1979) 821

[59] S. C. Pieper, R. B. Wiringa, and V. R. Pandharipande, Phys. Rev. **C46** 1741 (1992).

[60] M. B. Leuschner et al., Phys. Rev. C **49** 955 (1994).

[61] D. Vautherin and D. M. Brink, Phys. Rev. C **5** 626 (1972).

[62] D. G. Ravenhall, C. D. Bennett, and C. J. Pethick, Phys. Rev. Lett. **28** 978 (1972).

[63] P. J. Siemens, and V. R. Pandharipande, Nucl. Phys. **A173** 561 (1971).

[64] H. Krivine, J. Treiner, and O. Bohigas, Nucl. Phys. **A336** 155 (1980).

[65] B. Friedman and V. R. Pandharipande, Nucl. Phys. **A361** 502 (1981).

[66] B. S. Pudliner, A. Smerzi, J. Carlson, V. R. Pandharipande, Steven C. Pieper, and D. G. Ravenhall Phys. Rev. Lett. (In press)

[67] J. Dobaczewski, I. Hamamoto, W. Nazarewicz, and J. A. Sheikh, Phys. Rev. Lett. **72** 981 (1994).

Vita

Brian Pudliner was born in Carthage, NY on August 12, 1969. He promptly moved to Pennsylvania and spent his formative years in Walnutport, PA. He thought college would be a good idea, so he enrolled in Pennsylvania State University in August of 1987 and graduated with an Honors B.S. in Physics in May of 1991. In August of that year he found himself at University Illinois and for surviving his first year there with a University Fellowship, he received a M.S. in Physics in October of 1992. In the Spring of 1995, he was awarded the Felix T. Adler Fellowship and eventually decided to skip off to Lawrence Livermore National Lab in June of 1996.